Yang–Mills on a finite torus along Bałaban's route: a verified reduction

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Abstract

Bałaban's renormalization-group construction of four-dimensional SU(2) lattice Yang–Mills theory rests on the control of the running coupling, his Theorem 2, whose proof was never published. We prove that a weak form of Theorem 2 (two-sided bounds on $1/g_k^2$ with an additive constant), which is enough for Bałaban's Theorem 1, follows from one uniform inequality (H2) on a two-step minimizer map together with named fixed-level and history inputs; the shooting lemmas, the arithmetic of Theorem K, and the bridge lemmas of the certifier are formalized in Lean 4. We prove that, on a fixed torus, a uniform moment bound (H-U) gives compactness of the lattice Schwinger functions of the centred plaquette density, and that every limit point satisfies the torus forms of the Osterwalder–Schrader axioms OS0–OS3. The bound (H-U) is reduced, not proved: it rests on sourced versions of Bałaban's bounds, on a hypothesis (FX) that is now reduced to a holomorphy bound (H-hol) for Bałaban's increments together with history bounds and three named inputs of the large-field step, and on a bridge from the source form to (H-U), and one dependency table records the status of each block. At $L = 13$, inside Bałaban's range $L > 11$, (H2) is a certified theorem: an explicit majorant and an outward-rounded interval branch-and-bound give $\rho_1(j, K)^2 \le 10 < 13$ for every level j (including $j = \infty$) and every momentum K, so $\rho_2 \le 10 < 13$, given our reading of Bałaban's averaging operator; the result is not a solution of the Clay Millennium problem.

Contents

1 Introduction **4**
- 1.1 The problem . . . 4
- 1.2 Bałaban's programme and the missing Theorem 2 . . . 4
- 1.3 The two targets and the main results . . . 4
- 1.4 Status conventions . . . 6
- 1.5 Relation to other work . . . 7
- 1.6 Organization . . . 8

2 Setup and notation **8**
- 2.1 Lattice, measure, and scales . . . 8
- 2.2 Range of L . . . 8
- 2.3 The coupling sequence . . . 9
- 2.4 One-loop notation . . . 9
- 2.5 Observables and Schwinger functions . . . 9
- 2.6 Names . . . 10

3 The coupling flow and Theorem 2 **10**
3.1 What Bałaban's Theorem 1 actually needs . . . 10
3.2 The weak form proved conditionally . . . 11
3.3 Literal (0.31) . . . 11
3.4 Shooting lemmas . . . 11
3.4.1 Lemma D′ (shooting without early positivity) [PROVED+VERIFIED; LEAN] . . . 12
3.4.2 Lemma D″ (tail-sum form; no index J) [PROVED+VERIFIED; LEAN] . . . 12
3.4.3 Lemma D″ meets Balaban's proof: the cross-scale sweep [PROVED, UNVERIFIED (SWEEP); 4 OF 19 TABLE ROWS VERIFIED] . . . 13
3.4.4 Lemma D and Lemma E [PROVED+VERIFIED; LEAN] . . . 14
3.4.5 History dependence (L5) [OPEN] . . . 14
3.5 The exact structure of the one-loop sums . . . 14
3.5.1 Telescoping [PROVED+VERIFIED, CORRECTED SCOPE] . . . 15
3.5.2 Anchor-free identity [PROVED+VERIFIED] . . . 15
3.5.3 The constant c [PROVED+VERIFIED] . . . 15
3.5.4 The decomposition $S_k = ck \log L + \kappa_W^F - \kappa_k$ . . . 15
3.5.5 Level-uniform symbol facts used below [PROVED+VERIFIED] . . . 16
3.6 Bounded window sums: two formally distinct conditional routes . . . 16
3.6.1 Route K: Theorem K [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS] . . . 17
3.6.2 Route B: Theorem B_{PW} [PROVED+VERIFIED, CONDITIONAL (LOGIC OF THEOREM B_{PW})] . . . 18
3.6.3 Route B inputs after `BETA_FP_U` [PROVED, UNVERIFIED; NO VERIFIER FILE] . . . 19
3.6.4 Comparison . . . 19
3.7 The hypothesis (B2) of Route K, and the single uniform inequality (H2) . . . 20
3.7.1 Proved . . . 20
3.7.2 Refuted, and the failed one-step bound . . . 21
3.7.3 (H2) and the two-step theorems [PROVED+VERIFIED, CONDITIONAL ON (H2)] 21
3.7.4 One-step sufficiency [ARITHMETIC PROVED; INPUT CERTIFIED AT $L = 13$] . . . 22
3.7.5 The precise remaining inequality . . . 22
3.8 Conclusion of the reduction . . . 22
3.9 The convergence route: what it adds, and its status . . . 23
3.9.1 Components and status . . . 23
3.9.2 Early steps and j_* . . . 24

4 The finite-torus existence theorem **24**
4.1 What "exists" means here . . . 24
4.2 Hypothesis delivered by the blocks . . . 25
4.3 Theorem (finite-torus existence, compactness form) . . . 25
4.4 Theorem (nontriviality), conditional on (B-NT) . . . 26
4.5 Proof modulo blocks . . . 26
4.5.1 (B-T2) Theorem 2: the running coupling . . . 26
4.5.2 (B-S) Sourced bounds E14-a..g . . . 27
4.5.3 (B-N1) Bridge: source-form bounds ⇒ (H-U) [OPEN] . . . 27
4.5.4 (B-C) Compactness [PROVED HERE, GIVEN (H-U)] . . . 28
4.5.5 (B-REG) Regularity (OS0) [PROVED HERE, GIVEN (H-U)] . . . 28
4.5.6 (B-TI) Translation and hypercubic invariance (OS1) [PROVED HERE, GIVEN (H-U)] . . . 28

4.5.7 (B-RP) Reflection positivity (OS2) . . . 29
4.5.8 (OS3) Symmetry and reality [PROVED HERE] . . . 30
4.5.9 (B-NT) Nontriviality: what is needed . . . 30
4.6 Dependency table . . . 31
4.7 What the theorem does not give . . . 40
4.8 Corrections after verification . . . 41

5 The sources block **42**
5.1 Target 1: inputs of Theorem 2w . . . 42
5.2 Target 2, step 2: the sourced Theorem 1 (ledger B11) . . . 43
5.3 Step 3: the (S) clauses and (FX) (ledgers B21, B19) . . . 48
5.4 Steps 4–5: N3, (H-hol), and the large-field inputs (ledgers B18, B16, B17) . . . 49
5.5 Step 6: thresholds and second history derivatives (ledgers B15, B14) . . . 52
5.6 Step 7: history bounds and the gain (ledger B13) . . . 54
5.7 Step 8: Bałaban-level propagator inputs (ledgers A1–A4) . . . 56
5.8 Step 9: the alternative (E_U) route . . . 61

6 Numerical and certified results **62**
6.1 First increment $\beta_1(0)$ . . . 62
6.2 Later increments ($L = 2$ only) . . . 63
6.3 Quantities of the Route K proof . . . 63
6.4 Values of (H2) [NUMERICAL; (H2) AT $L = 13$ IS CERTIFIED IN SECTION 6.7] . . . 64
6.5 One-step values . . . 65
6.6 A rigorous point enclosure at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED] . . . 65
6.7 (H2) at $L = 13$ [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)] . . . 66
6.8 Certification plan and j_0 [DESIGNED; SUPERSEDED AT $L = 13$ BY SECTION 6.7] . . . 67

7 What remains **68**
7.1 Certification . . . 68
7.2 Fixed-level and structural inputs of Theorem 2w . . . 68
7.3 Items of the sources block . . . 70
7.4 Verification debt . . . 72
7.5 Open for stronger statements . . . 72
7.6 Why the mass gap is not on this route . . . 73
7.7 Trust ledger . . . 73

A Lean formalization **77**

B Index of results by status **79**

C Label map **83**

D Revision history **85**

1 Introduction

1.1 The problem

The Clay problem on quantum Yang–Mills theory asks for a four-dimensional quantum gauge theory with compact simple gauge group that satisfies axioms at least as strong as those of Osterwalder and Schrader [19, 20], and that has a mass gap [15]. One constructive approach starts from Wilson's lattice action on a torus with lattice spacing ε and takes $\varepsilon \to 0$. The bare coupling $g_0(\varepsilon)$ must then go to zero at the rate that asymptotic freedom prescribes, so that the effective coupling at a fixed physical scale stays fixed. This paper treats the first limit only: lattice spacing to zero on a fixed torus. It does not treat the infinite-volume limit or the mass gap (Section 4.7).

1.2 Bałaban's programme and the missing Theorem 2

Between 1984 and 1989 Bałaban built a renormalization-group (RG) analysis of lattice gauge theories [1, 2, 3, 5, 4, 6, 7, 8, 10, 9, 11, 12]. His main output for $d = 4$ is Theorem 1 of CMP 122 II: uniform bounds on the effective densities at every RG scale, under the hypothesis that the sequence of effective couplings stays small. The hypothesis was to be discharged by his Theorem 2, which controls the running coupling. The two statements read as follows.

Theorem 2 (CMP 109 (1987), p. 259): "Let $d = 4$, $G = SU(2)$, and let γ be a sufficiently small positive constant, then for a sufficiently small positive g there exists a bare coupling constant $g_0 = g_0(\varepsilon, g)$ such that the sequence of the effective coupling constants g_k is contained in the interval $]0, \gamma]$, and $g_K = g$. Moreover, there exist constants β, β', $0 < \beta \leq \beta'$, such that g_k satisfy the inequality $1/g^2 + \beta \log(L^k\varepsilon)^{-1} \leq 1/g_k^2 \leq 1/g^2 + \beta' \log(L^k\varepsilon)^{-1}$. (0.31)" Balaban adds that the proof "will be given in a separate paper". It was never published (CMP 122 II p. 355, verbatim: "has not been published yet, so we have Theorem 1 with the assumption").

Theorem 1 (CMP 122 II p. 355): "If the sequence of the effective coupling constants is contained in an interval $]0, \gamma]$ with a sufficiently small positive γ, then the effective densities ρ_k have the form, and satisfy all the conditions and bounds, described in Sect. 2 [III]." The bound (0.1) there reads $\chi_k \exp[-(1/g_k^2)A(U_k) - E_-|T_\eta|] \leq \rho_k \leq \exp E_+|T_\eta|$ (corrected reading, `BALABAN_EXCERPTS_CHECK`[1] item 3).

Without Theorem 2, the hypothesis of Theorem 1 is an assumption on the unknown coupling flow. Every continuum statement built on Bałaban's bounds inherits this assumption.

1.3 The two targets and the main results

The paper has two targets. The first target is a proof of enough of Theorem 2 to remove the coupling hypothesis from Theorem 1. The second target is a finite-torus existence theorem for the continuum limit in compactness form, with the Osterwalder–Schrader properties proved on the limit. Both results are reductions: each holds under named hypotheses, and the paper records the status of every hypothesis.

Theorem A (weak Theorem 2; proved in Section 3)**.** *Let $G = SU(2)$, $d = 4$, and L odd, $L > 11$ (or L odd, $L \geq 3$, with Bałaban's inductive assumption (H0) and his construction of $\beta_j(g)$ at this L as explicit hypotheses).*

(i) Assume the uniform inequality (H2) (Section 3.7.3) [CERTIFIED AT $L = 13$ (THEOREM 6.1); CERTIFIED AT $L = 13$ (REFINED) AND AT $L = 15, 17, 21$ (COARSE GRID,

[1] Project files named in this paper live in the `notes/` directory of the `yang-mills` repository [14].

NUMERICS/H1_OTHER_L.MD); A HYPOTHESIS AT OTHER L], *the fixed-level inputs (PI), (F), (M_j), V1, (G2) (Section 3.6.1), the history hypothesis L5 (Section 3.4.5), and the bound 7′, $|\beta_{k+1}(h,g) - \beta_{k+1}(0)| \le C_\beta g$ on every (non-monotone) past (item 7′ of Section 7)* [PROVED+VERIFIED, CONDITIONAL ON (H0) (LEDGER A5) OR ON THE (G-GAIN) INPUTS (LEDGER B13)]. *Then the conclusions (b), (c′), (S) of Theorem 3.1 (Theorem 2w) hold. Status of the implication:* [PROVED+VERIFIED, CONDITIONAL], *by Route K (Theorem 3.8 and Lemma 3.4).*

(ii) Given the conclusions of Theorem 2w, Bałaban's Theorem 1 holds for the bare coupling $g_0(K,g)$, for every K and every $g \in]0,g_1]$, without his coupling assumption. Status: [PROVED, UNVERIFIED (SWEEP), CONDITIONAL ON ROUTE K OR ROUTE B INPUTS, ON L5, ON THE FINITE-VOLUME IDENTIFICATION OF THE STEP (R-TERM PART, LEDGER B8, OPEN), AND ON BALABAN'S PUBLISHED PROOFS] *(Section 3.8).*

A second, formally distinct route (Route B, Theorem B_{PW} in Section 3.6.2) also gives (i), under its own input set (items B1–B7 of Section 7.2). Both routes rest on (H2).

At $L = 13$ the hypothesis (H2) is discharged by Theorem 6.1, so the weak form of Theorem 2 rests only on the fixed-level inputs, L5, item 7′, and the Bałaban-level items of the trust ledger (with the residue of Theorem 6.1, Section 6.7).

Theorem B (finite-torus existence; proved in Section 4)**.** *Let $G = SU(2)$, $d = 4$, L odd and large, and $\ell \in 2\mathbb{N}$ fixed. Let $g_0 = g_0(\varepsilon, g)$ be a shooting solution whose effective couplings satisfy $g_k \in]0,\gamma]$ for $0 \le k \le K$ and $g_K = g$. Assume the uniform bound (H-U) of Definition 4.1. When (H-U) is taken from the blocks rather than assumed, the chain also uses the $\kappa \to \kappa/2$ decay recovery of the sourced small-field step (ledger B22), a* [VERIFIED REDUCTION] *by $L/2$-rescaling that rests on the unprinted large-field T-step rate loss (ledger A23). Then the lattice Schwinger functions of the centred plaquette density are relatively compact as $\varepsilon = L^{-K} \to 0$, and every limit family satisfies the torus forms of (OS0)–(OS3). This is Theorem 4.2,* [PROVED MODULO NAMED BLOCKS]. *Nontriviality is the separate Theorem 4.3, conditional on (NT).*

The sourced step also uses the unit-scale condition (U), $\rho_1 \le c_* r_0 / M^a$ at $\lambda_K = 1$ (C23): this is a choice of the source-gradient constant, made once after r_0 and before K, not a hypothesis on test functions, and the bridge B-N1 meets it by rescaling at a C^n cost (this (U) is not the Route B hypothesis (U)).

The trust boundary of the two theorems is recorded in the trust ledger (Section 7.7): Theorem A rests on 4 Bałaban-level statements and one external result, and Theorem B rests on 17 Bałaban-level statements and four external results. Of the 20 Bałaban-level statements in the ledger, 13 are printed by Bałaban without a proof (3 of the 4 behind Theorem A and 11 of the 17 behind Theorem B). Four of these (A1–A4, the propagator and covariance bounds) are now verified reductions to published theorems of CMP 99 and CMP 102; their joint residue is Bałaban's walk expansions of the final propagators (through the walk-expansion form A1w of A1), the complex walk bound (W^c) in the exponential gauge and the decay of $C^* \Delta_k C$ at real U (A7) under A2, the regular-class identification A18, two unprinted conventions, and a stated torus-side hypothesis.

The hypothesis (H-U) is the output of three blocks: Theorem 2 (B-T2, that is, Theorem A), sourced versions of Bałaban's bounds (B-S), and a bridge from the source form to (H-U) (B-N1). The dependency table, Section 4.6, gives the status of each block and is the single status table of the paper. Section 7 lists what remains.

What the paper does not give: literal (0.31) at every K (Section 3.3), the limit of $\beta_j(0)$ and a rate (Section 3.9), uniqueness of the limit, infinite volume, rotation invariance, and a mass gap

(Section 4.7). Theorem B is a compactness ("weak-existence") result. Under Jaffe–Witten footnote 2 it is not an admissible solution of the Millennium problem (Section 4.1).

1.4 Status conventions

Every result carries a status tag in small capitals. The tags mean the following.

- [PROVED+VERIFIED]: proved in a project file and confirmed by an independent verifier file.
- [PROVED+VERIFIED, CONDITIONAL ON X]: the implication is confirmed; X is open or assumed.
- [PROVED, UNVERIFIED]: proved in a project file; no verifier has graded it yet.
- [LEAN]: formalized in Lean 4 + Mathlib, no `sorry`, axioms propext, Classical.choice, Quot.sound only.
- [VERIFIED QUOTE]: read on the page image of Balaban's paper (`reports/BALABAN_EXCERPTS_CHECK.md` or a verifier file).
- [NUMERICAL, REPRODUCED]: floating-point result reproduced by a verifier. [NUMERICAL, RUN BY THE VERIFIER]: run once, by a verifier. [NUMERICAL]: not reproduced. None is certified.
- [ARGUED]: a plausibility argument, not a proof. [OPEN]: no proof. [REFUTED]: shown false.
- [BALABAN]: an input taken from Balaban's published papers, used as stated there.
- [NOT WRITTEN]: a proof plan; no project file contains the proof.
- [CITED]: a standard result; its text was not re-read. [CITED, SOURCE CHECKED]: the source was read, the transfer to the lattice is standard and not proved here.
- [PROVED HERE]: a short proof given in full in this paper (Section 4). [HYPOTHESIS]: assumed, and owned by a named block. [FORMAL]: a formal step whose analytic justification is not written. [DERIVED SKETCH]: a route with steps listed, not a proof.
- [INTERVAL ENCLOSURE, VERIFIED]: a rigorous interval-arithmetic enclosure at a point, rerun by a verifier.
- [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]: a statement over a continuum of parameters (all K, all j), proved by an analytic majorant plus a branch-and-bound in interval arithmetic with outward rounding, and confirmed by an independent verifier file. It differs from [INTERVAL ENCLOSURE, VERIFIED] (one point) and from [NUMERICAL] (floating point, no rounding control).
- [VERIFIED REDUCTION]: a reduction confirmed by a verifier file; the inputs stay open.
- [PROVED MODULO NAMED BLOCKS]: proved given the hypotheses owned by the named blocks.

A compound tag such as [PROVED+VERIFIED, CONDITIONAL ON (H2)] combines these meanings. Tags refer to the project files cited next to them. In [VERIFIED QUOTE], a quote of a source other than Bałaban's papers was checked verbatim against a copy of that source in the project. Project files are cited as `NAME.md`; a section sign after a file name (for example `KAPPA_BOUNDED_VERIFY` §4) refers to a section of that file, not of this paper. "Grade n" and "item n" after a verifier file name refer to the numbered verdicts of that file. The paper as a whole is not verified as one document.

1.5 Relation to other work

Two recent works overlap with this paper. The record of how their content reached this project is given below; it is kept in the form fixed when the overlap was found.

Related work 1. E. Sussman and L. Alpöge (with Claude; README: "written largely by LLM ... Novel mathematical ideas should be attributed to Claude (under the guidance of LA)"), "The Yang–Mills continuum limit along Bałaban's renormalization group: a complete account of Theorem 0" (LaTeX, files `theorem_0_complete_account_raw_{master,I,II,III}.tex`, 424k lines), and "SU(2) Yang–Mills theory in four dimensions via Bałaban's renormalisation group: construction, axioms and non-triviality" (`ym4_manuscript_zeroth_draft.pdf`, dated 6 October 2026) [26, 25]. Repository: `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, commit `1b92bd0347e3b8763cd8ecc1f96dc3c2a92dd310` (2026-10-27T05:45:28-05:00), accessed 2026-10-27. Repository created 2026-10-06. Disclaimer in README: "This repository contains work-in-progress, written largely by LLM. Its content has not been satisfactorily vetted. All claims should be understood as tentative, pending verification." Suggested keys: SA26-T0 (account), SA26-YM4 (draft).

Related work 2. OpenAI, "The canonical massive continuum limit of the two-dimensional O(3) model" (October 4, 2026), `github.com/openai/math`, family 215 (preprints folder; read locally from the PDF, `reports/ONLINE_SWEEP.md` §1.2; text in `reports/o3_continuum.txt`) [18]. Suggested key: OAI26-O3.

How content from these works reached this project. The record (file times, `PLAN.md`, read lists; `THEOREM2_PROOF_v2_VERIFY` §I) shows the following.

1. Section 3 was written without reading the Sussman–Alpöge repository. Only the citation block above (`reports/SUSSMAN_ALPOGE_MAP.md` §5.1) was read.

2. First channel. On the night of 2026-10-27, `reports/ONLINE_SWEEP.md` §2 (about 20:20) read the Sussman–Alpöge repository directly. It reported that they prove a weak two-sided flow with an additive constant, never need per-step positivity, use a first-passage cutoff, and invoke only containment $g_k \le \gamma$. This is the first project file with Sussman–Alpöge content.

3. Second channel. `reports/SUSSMAN_ALPOGE_MAP.md` (about 20:28) was written by a separate agent under a firewall. Through it, the bounded-κ idea entered the project: the weak form of Theorem 2 needs only κ_k bounded, not convergence (their Lemma 4.19 structure; `PLAN.md` line 146). `KAPPA_BOUNDED.md` (about 20:39) received this idea through its prompt. Its read list excludes the map and ONLINE_SWEEP.

4. Lemma D′ and the cross-scale sweep predate any Sussman–Alpöge content in project files. The shooting-without-positivity lemma and the band with additive constant appear in `THEOREM2_PROOF_VERIFY` Point 5(b) (about 20:05) and `LEMMA_DPRIME.md` (about 20:10). `LEMMA_DPRIME_SWEEP.md` (about 20:16–20:23) overlaps ONLINE_SWEEP in time, but its read list is LEMMA_DPRIME and its verifier only; it did not read ONLINE_SWEEP. Lemma D″ is in `KAPPA_BOUNDED.md`, written after the map.

5. Theorem K. Its result form, $|S_k - ck \log L| \le \kappa_*$, matches Sussman–Alpöge Lemma 4.2(b), $|\sum_l \beta_l^0 - kc_0| \le c_2$ (as reported in `PLAN.md` line 146; cited there, not proved). The proof in this project (Prop. 1.1, Lemma 2.1, Prop. 2.2 of `KAPPA_BOUNDED`) was written by firewalled agents from Balaban's papers and project files.

6. O(3) paper. Route B's doubling comparison follows the template of OAI26-O3 Lemma 6.1 (doubling difference; `BETA_FP_PW` uses the "O(3) Lemma 6.1 form"). The idea of a cou-

pling band in place of per-step positivity also appears there (ONLINE_SWEEP §1.2(a)). `BETA_FP_PW` read ONLINE_SWEEP §1.2, §5, §8 and o3_continuum Lemma 6.1, and did not read ONLINE_SWEEP §2.

7. Route B (other than the O(3) template), the numerics, and the symbol results of Sections 3.5.5 and 3.7 were derived by firewalled agents from Balaban's papers and project files.

In short: two agents read the Sussman–Alpöge repository, the web sweep (item 2) and the mapping agent (item 3), and neither wrote a derivation. The derivation agents were firewalled; their read lists exclude the repository, the map, and ONLINE_SWEEP §2. One idea crossed the firewall: the bounded-κ route reached `KAPPA_BOUNDED.md` through the map (item 3). Their public, timestamped account predates this one. Any write-up must cite SA26-T0, SA26-YM4, and OAI26-O3, and state these overlaps.

1.6 Organization

Section 2 fixes the notation for both parts of the paper. Section 3 proves the reduction of Theorem 2 (Theorem A). Section 4 proves the finite-torus existence theorem modulo named blocks (Theorem B) and contains the dependency table. Section 5 gives one paragraph for each verified source file of the sources block. Section 6 collects the numerical and certified results. Section 7 lists what remains. Appendix A lists the Lean formalization, Appendix B indexes the results by status, Appendix C maps the numbering of this paper to the names used in the project files, and Appendix D records the revision history.

2 Setup and notation

This section fixes the notation for the whole paper. Sections 2.1–2.4 serve the coupling flow of Section 3. Section 2.5 serves the existence theorem of Section 4. Section 2.6 lists the names that would otherwise collide.

2.1 Lattice, measure, and scales

Notation. L is odd, $L > 11$, fixed (Balaban, CMP 109 p. 256 [8]). $K \in \mathbb{N}$, $\varepsilon = L^{-K}$. The physical side $\ell \in \mathbb{N}$ is in units of the unit lattice. $T_\ell^4 = (\mathbb{R}/\ell\mathbb{Z})^4$ and $T_\varepsilon = (\varepsilon\mathbb{Z}/\ell\mathbb{Z})^4$, so $N_\varepsilon = \ell L^K$ sites per direction. The Wilson action is $A(U) = \sum_p P_p(U)$, $P_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U(\partial p) \in [0, 2]$, over unoriented plaquettes p. The measure is $\mu_\varepsilon = Z_\varepsilon^{-1} \exp(-g_0^{-2} A(U)) \prod_b dU_b$ (Haar). x_p is the centre of p.

The bare coupling $g_0 = g_0(\varepsilon, g)$ is any solution of the shooting problem of Lemma 3.4 (Lemma D″) with terminal coupling $g_K = g$. Lemma D″ gives existence, not uniqueness. Any choice works below.

2.2 Range of L

Balaban assumes L odd, $L > 11$, as a standing assumption for the whole of CMP 109 (pp. 251–252; OCR line 124, verbatim confirmed in `THEOREM2_PROOF_VERIFY` Step 6) [8]. His construction of $\beta_j(g)$ for $g > 0$ and his inductive assumption (H0) are therefore available only at odd $L > 11$. Every statement below that uses (H0) or $\beta_j(g)$ at $g > 0$ is stated for odd $L > 11$, or carries "(H0) and Balaban's construction at this L" as an explicit hypothesis. Statements about the one-loop quantities $\beta_j(0)$, S_k, and the symbols σ_j hold for odd $L \geq 3$.

2.3 The coupling sequence

$\varepsilon = L^{-K}$, so $\log(L^k\varepsilon)^{-1} = (K-k)\log L$. $u_k := 1/g_k^2$. Recursion (CMP 109 (0.18)/(0.20)): $u_{k+1} = u_k - \beta_{k+1}(g_k)$. The increment β_{k+1} depends on the whole history $g_0, \dots, g_k$ (CMP 109 p. 298, [VERIFIED QUOTE]). $\beta_j(0)$ is the one-loop (Gaussian) increment; it is history-independent by definition.

Shooting notation: $u_k = 1/g_k^2$, $t = 1/g^2$, $T(k,K) := \sum_{i=k+1}^{K} \beta_i(g_{i-1})$. Clamp $\varphi(u) := \max(u, 1/\gamma^2)^{-1/2}$.

Here γ is the upper bound for the couplings in Bałaban's Theorem 1. Bałaban's small constant β_0 (1/7 in CMP 122 II p. 389) is not a coupling (Section 3.1). In Lemma 3.6 (Lemma E) the symbols β_0, β_0' instead denote lower and upper bounds on $\beta_j(0)$, as in `LEMMAS.md`.

2.4 One-loop notation

The letter K has two uses. As an integer it is the number of RG steps, $\varepsilon = L^{-K}$. As a vector $K \in \mathrm{BZ} = [-\pi, \pi]^4$ it is a momentum in the Brillouin zone of the coarse unit lattice; then $k_l = (K + 2\pi l)/L$ are the fine momenta over K. The meaning is always clear from context. The other one-loop objects are defined where they are first used:

- S_k, the $p \to 0$ F^2-coefficient of the composed k-step one-loop integral, and its telescoping into $\beta_j(0)$: Section 3.5.1;
- $c = 11N^2/(12\pi^2)$ and $b := c\log L$: Section 3.5.3;
- the f^m Taylor symbols $\sigma_j^{(m)}(K)$ of the coarse kernel at level j, and d_j, κ_j, κ_W^F: Section 3.5.4;
- Δ_M (Gaussian RG kernel at block size $M = L^j$), Δ_∞, and the bordered matrix $\mathfrak{B}_j(K)$: Sections 3.5.5 and 3.9;
- the one-step minimizer map $H_j(K)$, ρ_1, ρ_2, and the inequality (H2): Section 3.7;
- the plane-wave quantities $\beta_{j+1}(p)$, Γ^{ex}, θ_* of Route B: Section 3.6.2.

2.5 Observables and Schwinger functions

Observable (class C1 of `SOURCES.md` §1.2). $\varphi_p := P_p - c_\varepsilon$, $c_\varepsilon := \langle P_p \rangle_{\mu_\varepsilon}$. By hypercubic invariance (Section 4.5.6) c_ε does not depend on p.

Test space. $\mathcal{D}_n := \{F \in C^\infty(T^{4n}) : \text{every derivative of } F \text{ vanishes on every diagonal } \{x_i = x_j\}, i \neq j\}$. The norm is $\|F\|_m := \max_{|\alpha|\le m} \sup |\partial^\alpha F|$. On the compact torus this is the Osterwalder–Schrader (OS) Schwartz seminorm without weights.

Lattice Schwinger functions (`SOURCES.md` §1.3). $S_{n,\varepsilon}(F) := \sum_{p_1,\dots,p_n} F(x_{p_1}, \dots, x_{p_n}) \langle \varphi_{p_1} \cdots \varphi_{p_n} \rangle_{\mu_\varepsilon}$, $F \in \mathcal{D}_n$. These are moments, not cumulants (`SOURCES_VERIFY` item 5(a)). No field renormalization factor is inserted: `SOURCES_VERIFY` item 8 gives $B_K = \prod c_j$ bounded above and below ($c_j = 1 + O(g_{j-1}^3)$, $\sum g_j^3 < \infty$), so a bounded factor B_K^n changes only the normalization of the limit.

Scaling check (`SOURCES_VERIFY` item 4, derived there). $\langle \varphi_p \varphi_{p'} \rangle \sim \varepsilon^8 g^4 |x_p - x_{p'}|^{-8}$ at tree level. So $S_{2,\varepsilon}(F) = O(1)$ for $F \in \mathcal{D}_2$ with no rescaling, while $\sum_p f(x_p)\varphi_p$ has variance $\sim \varepsilon^{-4}$. The limit object is a family of distributions off the diagonals, not a random field.

Extra hypothesis found while writing the source. ℓ is even (sufficient for the reflection argument; see Section 4.8, item 1). For a reflection of the periodic lattice to exchange two halves with both seams being link planes, N_ε must be even. With L odd, $N_\varepsilon = \ell L^K$ is even iff ℓ is even.

2.6 Names

Hypotheses are named by short labels in parentheses, such as (H2), (PI), or (FX); the dependency table (Section 4.6) lists each one with its status. Three names of the project files collide in one document and are changed here; Appendix C records the old names.

- The plane-wave theorem of `BETA_FP_PW` (its Theorem 3.2, called "Theorem B" there) is called Theorem $\mathrm{B_{PW}}$ (Theorem 3.9), so that "Theorem B" means the main Theorem B.
- The three global conditions of the dependency table, called (G1)–(G3) in `ASSEMBLY.md`, are called (GC1)–(GC3). The labels (G1) (constraint conditions of Proposition 3.7) and (G2) (per-cell infinite-volume densities) keep their meaning from the Theorem 2 files.
- The final sourced bound of `SOURCES_D4D5.md`, called (S) there, is called ($\mathrm{S_{src}}$). The label (S) means the smallness condition of Section 3.1.

3 The coupling flow and Theorem 2

This section proves Theorem A. Section 3.1 lists what Bałaban's proof of Theorem 1 uses of the coupling sequence; the answer is weaker than his (0.31). Section 3.2 states the weak form, Theorem 2w, that supplies exactly this. Sections 3.4–3.7 prove Theorem 2w from bounded window sums of the one-loop increments, give two routes to these window sums, and reduce the uniform part of both routes to the single inequality (H2). Section 3.8 assembles the result. Its output, the trajectory conditions (b), (c′), (S), is the block (B-T2) of Section 4.

3.1 What Bałaban's Theorem 1 actually needs

Source: `LEMMA_DPRIME_SWEEP` §2, §7. The statements of Theorems 1 and 2 are quoted in Section 1.2.

The quoted hypothesis of Theorem 1 is only $g_k \in]0, \gamma]$. The proof uses more. CMP 119 Theorem 1 (p. 262) assumes "(I.0.33)", a misnumbered reference to (0.31) of CMP 109. Its working form is CMP 119 p. 255–256 (2.6)–(2.9), for scales $m < n$:

- (2.6) $g_n \le (1 + g_n^2 \beta'(n-m))^{1/2} g_m$ and $g_m \le (1+\beta_0) g_n$;
- (2.7) $(\log g_n^{-2})^p \le (1+\beta_0)(\log g_m^{-2})^p$ and $(\log g_m^{-2})^p \le (1 + g_n^2\beta'(n-m))^{\beta_0} (\log g_n^{-2})^p$;
- (2.8), (2.9) the same for $\varepsilon_j = g_j p_0(g_j)$ and R_j.

Here β_0 is Balaban's small constant (1/7 in CMP 122 II p. 389), not a coupling. [VERIFIED QUOTE] for (2.6)–(2.9): page images, `LEMMA_DPRIME_SWEEP` §1; the sweep itself is single-author, see Section 3.4.3.

In u-variables these follow from two pairwise bounds:

- (P-up) $u_m \le u_n + B'(n-m)$, with no additive constant;
- (P-low) $u_m \ge u_n + \beta(n-m) - D'$, with $\beta > 0$, $D' \ge 0$ fixed,

plus the smallness condition

- (S) $\gamma^{-2} \ge c_{sw} D'$, $c_{sw} = \max(2, 1/(1-(1+\beta_0)^{-2})) = 64/15$ at $\beta_0 = 1/7$, and $\gamma \le \gamma_*(\beta_0, \beta, B', p_0, p_1, r, \nu, \kappa_0)$ (the conditions that literal (0.31) also needs). γ_* is free of the D'-ratio conditions, but it depends on R through B' (it contains $\gamma^2 B' \le (1+\beta_0)^2 - 1$ with $B' = 5b/4 + 2R$). The order of constants in Section 3.8 (R before γ) handles this.

β_0 must be fixed before γ.

3.2 The weak form proved conditionally

Theorem 3.1 (Theorem 2w, weak form)**.** *Let $G = SU(2)$, $d = 4$, L odd, $L > 11$ (equivalently: L odd ≥ 3 with (H0) and Balaban's construction of $\beta_j(g)$ at this L as an explicit hypothesis). There are $\gamma > 0$, $g_1 \in]0, \gamma]$, $\beta > 0$, $B' \geq \beta$, $D' \geq 0$ such that for every $K \geq 1$ and every $g \in]0, g_1]$ there is a bare coupling $g_0 = g_0(K, g)$ with*

- *(b) $g_k \in]0, \gamma]$ for $0 \leq k \leq K$, and $g_K = g$;*
- *(c′) for all $0 \leq m \leq n \leq K$: $u_n + \beta(n - m) - D' \leq u_m \leq u_n + B'(n - m)$;*
- *(S) holds for γ, D'.*

Consequence: Balaban's (0.31) holds up to the additive constant D' on the lower side, with constants $\beta/\log L$ and $B'/\log L$. The two-sided bound fixes $1/g_k^2$ up to constants, as required.

Status: [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS OF SECTION 3.6] by Route K (Theorem K + Lemma D″). (B2) is reduced to the single inequality (H2) of Section 3.7 [PROVED+VERIFIED REDUCTION: `B2_TWOSTEP_VERIFY` GRADE 3]. Second route: Route B (`BETA_FP_PW` Theorem 3.2 + Lemma D″). The logic of Theorem B_{PW} is [PROVED+VERIFIED, CONDITIONAL] (`BETA_FP_PW_VERIFY` §6 items 1–4). Its inputs (U), (DF), (T1) from `BETA_FP_U` are [PROVED, UNVERIFIED]; the route is conditional on (U), ($B\theta$), and the plane-wave defects. `BETA_FP_U` reduces (U) to (T1) + (H2) [PROVED, UNVERIFIED]. Both routes also need (H0) at this L and L5 (Section 3.4.5). See Section 3.6.

3.3 Literal (0.31)

- (L1) Literal (0.31) for every $K \geq K_0$ and every $g \in]0, \gamma]$. Needs eventual uniform positivity: there are J and $\beta_* > 0$ with $\beta_j(0) \geq \beta_*$ for all $j > J$, plus $\sup_j |\beta_j(0)| < \infty$ (Lemma 3.2 (d)). [PROVED+VERIFIED; LEAN] as an implication. Theorem K and Theorem B_{PW} do NOT give eventual positivity: they bound window sums, not single increments. Eventual positivity follows from the convergence route (Section 3.9), which is open.
- (L2) Literal (0.31) for every $K \geq 1$. Needs $\inf_j \beta_j(0) > 0$ and $\sup_j \beta_j(0) < \infty$ (Lemmas D + E). If $\beta_j(0) \to \beta_\infty > 0$, then "every $\beta_j(0) > 0$" is equivalent to $\inf_j \beta_j(0) > 0$ (Lemma D′ Step 8). [PROVED+VERIFIED; LEAN] as an implication. Inputs open.
- Literal (0.31) is not needed for Theorem 1: Section 3.1 shows the proof uses (c′) + (S). Literal (0.31) is used directly in one place only, the alternative choice of ε_0 in CMP 119 p. 248, which Balaban does not adopt.
- Necessity. Literal (0.31) at every $K \geq 1$ forces $\beta_K(0) \geq \beta \log L > 0$ for every K (window $k = K - 1$, let $g \to 0$ using (H0); so L odd > 11). [PROVED, UNVERIFIED]; elementary; `THEOREM2_PROOF_VERIFY` Point 5(a) agrees. So certified early steps are needed only for (L2), not for Theorem 1.

3.4 Shooting lemmas

The shooting lemmas turn bounds on the increments β_j into a trajectory with prescribed terminal coupling. Lemma 3.2 needs eventual positivity of $\beta_j(0)$; Lemma 3.4 needs only bounded window sums, and is the one used for Theorem 2w. The notation is that of Section 2.3. Lemmas D′, D″, D, E are abstract statements about a sequence of functions β_j. Applied to Balaban's $\beta_j(g)$, they need L odd > 11 (or (H0) at this L) and L5.

3.4.1 Lemma D′ (shooting without early positivity) [PROVED+VERIFIED; LEAN]

Sources: `LEMMA_DPRIME.md` §1–§2; `LEMMA_DPRIME_VERIFY.md` grades 1, 2, 5 (CONFIRMED). Step 7 of that file also holds grades 3, 4, 6, which are CORRECTED (Remark 1 overclaim, misreads in §5, ledger); this chapter uses none of them except through the sweep (Section 3.4.3). Lean: `lean/YmLean/LemmaDPrime.lean` (`YmLean.lemma_DPrime`, 405 lines), statement check `lean/VERIFY_LemmaDPrime.md`.

Lemma 3.2 (Lemma D′)**.** Hypotheses. *(i) β_{k+1} continuous on $[0,\gamma_0]$ and $|\beta_{k+1}(g)-\beta_{k+1}(0)| \le C_1 g$, C_1 independent of k, K. (ii) $\sup_j |\beta_j(0)| \le B$. (iii′) there is J with $\beta_j(0) \ge \beta_\infty/2$ for all $j > J$ (this is all the proof uses of "$\beta_j(0) \to \beta_\infty > 0$").*

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, \beta_\infty/(4C_1))$; $B' := B + \beta_\infty/4$; $D := J \cdot B'$; $D' := J(B + \beta_\infty/2)$; $g_1 := (1/\gamma^2 + D)^{-1/2}$; $K_0 := J + \lceil 8D'/\beta_\infty \rceil$.

Conclusions.

- *(a) $T(k,K) \ge -D$ for all $k < K$ along any couplings in $]0,\gamma]$.*
- *(b) For every $K \ge 1$ and $g \in]0,g_1]$ there is g_0 with all $g_k \in]0,\gamma]$ and $g_K = g$.*
- *(c) $u_K + (\beta_\infty/4)(K-k) - D' \le u_k \le u_K + B'(K-k)$. The upper side has no constant.*
- *(d) For $K \ge K_0$: literal (0.31) with lower constant $\beta_\infty/8$, and shooting for every $g \in]0,\gamma]$. If every $\beta_j(0) > 0$: literal (0.31) for all K (Step 8 = Lemmas D + E).*

Proof idea. Clamped flow $\tilde{u}_{k+1} = \tilde{u}_k - \beta_{k+1}(\varphi(\tilde{u}_k))$ is continuous in u_0. Its tail sums lie in $[-D, B'(K-k)]$ for every u_0, because clamped couplings lie in $]0,\gamma]$. IVT on $[t-D, t+B'K]$ gives $\tilde{u}_K = t$. Then $\tilde{u}_k \ge t - D \ge 1/\gamma^2$ for $g \le g_1$, so the clamp is never active. The bound is derived for the clamped flow first; there is no circularity. For history-dependent β this proof needs β_{k+1} defined and continuous on the whole cube $]0,\gamma]^{k+1}$; see L5 in Section 3.4.5.

Lean. The Lean hypotheses are weaker (ContinuousOn instead of C^1; (iii′) instead of convergence). Step 8 is `lemma_DPrime_step8` (`lean/YmLean/LemmaDE.lean`). Lean covers one-argument $\beta_j(g)$ only, not the history form.

Remark 3.3 (Remark 4 of `LEMMA_DPRIME.md`). Without $g \le g_1$, (b) can fail at small K ($\beta_1 = \beta_2 = \beta_3 = -0.8$, $K = 3$, $g = \gamma$). [PROVED+VERIFIED]

3.4.2 Lemma D″ (tail-sum form; no index J) [PROVED+VERIFIED; LEAN]

Source: `KAPPA_BOUNDED.md` §5.3; `KAPPA_BOUNDED_VERIFY` §6 (CONFIRMED: "No J needed"); hand rederivation in `THEOREM2_PROOF_v2_VERIFY` §A.

Lemma 3.4 (Lemma D″)**.** Hypotheses. *(i) as in 2.1, and the window bound*

$$(5.1)\quad T_0(k,K) := \sum_{i=k+1}^{K} \beta_i(0) \in [\, b(K-k) - 2R,\ b(K-k) + 2R\,] \quad \text{for all } 0 \le k < K,$$

with $b := c \log L > 0$ and some $R \ge 0$. (Route K gives $R = \kappa_*$; Route B gives $R = C_B$.)

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, b/(4C_1))$; $\beta := 3b/4$; $B' := 5b/4 + 2R$; $D = D' := 2R$; $g_1 := (1/\gamma^2 + D)^{-1/2}$.

Conclusions. *(b) of 2.1, and the pairwise bound*

$$(c')\quad u_n + \beta(n-m) - D' \le u_m \le u_n + B'(n-m), \quad 0 \le m \le n \le K,$$

upper side without constant.

Proof. $|T - T_0| \le C_1\gamma(K-k) \le (b/4)(K-k)$. Add (5.1). Steps 2–5 of Lemma D′ use only continuity and the bracket $\tilde{T} \in [-D, B'(K-k)]$; this bracket now comes from (5.1). The pairwise form is the same window argument with $K \to n$. The proof is the clamped-flow proof of 2.1; for history-dependent β the L5 caveat of Section 3.4.5 applies.

Lean: `lean/YmLean/LemmaDDoublePrime.lean` (`YmLean.lemma_DDoublePrime`, 146 lines, no `sorry`, axioms propext, Classical.choice, Quot.sound; `lean/README.md`). It proves (a″), (b) with the clamp inactive on the solution, (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, and (d) below. The Lean hypotheses are weaker than the note: continuity instead of C^1; any γ with $0 < \gamma \le \gamma_0$ and $C_1\gamma \le b/4$; $R \ge 0$ derived; $K = 0$ allowed; (c′) and (d) hold for every trajectory with couplings in $]0, \gamma]$. One-argument β only (L5 not formalized).

(d) [LEAN]: for $n - m \ge 8R/b$, $u_n + (b/2)(n-m) \le u_m \le u_n + B'(n-m)$. So literal (0.31) from (5.1) alone holds, with lower constant $b/2$, only for windows with $n - m \ge 8R/b$; short windows near K are not controlled (see Section 3.3).

3.4.3 Lemma D″ meets Balaban's proof: the cross-scale sweep [PROVED, UNVERIFIED (SWEEP); 4 OF 19 TABLE ROWS VERIFIED]

Source: `LEMMA_DPRIME_SWEEP.md` (all 12 Balaban files; OCR grep plus page image for every non-single-scale hit). `LEMMA_DPRIME_VERIFY` Steps 4–5 checked comparisons A–E on page images (CMP 122 I p. 192, p. 198 (1.94); II p. 381 ×2, p. 386) and corrected two misreadings of A (factor $(1+\beta_0)^2(k-j)^{\beta_0}$; $\delta'_j = g_j A_1 p_1(g_j)$). The five comparisons map to four table rows: A = row 11, E = row 12, C and D = row 16, B = row 18.

Result. 19 cross-scale uses of the coupling sequence (table in SWEEP §7), all in CMP 119 and CMP 122 I–II; none in CMP 95, 96, 98, 99, 102, 109, 116. Each is covered by (b) + (c′) + (S) of Section 3.1:

- items 1, 7, 10, 12, 17 need (P-up) with no constant (e.g. (2.6) line 1 is exactly (P-up));
- items 2, 3, 11, 13, 15, 16, 18 need (P-low) with fixed multiplicative slack $(1+\beta_0)$ or $(1+\beta_0)^2$; item 2 ((2.6) line 2) gives the constant 64/15; item 11 needs only $2.42D'$;
- items 4 ((2.8)–(2.9), ε and R ratios), 6 (CMP 119 (3.7), (3.8), (3.12), (3.13)), and 9 (CMP 122 I (1.52)), the last two of the form $\varepsilon_{k+1} \le (1+\beta_0)\varepsilon_k$, use both sides; they follow as items 1 and 3 do (item 4 from items 1–3);
- item 14 (CMP 122 I p. 200, R_{k-N_0+1} vs R_k) uses the lower side; it follows as item 3 does;
- item 5 (CMP 119 (2.31), (2.46): $\sum_j g_j^{\kappa_0} \le g_n^{\kappa_0 - 6}$) needs linear growth (P-low) and $\gamma^{-2} \ge 2D'$;
- item 0 is CMP 119 Theorem 1 assuming (0.31); its proof works with (2.6)–(2.9);
- item 8 (CMP 119 p. 284) is in a sketch, not load-bearing.

Caveats [OPEN]: (1) the sweep is a grep plus page images, not a line-by-line read; (2) "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) was not located; (3) Balaban says β_0 "can be chosen arbitrarily small, if g is sufficiently small"; under (P-low) the floor is $\beta_0 \gtrsim D\gamma^2/2$, so β_0 must be fixed first; no step was found that needs $\beta_0 \to 0$; (4) the sweep has no independent verifier beyond rows 11, 12, 16, 18; (5) `BALABAN_EXCERPTS_CHECK` item 11 lists readings not checked on page images: Π (4.37)/(5.1), CMP 119 (2.42), the (2.12)/(2.23) ambiguity, the "L odd" page, the Sect. 3 estimates, and the 1989 bound lists.

3.4.4 Lemma D and Lemma E [PROVED+VERIFIED; LEAN]

Sources: `LEMMAS.md` (corrected), `LEMMAS_DE_VERIFY.md` §1.4, §1.5, §2.4; `lean/YmLean/LemmaDE.lean` (`lemma_D`, `lemma_E`, `lemma_DE`).

Lemma 3.5 (Lemma D)**.** *If $b \le \beta_j(g) \le b'$ on $]0,\gamma]$ with $b > 0$, then for every K and $g \in]0,\gamma]$ shooting succeeds and $t + b(K-k) \le u_k \le t + b'(K-k)$.*

This is literal (0.31). An earlier draft used a wrong first IVT interval, which the verifier refuted; the corrected interval is $[t+bK,\ t+b'K]$. The nested-interval variant for β defined only on admissible histories (`LEMMAS_DE_VERIFY` §1.5) is [PROVED+VERIFIED], not in Lean.

Lemma 3.6 (Lemma E)**.** *If $\beta_0 \le \beta_j(0) \le \beta_0'$ and $|\beta_j(g) - \beta_j(0)| \le Cg^2$, then for $\gamma \le \sqrt{\beta_0/(2C)}$: $\beta_0/2 \le \beta_j(g) \le \beta_0' + \beta_0/2$.*

The verifier checked this Cg^2 form; the C^1 form ($C_1 g$, $\gamma \le \beta_0/(2C_1)$) is the same two-line proof [PROVED, TRIVIAL VARIANT]. Lean has the Cg^2 form.

3.4.5 History dependence (L5) [OPEN]

Balaban's β_{k+1} depends on $(g_0, \dots, g_k)$. Lemma D′ Remark 3 (`LEMMA_DPRIME.md`) states the cube-continuity version (same proof). The gap is as follows (`THEOREM2_PROOF_v2_VERIFY` §F).

- *Domain.* Balaban defines β_{k+1} only on histories for which his Theorem 1 induction has run to step k (`LEMMAS_DE_VERIFY` §2 item 1). By the sweep (§2.3), that induction uses (2.6)–(2.9), that is (c′) + (S), on the history, not only $g_i \in]0,\gamma]$.
- *The new gap.* The clamped shooting of Lemmas 3.2 and 3.4 evaluates β on clamped histories $\tilde{g}_i = \varphi(\tilde{u}_i)$. A clamped history can violate the growth side of (c′): from $\max(a,c) - \max(b,c) \ge \min(a-b, 0)$ one gets only $u_m \ge u_n - D'$, not $u_m \ge u_n + \beta(n-m) - D'$. So the clamped shooting evaluates β on histories outside Balaban's domain. The clamp proof is valid only under the cube hypothesis, which Balaban does not supply.
- *A variant that avoids it* [NOT WRITTEN]. Use a nested-interval proof in the style of `LEMMAS_DE_VERIFY` §1.5, with domain $D_k = \{u_0, \dots, u_k \ge \gamma^{-2}\}$ of true (unclamped) trajectories. On such trajectories, (c′) up to step k follows from the tail-sum bound (a″) of Lemma D″, and (S) holds by the choice of γ. So every evaluation of β stays inside Balaban's domain. At each step k, keep the set of u_0 whose trajectory stays in D_k; show by IVT, step by step, that this set contains a closed interval on which u_k sweeps a range that contains the next target interval. For Lemma D (all increments positive) this is written. For Lemma D″ the increments can be negative, so the nested intervals need new envelopes, built from the window bounds $[b(n-m) - 2R - (b/4)(n-m), \dots]$. The adaptation is plausible; it is not written and not verified.
- Lean is one-argument only.

3.5 The exact structure of the one-loop sums

The window sums of the shooting lemma are sums of one-loop increments $\beta_j(0)$. This section expresses these sums as level-k quantities of a coarse Gaussian theory and isolates the remainder κ_k whose boundedness is the content of Theorem K. It also lists the symbol facts, uniform in the level, that Section 3.6 uses.

3.5.1 Telescoping [PROVED+VERIFIED, CORRECTED SCOPE]

$S_k := p \to 0$ F^2-coefficient of the composed k-step gauge-fixed one-loop integral (U1 (1.2)). Then $S_k = \sum_{j=1}^{k} \beta_j(0)$, $S_0 = 0$ (`BETA_FP` Prop. 1.1; `BETA_FP_VERIFY` item 1). Scope: exact for the $p \to 0$ coefficient at fixed k, not for $\log Z_k$ at fixed p (the plane wave is off-shell).

3.5.2 Anchor-free identity [PROVED+VERIFIED]

Source: `BETA_FP_LEMMA43` Prop. 2.1; `BETA_FP_LEMMA43_VERIFY` items 1, 3–4 (hand rederivation; lattice check with the real CMP 109 averaging at random SU(2) backgrounds, relative difference 10^{-14}).

Proposition 3.7 (anchor-free identity; Prop. 2.1 of `BETA_FP_LEMMA43`)**.** *Under (G1) = $\{\Delta D = 0,\ \tilde{Q}D = D_c \iota_a^T\}$:*

$$\log Z_\Lambda(V) = \big[\log\det X - \tfrac{1}{2}\log\det K\big] - \big[\log\det(D_c^T D_c) - \tfrac{1}{2}\log\det(\Delta_\Lambda(V) + D_c D_c^T)\big].$$

In words: constrained one-loop = fine Wilson one-loop in background Feynman gauge minus the one-loop of the coarse Gaussian theory with kernel $\Delta_\Lambda(V)$. All point-anchor objects cancel; the constant is exactly 0.

- $\tilde{Q}D = D_c \iota_a^T$ is exact at any background (lattice 10^{-15}). $\Delta D = 0$ is load-bearing (raw Hessian gives $O(1)$ errors).
- At the constant abelian field V_f the background is critical, so $\Delta D = 0$ is exact and Prop. 2.1 holds exactly (`BETA_FP_H_VERIFY` §2, CONFIRMED). At a plane wave it holds up to the defect $\mathfrak{d}_Y$ (`BETA_FP_PW` Lemma 2.3) [PROVED, UNVERIFIED].
- Per-cell infinite-volume densities only: on a torus the constant zero modes must be removed. This is input (G2) [FORMAL].

3.5.3 The constant c [PROVED+VERIFIED]

$c = 11N^2/(12\pi^2) = 2N \cdot b_0/(8\pi^2)$, $b_0 = 11N/3$; SU(2): $c = 0.371511$. Normalization $1/g_B^2 = 2N/g^2$ (`R2_VERIFY`; `BETA_FP_LEMMA43_VERIFY` §5). The log coefficient of the fine piece at constant field is $c_2 = c/2$ per $\ln(1/\varphi)$, checked against Savvidy, arXiv:1910.00654 eq. (1.4) [23] (`BETA_FP_H_VERIFY` §4) = input (F) [CITED, SOURCE CHECKED; LATTICE = CONTINUUM LOG COEFFICIENT IS STANDARD, NOT PROVED IN PROJECT FILES].

$b := c \log L$: 0.25751 ($L = 2$), 0.40815 (3), 0.51502 (4), 0.59793 (5), 0.72293 (7).

3.5.4 The decomposition $S_k = ck \log L + \kappa_W^F - \kappa_k$

- As bookkeeping the split is empty: any c' gives an exact identity if κ_k is defined as the remainder (`U1_VERIFY`). Boundedness of a remainder-defined κ_k is a restatement, not progress.
- Content comes from defining κ_k by the coarse theory alone. `KAPPA_BOUNDED` Prop. 1.1 [PROVED+VERIFIED, CONDITIONAL ON (PI), (F), (M_j), V1, (G2)]: with the Wilson level $j = 0$ as reference,

$$S_j = cj \log L - d_j, \qquad \kappa_j = \kappa_W^F + d_j,$$

$$d_j := -\frac{1}{2n_2} \int_0^1 dt \int_{\mathrm{BZ}} \frac{dK}{(2\pi)^4}\ \mathrm{tr}\big[b_{t,2}\,\delta\sigma_j^{(0)} + b_{t,1}\,\delta\sigma_j^{(1)} + b_{t,0}\,\delta\sigma_j^{(2)}\big],$$

where $\delta\sigma_j^{(m)} = \sigma_j^{(m)} - \sigma_0^{(m)}$ are differences of the f^m Taylor symbols of the coarse kernel at level j and level 0, and $b_{t,m}$ are the Moyal inverse coefficients of the interpolated coarse operator. No limit symbols σ_∞ are used.

- Checks [NUMERICAL, REPRODUCED]: $\rho(1) = \kappa_W^F$ (Lüscher–Weisz (5.7) [16]); $L = 2$: $d_1 = -0.372$, $\kappa_1 = 1.114$; $L = 3$: $d_1 = c\log 3 - \beta_1 = -0.256 \pm 0.01$ (`KAPPA_BOUNDED_VERIFY` §4: fits give $-0.254 \cdots - 0.258$). This checks the definition of d_j through S_j; it does not evaluate (1.1) from the symbols independently.

- Cor. 3.1 of `BETA_FP_LEMMA43` ($\mathcal{C}(\Lambda) = c\log\Lambda + \kappa_W^F - \rho(\Lambda)$) is exact as an identity. "The whole $\log\Lambda$ sits in the fine piece" is equivalent to $\rho(\Lambda) = o(\log\Lambda)$. It is not a separate fact (`BETA_FP_LEMMA43_VERIFY` item 2; an earlier overclaim is removed).

- Given convergence $\beta_j(0) \to \beta_\infty$, Cesàro gives $\beta_\infty = c\log L - \lim\kappa_k/k$. So the hypothesis (H′) "$\kappa_k = o(k)$" of the first version of these notes is EQUIVALENT to $\beta_\infty = c\log L$, not weaker (`THEOREM2_PROOF_VERIFY` Point 4). Theorem K (Theorem 3.8) gives κ_k bounded, hence $\kappa_k = o(k)$ trivially.

3.5.5 Level-uniform symbol facts used below [PROVED+VERIFIED]

Source: `U3_AW.md`, `U3_AW_VERIFY.md` (grades 1–4 CONFIRMED); `U3_REPAIRS_VERIFY`.

- O5/A0: Δ_M real-analytic on a fixed strip, $|\partial^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, uniform to $K = 0$ (odd L, centered blocks). Prop. B: $c_0|\hat{K}|^{-2}P_T' \le P_T' S_M P_T' \le C_1|\hat{K}|^{-2}P_T'$, $c_0 = (2/\pi)^{12}$, $C_1 \le 1363$. Existential constants.

- Prop. E (evenness): $\sigma_j(f;-K) = \sigma_j(f;K)$ in the midpoint trivialization, all f, j. Tested in $d = 2$ only.

- Theorem W: $\sigma_j^{(1)}(0) = \tau_c = (3ie/2)(E_{10} - E_{01})$ for every $j \ge 0$, from the exact Ward identity $\Delta_j(f)D_f = 0$. Numerically 0.750000, 0.75015, 0.75051 at $j = 0, 1, 2$ vs 0.75 ($d = 2$ proxy).

- Theorem A3: $|\sigma_j^{(1)}(K) - \tau_c| \le C_A|K|^2$, real K, uniform in j (C^0 only).

- Prop. A2: $\sigma_j^{(m)}$, $m \le 2$, analytic on the same strip at every fixed j (not uniform in j for $m = 1$ derivatives, $m = 2$).

- Lemma M: Maxwell coefficient exactly 1 at every level.

3.6 Bounded window sums: two formally distinct conditional routes

This section gives the two routes to the window bound (5.1) that Lemma 3.4 needs.

Both routes give the window bound (5.1) of Lemma 3.4, $|S_K - S_k - b(K-k)| \le 2R$, with $b = c\log L$. Lemma D″ then gives Theorem 2w (at odd $L > 11$, or with (H0) at this L as a hypothesis, and with L5). Neither route uses convergence of $\beta_j(0)$, a rate, (N), or certified early steps. The two routes have formally distinct named hypotheses. Their mathematical content is not independent: both reduce their uniform-in-level content to the same inequality (H2) on the two-step minimizer map (Section 3.7), plus A3-type facts (`BETA_FP_U` §3.5, §4.3 [PROVED, UNVERIFIED]; Route K side: `B2_TWOSTEP_VERIFY` grade 3 [PROVED+VERIFIED]).

3.6.1 Route K: Theorem K [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS]

Source: `KAPPA_BOUNDED.md`; `KAPPA_BOUNDED_VERIFY.md` (items (i)–(v) and (vii) CONFIRMED; (vi) CORRECTED in reasoning, conclusion unchanged).

Theorem 3.8 (Theorem K)**.** *L odd ≥ 3. Assume (B2) (Section 3.7; (B2) $\Leftarrow$ (H2) [PROVED+VERIFIED]) and the fixed-level inputs below. Then $\sup_k |d_k| \leq \kappa_* < \infty$, so $\sup_k |\kappa_k| \leq |\kappa_W^F| + \kappa_*$ and $|S_k - ck \log L| \leq \kappa_*$ for all k. Window sums satisfy (5.1) with $R = \kappa_*$. (Theorem K is a statement about one-loop quantities. Its use in Lemma D′ for Balaban's $\beta_j(g)$ needs L odd > 11 or (H0) at this L.)*

Proof structure.

1. Prop. 1.1 (Section 3.5.4): $S_j = cj \log L - d_j$. [PROVED+VERIFIED, CONDITIONAL ON (PI), (F), (M_j), V1, (G2)]
2. Lemma 2.1: the only K-derivative of $\sigma^{(1)}$ in d_j enters linearly with f^0 coefficients. Integration by parts moves it onto f^0 data; boundary terms vanish (periodic up to a common constant conjugation; inner sphere $O(r^2)$). So $\sigma^{(1)}$ is needed only in C^0, which Theorem A3 gives. [PROVED+VERIFIED]
3. Prop. 2.2: with (E0) f^0 bounds (Prop. B, O5/A0, Lemma M), (E1) Theorem A3 + Theorem W, and (B2), every term of the d_j integrand is $\leq C|K|^{-2}$, uniformly in j and t. In $d = 4$ this has two powers of infrared margin. $|d_j| \leq \kappa_* := (C(L)/n_2)(1 + Y_1^2 + C_A(1 + Y_1) + Y_2) \cdot I_2$. [PROVED+VERIFIED, CONDITIONAL ON (B2)]
4. Infrared: the degree -4 part of the coarse f^2 integrand is the same at every level (Lemma M + Theorem W), so the $\tau_c \log$ cancels exactly in level differences. [PROVED+VERIFIED]

Constants are existential; κ_* is not evaluated.

Fixed-level inputs of Route K (needed at each j separately; no uniformity in j):

- (PI) probe independence: plane-wave p^2 coefficient = constant-field f^2 coefficient [ARGUED; PER-CELL LIMIT FOR THE LINEAR POTENTIAL NOT WRITTEN].
- (F) fine log coefficient c_2 at constant field [CITED, SOURCE CHECKED], Section 3.5.3.
- (M_j) the formal f^2 coefficient d_j equals the true one at fixed j. The resolvent step of the first argument was false (Nielsen–Olesen zero at $k_\perp^2 = |ef|$; `BETA_FP_H_VERIFY` §5). `BETA_FP_M` repairs it with a direct $\log|\det|$ bound given spectral bounds (S1)–(S3) [OPEN ON THE LATTICE]; Prop. 4.4 there is [ARGUED]; `BETA_FP_M` has no verifier file. The t-subdivision of KAPPA §1.3 removes the small-ε requirement [ARGUED, CONFIRMED AS PLAUSIBLE]. Status: [ARGUED].
- V1: the Maxwell coefficient of the per-step determinant equals $\beta_{j+1}(0)$, i.e. order-g^0 completeness of CMP 109 (1.3)–(1.5) [OPEN; SUPPORTED AT ORDER p^2 BY `U3_VERIFY` §6].
- (G2) per-cell infinite-volume densities [FORMAL].
- (H0) Balaban's inductive assumption: β_{j+1} smooth on $[0, \gamma]$, "uniformly bounded on this interval together with all derivatives" (CMP 109 pp. 263–264, (1.22)) [BALABAN; AVAILABLE AT ODD $L > 11$ ONLY]. CMP 116 has no theorem statement and no C^m-in-g bound in words (`BALABAN_EXCERPTS_CHECK` item 10); the smoothness in g is CMP 109's inductive assumption, asserted by CMP 109 Theorem 3, whose proof CMP 116 completes.

- L5 history-dependent β, its domain, and its continuity (Section 3.4.5) [OPEN].

3.6.2 Route B: Theorem $\mathbf{B}_{\mathrm{PW}}$ [PROVED+VERIFIED, CONDITIONAL (LOGIC OF THEOREM B_{PW})]

Source: `BETA_FP_PW.md` §1–§5 (Theorem 3.2 = Theorem B_{PW}, Theorem 5.1). `BETA_FP_PW_VERIFY.md` §6 grades it: items 1–4 CONFIRMED (doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B as conditional logic; Lemma 4.1, conditional), items 5 (comparison) and 6 (numerics) CORRECTED. The tag follows the decision recorded in `PLAN.md` (2026-10-28): the logic of Theorem B_{PW} is [PROVED+VERIFIED, CONDITIONAL] on its hypotheses, citing `BETA_FP_PW_VERIFY` §6 items 1–4. Its inputs (U), (DF), (T1) from `BETA_FP_U` stay [PROVED, UNVERIFIED]. The file rederives the Lemma 1.3 arithmetic and the Lemma 2.3 / Prop. 2.2 algebra correctly, confirms the Prop. 2.2 determinant identity on a 4^4 lattice (4e−13), and notes that identity 3.0 is a tautology ($\Gamma^{ex} := \hat{\Pi} - S$), so all content sits in $(B\theta)/(C\theta)$.

Theorem 3.9 (Theorem B_{PW}; Theorem 3.2 of `BETA_FP_PW`)**.** *L odd ≥ 3. Assume*

- *(U) there are $p_* > 0$, C_U with $|\beta_{j+1}(p) - \beta_{j+1}(0)| \leq C_U p^2$ for all $j \geq 0$, $0 < p \leq p_*$* [HYPOTHESIS; ARGUED FROM UNIFORM ANALYTICITY IN p OF THE ONE-STEP PLANE-WAVE DATA PLUS ONE-STEP O6]*;*
- *($B\theta$) there is $\theta_* \in (0, \min(p_*, 1))$ with $B_* := \sup_k |\Gamma^{ex}(\theta_*; L^k)| < \infty$, where Γ^{ex} is the coarse remainder at fixed coarse momentum θ_** [HYPOTHESIS]*;*
- *Lüscher–Weisz (5.7) [16]* [CITED].

Then $|S_k - ck \log L| \leq C_B := B_* + c_2 \ln(1/\theta_*^2) + |\kappa_W^F| + \sup_{p \leq \theta_*} |\eta(p)| + C_U \theta_*^2/(L^2 - 1)$, and window sums satisfy (5.1) with $R = C_B$. $\theta_* < 1$ is required so that $c_2 \ln(1/\theta_*^2) \geq 0$; otherwise the term must be replaced by its absolute value (`BETA_FP_PW_VERIFY` §3).

Key step. Lemma 1.3: $|S(\theta; L^k) - S_k| \leq C_U \theta^2/(L^2 - 1)$ from (U). The massless coarse side is then needed only at a fixed $\theta_* > 0$, where there is no infrared problem. No (W), (PI), (M), IR weights, or spectral theory.

What ($B\theta$) reduces to (`BETA_FP_PW` §4) [ARGUED]: Lemma 4.1(i) (trivial-background convergence, from O5, odd L) [PROVED]; vertex bounds at fixed θ_* (the uniform bounds cited by U3 Theorem 7.1 to Balaban; the fixed-θ data hypothesis ($D\theta$) is needed only in its "bounded" form); and boundedness in k of the defect $\mathfrak{d} = \mathfrak{d}_Y + \mathfrak{d}_{gf}$.

- $\mathfrak{d}_Y$ (Prop. 2.1 defect at the non-stationary plane wave): $O(a\Lambda^{-2})$ relative at fixed θ, from zero capacity of point anchors in $d = 4$ [ARGUED; REGION $|K| \lesssim \Lambda^{-1}$ NOT WRITTEN].
- $\mathfrak{d}_{gf}$ (composite gauge vs D_k^T gauge at the plane wave): $O(\theta^2)$ relative; uniformity in k not shown [ARGUED]. Data: 0.953 (direct) vs 0.9595 (recursion) at $L = 2$, $p_0 = 0.2$; difference below the tabulation error, so unresolved.

Identification. Route B's $\beta_j(0)$ is the per-level plane-wave increment (the quantity the code computes). It matches Balaban's second-moment β up to the expansion point (fine wave vs constrained minimizer), argued to change S by $O(\theta^2)$ relative (`SOURCES_D1_VERIFY` item 8) [ARGUED]. This identification and the $\mathfrak{d}_{gf}$ bound need expansion-point bounds uniform in k. So the (PI)-type content of Route K is not removed in Route B; it moves into a uniform-in-k input (`BETA_FP_PW_VERIFY` §3).

Three points from `BETA_FP_PW_VERIFY` §3 (item 5 of its §6 grades them CORRECTED):

1. $\theta_* < 1$ is required (above).

2. The β identification (B4) is a uniform-in-k input, not a fixed-level input.

3. (U) is a uniform-in-j analyticity of the level data. It plausibly needs the plane-wave analogue of a uniform C^1 bound on $\sigma^{(1)}$: uniform p-smoothness of the a^1 vertex. That is the same uniform first-order vertex smoothness as Route K's open core ($C1\sigma$).

3.6.3 Route B inputs after `BETA_FP_U` [PROVED, UNVERIFIED; NO VERIFIER FILE]

Source: `BETA_FP_U.md` §2–§4. It reduces the two uniform hypotheses of Theorem $\mathrm{B_{PW}}$.

- (U) $\Leftarrow$ (T1) + (H2) (Theorem 2.4, odd $L \geq 3$, with A0, A1, Lemma 2.2). (T1): $\sup_j \sup_q |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of Theorem A3 [ARGUED]. (H2) is the inequality of Section 3.7 (`BETA_FP_U` writes it as $\tilde{h}_{(2)} < L$).

- ($B\theta$): $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ exactly (Prop. 3.1). γ_{rec} is bounded in k given Theorem 2.4 (Prop. 3.2). $\mathfrak{d}_F$ is bounded under (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$ [HYPOTHESIS, ARGUED]. `BETA_FP_U` corrects `BETA_FP_PW` §4.3(b): $\mathfrak{d}_F$ is $O(1)$ relative at fixed θ, not $O(\theta^2)$; only boundedness is used.

- Corollary 4.1: given (T1), (H2), (DF), LW (5.7), $|S_k - ck \log L| \leq C_B$, so (5.1) holds with $R = C_B$.

- Theorem 2w by Route B then needs in addition A6 (plane-wave β = Balaban's β up to the expansion point, uniform in k) [ARGUED] and (H0) (Lemma D″ (i)).

Route B hypothesis set: (T1), (H2), (DF), LW (5.7), A6, (H0) (plus L5, (G2), odd L, as for Route K). The fixed-level inputs (PI), (M_j), V1 of Route K do not appear; (DF) and A6 replace them.

3.6.4 Comparison

	Route K	Route B
uniform-in-level input	(B2): $\sup_j \sup_K \|\sigma^{(2)}_j\| < \infty$, $\Leftarrow$ (H2) [PROVED+VERIFIED]	(U), ($B\theta$), and the β identification and $\mathfrak{d}_{gf}$ bound, both with k-uniform expansion-point bounds; (U) $\Leftarrow$ (T1) + (H2), ($B\theta$) $\Leftarrow$ Theorem 2.4 + (DF) [PROVED, UNVERIFIED]
single uniform inequality	(H2)	(H2)
fixed-level inputs	(PI), (F), (M_j), V1, (G2)	LW (5.7), $\mathfrak{d}_Y$, $\theta_* < 1$
uses (W), Theorem A3	yes	no; `BETA_FP_U` uses (T1), the plane-wave analogue of A3 [ARGUED]

verification	verified (conditional)	logic of Theorem B_{PW} verified (conditional): `BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED [PROVED+VERIFIED, CONDITIONAL]; inputs from `BETA_FP_U` [PROVED, UNVERIFIED], no verifier file
common	(H0) at odd $L > 11$, L5, (G2), V1-type identification, Balaban's proofs, SU(2)	

Shared inputs: (H0), L5, (G2), odd L (> 11 for (H0)), and the identification of the computed $\beta_j(0)$ with Balaban's $\beta_j(0)$ (V1; in Route B also the expansion point). The uniform-in-level hypotheses ((B2) vs (U) + $(B\theta)$) are formally distinct as named hypotheses. Their content overlaps: both rest on (H2) (Section 3.7). In Route K, (H2) gives $(C1\sigma)$ (Theorem C1) and then (B2) (Theorem B2). In Route B, (H2) gives (U) with (T1), and the vertex bound behind $(B\theta)$. So a proof of (H2) serves both routes. One closed route suffices for Theorem 2w.

3.7 The hypothesis (B2) of Route K, and the single uniform inequality (H2)

Both routes need a bound on the coarse symbols that is uniform in the level. This section states that bound for Route K, records what is proved and what is refuted, and reduces it to the inequality (H2). The numerical values of (H2) are in Sections 6.4–6.8. At $L = 13$, (H2) is a certified theorem (Theorem 6.1, Section 6.7).

(B2): $\sup_{j\geq 0}\sup_{K\in \mathrm{BZ}} |\sigma_j^{(2)}(K)| < \infty$. Weaker form that suffices, (B2w): $\sup_j \int_{\mathrm{BZ}} |K|^{-2}|\sigma_j^{(2)}(K)|\, d^4K < \infty$ ($\sigma^{(2)}$ enters d_j only undifferentiated; `KAPPA_BOUNDED` Remark 2.3) [PROVED+VERIFIED: NOT GRADED BY NAME; IMPLIED BY `KAPPA_BOUNDED_VERIFY` §9(IV)].

$\sigma^{(2)}$ obeys the one-step recursion (`U3_AW` (1.2)): $\sigma_j^{(2)} = L^{-2}\mathcal{A}_j\sigma_{j-1}^{(2)} + Q_j[\sigma_{j-1}^{(1)}] + g_j^{(2)}$, with $\mathcal{A}_j s = L^{-2}H_j^\dagger(\oplus_l s(k_l))H_j$ and $H_j(K)$ the one-step gauge-fixed minimizer map. Notation: $\rho_1 := \sup_{j,K}\|H_j(K)\|/L$ (one step); $H_{(2),j}(K) := (\oplus_l H_{j-1}(k_l))H_j(K)$ (two steps), $\rho_2 := \sup_{j\geq 2}\sup_{K \text{ real}}\|H_{(2),j}(K)\|/L^2$.

3.7.1 Proved

- $K = 0$ bound [PROVED+VERIFIED]: $\sup_j |\sigma_j^{(2)}(0)| \leq C(|e|C_A + e^2)$. Theorem A3 + scaling give $|\partial^2\sigma_j^{(1)}(0)| \leq 2C_A$ (polarization; `KAPPA_BOUNDED_VERIFY` §7), and the slaving identity (W-2) expresses $\sigma_j^{(2)}(0)$ through these second derivatives.

- Each fixed j [PROVED+VERIFIED]: $\sigma_j^{(2)}$ is analytic on a fixed strip (Prop. A2), hence bounded. The content of (B2) is uniformity in j.

- Prop. 4.1 [PROVED+VERIFIED, CONDITIONAL]: if (h) $h := \sup_{j,K}\|H_j(K)\| < L^2$ and $(C1\sigma)$ $\sup_j\sup_K|\partial_K\sigma_j^{(1)}(K)| < \infty$, then (B2) holds. Proof: in the form norm $\pm s \leq N(\Delta + 1)$, $N_j \leq (h^2/L^4)N_{j-1} + Z_2$, and Z_2 is finite by Theorem A3, $(C1\sigma)$, and fixed f^0 vertices.

- Why the Lemma A4 mechanism alone fails [PROVED+VERIFIED, REASON CORRECTED BY `KAPPA_BOUNDED_VERIFY` §7]: the obstruction is the missing uniform C^1 bound on $\sigma^{(1)}$, which

enters twice: in Q_j (term $L^{-3}\partial_K \sigma^{(1)}_{j-1}$) and in the longitudinal part of $\sigma^{(2)}$ given by (W-2). Theorem A3 is C^0 only.

3.7.2 Refuted, and the failed one-step bound

- (HS) $h < L\sqrt{L-1}$ (named (H′) in `U3_AW`) [REFUTED AT $L = 3$]. Measured $\max_K \|H_j(K)\|/L = 1.652, 1.736, 1.757, 1.760$ for $j = 1, 2, 3, \infty$ (`U3_AW_VERIFY` §6); the threshold is $\sqrt{2} = 1.414$. Gauge choice cannot help: H_j is already minimal-norm in its gauge orbit.

- One-step sup-norm C^1 recursion at $L = 3$ [NUMERICAL, REPRODUCED; `B2_TWOSTEP_VERIFY` GRADE 2]. Its factor is $\rho_1^2/L = 0.909, 1.0046, 1.0289, 1.0325$ for $j = 1, 2, 3, \infty$, so it exceeds 1 for $j \geq 2$ ($\rho_1 > \sqrt{3} = 1.7321$). At $K = 0$ the factor is L^{-1}; the bad region is the face $K_3 = K_4 = \pi$, $K_1 = K_2 \approx 1.86$–1.92. This shows that the one-step sup-norm bound fails at $L = 3$. It does not show that a C^1 bound fails; a weighted norm is not excluded.

3.7.3 (H2) and the two-step theorems [PROVED+VERIFIED, CONDITIONAL ON (H2)]

Source: `B2_TWOSTEP.md` §1–§3; `B2_TWOSTEP_VERIFY.md` grades 1, 3, 4 (CONFIRMED).

$$\text{(H2)} \quad \rho_2 = \sup_{j\geq 2} \sup_{K\in \mathrm{BZ\ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L.$$

(H2) is a plain operator-norm inequality on real K. It uses no strip, no Δ-relative weight, and only $n = 2$.

Theorem 3.10 (Theorem C1 of `B2_TWOSTEP`). *(H2) ⇒ (C1σ), with contraction factor $q_1 = \rho_2^2/L^2$ on the derivative.*

Theorem 3.11 (Theorem B2 of `B2_TWOSTEP`). *(H2) ⇒ (B2), hence (B2w), in plain sup norm, with factor $q_2 = \rho_2^2/L^4$.*

Here $(C1\sigma)$ is $\sup_j \sup_K |\partial_K \sigma_j^{(1)}(K)| < \infty$ (Section 3.7.1). The verified points are these.

- Two-step map. $\mathcal{A}_j\mathcal{A}_{j-1}s = L^{-4}H^{\dagger}_{(2)}(\oplus s)H_{(2)}$, with $dk_{l,l'}/dK = L^{-2}$ [CONFIRMED, grade 1].

- Theorem 3.10. The chain-rule gain is exactly L^{-2} per two steps on real K. So the threshold is $\rho_2 < L$, not $\sqrt{L^2-1}$: no strip-Cauchy loss enters. The large constants ($b(L)$ of O6, η_1, η_2, G_1) enter only the source term, not the contraction factor [CONFIRMED, grade 3].

- Theorem 3.11. This contraction needs only $\rho_2 < L^2$, which (H2) implies [CONFIRMED, grade 3]. In one step it is Prop. 4.1's (h).

- (H2) vs (H″). (H2) is the plain operator norm of the same matrix as `U3_AW` (H″) at $n = 2$. It is weaker: threshold L (3 at $L = 3$) against $\sqrt{L^2-1}$ (2.828), real K only, $n = 2$ only [CONFIRMED, grade 4]. (H″) ⇒ (B2) stays [ARGUED] and is no longer needed.

- Form of Q_j (G3) [CONFIRMED by read-off of `U3_AW` (1.2)/(3.3) and P10]: $\partial\sigma^{(1)}$ enters linearly. `U3_LEMMA51` (3.3) itself was not re-derived from the Moyal rule by either party.

- Not covered: C^2 and C^3 bounds on $\sigma^{(1)}$ under (H2) (`B2_TWOSTEP` §5, "look routine", not written, not graded). They are not needed for (B2).

- Submultiplicativity: $\rho_2 \leq \rho_1^2$ always. At $L = 3$ the gain of the two-step map is that this is strict by a factor 1.50 (2.064 vs 3.098) [CONFIRMED].

3.7.4 One-step sufficiency [ARITHMETIC PROVED; INPUT CERTIFIED AT $L = 13$]

$\|H_{(2),j}(K)\| \le \max_l \|H_{j-1}(k_l)\| \cdot \|H_j(K)\| \le L^2\rho_1^2$. So $\rho_1^2 < L$ implies (H2), and then the one-step C^1 recursion of `B2_TWOSTEP` (2.2) contracts with factor ρ_1^2/L. At $L = 3$ the measured ρ_1^2 exceeds L, so the one-step route fails and the two-step map is needed; at $L = 5$ and $L = 7$ the measured ρ_1^2/L is below 1. The values are in Section 6.5. At $L = 13$ the input is certified: $\sup_{j,K} \rho_1^2 \le 10 < 13$ (Theorem 6.1), so (H2) holds there with $\rho_2 \le 10$.

3.7.5 The precise remaining inequality

At a fixed odd $L \ge 3$ the single uniform hypothesis is (H2).

- Route K: (H2) $\Rightarrow$ ($C1\sigma$) (Theorem C1) $\Rightarrow$ (B2) (Theorem B2) [PROVED+VERIFIED, CONDITIONAL ON (H2)].
- Route B: (H2) + (T1) $\Rightarrow$ (U), and (H2) $\Rightarrow$ the vertex bound for γ_{rec} in ($B\theta$) (`BETA_FP_U` Theorem 2.4, Prop. 3.2) [PROVED, UNVERIFIED].
- Status of (H2): at $L = 13$, inside Bałaban's range $L > 11$, [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]: $\rho_1(j, K)^2 \le 10 < 13$ for all $j \ge 1$ (including ∞) and all K, hence $\rho_2 \le 10 < 13$ (Theorem 6.1; residue in Section 6.7). At $L = 3, 5, 9$: [NUMERICAL] only (at $L = 9$ one point).
- Kept as alternatives: ($C1\sigma$) + (h) $\Rightarrow$ (B2) [PROVED+VERIFIED]; (H″) $\Rightarrow$ (B2) [ARGUED]; (B2w) directly. No per-momentum $d = 4$ constant-field code exists, so (B2) itself has not been scanned numerically (`KAPPA_BOUNDED_VERIFY` §8).

3.8 Conclusion of the reduction

Theorem 2w $\Leftarrow$ (Route K inputs) or (Route B inputs). Precisely:

Route K. (B2) [OPEN; REDUCES TO (H2) (VERIFIED REDUCTION, SECTION 3.7.3); (H2) CERTIFIED AT $L = 13$ (THEOREM 6.1)] + (PI), (F), (M_j), V1, (G2) [FIXED-LEVEL] + (H0) [BALABAN, ODD $L > 11$] + L5 [OPEN] $\Rightarrow$ Theorem K ($R = \kappa_*$) $\Rightarrow$ (5.1) $\Rightarrow$ Lemma D″ $\Rightarrow$ Theorem 2w. Status of the implication: [PROVED+VERIFIED, CONDITIONAL].

Route B. (U) + ($B\theta$) with $\theta_* < 1$ + LW (5.7) + bounded $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ + identification of β (uniform in k) + (G2) + (H0) [ODD $L > 11$] + L5 $\Rightarrow$ Theorem $\mathrm{B_{PW}}$ ($R = C_B$) $\Rightarrow$ (5.1) $\Rightarrow$ Lemma D″ $\Rightarrow$ Theorem 2w. Status: Theorem $\mathrm{B_{PW}}$ logic [PROVED+VERIFIED, CONDITIONAL]; inputs [PROVED, UNVERIFIED]. With `BETA_FP_U` (Section 3.6.3): (T1) + (H2) + (DF) + LW (5.7) $\Rightarrow$ Corollary 4.1 ($R = C_B$); with A6 + (H0) + L5 + (G2) $\Rightarrow$ Theorem 2w. Status: [PROVED, UNVERIFIED, CONDITIONAL] (the inputs (T1), (DF), A6 and the `BETA_FP_U` reductions are unverified).

Single uniform hypothesis. Both routes need (H2). Theorem 6.1 certifies (H2) at $L = 13$, an odd $L > 11$. This discharges (B2), the only uniform-in-level hypothesis of Route K (Theorems C1, B2). In Route B it reduces (U) to (T1); (DF) and A6 remain.

Then Balaban's Theorem 1. Theorem 2w gives (b) and (c′). With γ shrunk so that (S) holds, every one of the 19 cross-scale uses in CMP 119 and CMP 122 I–II is covered (Section 3.4.3), and the single-scale uses need only $g_k \le \gamma$. So Theorem 1 holds for the bare coupling $g_0(K, g)$, for every K and every $g \in]0, g_1]$, without Balaban's coupling assumption. This needs L5 in the sharpened form of Section 3.4.5: β must exist on every history the shooting visits. Status: [PROVED, UNVERIFIED (SWEEP), CONDITIONAL ON ROUTE K OR ROUTE B INPUTS, ON L5, AND ON BALABAN'S PUBLISHED PROOFS].

Order of constants. Fix $\beta_0, p_0, p_1, r, \nu, \kappa_0$ (Balaban). Fix L (odd, > 11), then R from Route K or B, then $b = c \log L$, then $\gamma \le \min(\gamma_0, \gamma_{Bal}, b/(4C_1), \gamma_*)$ with $\gamma^{-2} \ge c_{sw} \cdot 2R$. Then $g_1 = (\gamma^{-2} + 2R)^{-1/2}$. Balaban's phrase "for a sufficiently small positive g" fits $g \le g_1$.

What this does not give.

- Literal (0.31): only for windows of length $\ge 8R/b$. Literal (0.31) at every K needs (L1) or (L2) of Section 3.3, which need individual increments, not window sums. Not needed for Theorem 1.
- The value $\lim \beta_j(0)$ and any rate: Section 3.9.
- Even L, or odd $L \le 11$ without (H0) as an extra hypothesis (Balaban assumes L odd > 11 throughout CMP 109); SU(N) with $N > 2$.
- Monotone couplings: g_m can exceed g_n for $m < n$ by the bounded amount $u_m \ge u_n - D'$. Section 3.4.3 shows Balaban's proof tolerates this under (S); a line-by-line read was not done.
- A continuum limit (`ASSEMBLY.md` treats the assembly separately; not claimed here).

3.9 The convergence route: what it adds, and its status

Not needed for Theorem 1. It adds: (i) existence of $\beta_\infty = \lim \beta_j(0)$; (ii) its value $c \log L$; (iii) a rate; (iv) eventual positivity, hence literal (0.31) for $K \ge K_0$ (Lemma D′ (d)); (v) with certified early steps, literal (0.31) at every K (Section 3.3 (L2)). It also implies (B2), since convergence of $\sigma_j^{(2)}$ implies boundedness. So it needs strictly more than Route K.

3.9.1 Components and status

- U3 Theorem 4.1 with O5: $|\partial_K^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, $|\alpha| \le 2$, uniform to $K = 0$, odd L [PROVED+VERIFIED] (`U3_VERIFY`, `U3_REPAIRS_VERIFY`). Numerically $|\Delta_M - \Delta_\infty| =$ 1.007, 0.129, 0.0147 at $M = 3, 9, 27$.
- O6: $\|\mathfrak{B}_j(K)^{-1}\| \le b(L)$ for all j, K, one-step constraint, trivial background [PROVED+VERIFIED]. Crude constant ($b(3) \approx 7.5 \cdot 10^{33}$); measured corner values 103.5, 176.7, 193.0, 195.3 ($j = 0, 1, 2, \infty$, $L = 3$) use a surrogate Λ (`U3_REPAIRS_VERIFY` §5).
- Theorem 6.1: $\beta_{j+1}(0) = \Phi(\mathcal{D}_j)$ is Lipschitz in the level data [PROVED+VERIFIED, CONDITIONAL ON V1]. The norm is the Taylor-data norm (`U3_VERIFY` P8 correction).
- Lemma 5.1 (convergence of $\sigma_j^{(1)}$, $\sigma_j^{(2)}$ in C^2, $|\alpha| \le 2$):
  - neutral sector, rate L^{-4j} [PROVED+VERIFIED FOR THE VALUE BOUND; THE K-DERIVATIVE PARAGRAPH IS PROVED, UNVERIFIED];
  - (W) [PROVED+VERIFIED] (`U3_AW` Theorem W); it removes the marginal direction of the f^1 recursion;
  - (A) qualitative and Theorem A3 [PROVED+VERIFIED]; (A) in full (strip/C^3 for $m = 1$, all of $m = 2$) $\Leftarrow$ (H″) [OPEN]; the route via (HS) is [REFUTED] at $L = 3$;
  - $\sigma^{(1)}$ convergence at rate jL^{-2j} [SKETCH IN `U3_AW` §4.7, UNVERIFIED]; $\sigma^{(2)}$ convergence [OPEN, NEEDS (A) IN FULL];

– the single-block-constraint version of Lemma 5.1 is false for $m = 2$ in $d = 4$ [PROVED+VERIFIED]; Balaban's composed constraint is not affected.

- Prop. 3.6: $|\beta_{j+1}(0) - \beta_\infty| \le C(1+j)^2 L^{-2j}$ [PROVED, UNVERIFIED AS AN ASSEMBLY; CONDITIONAL ON V1, (A) IN FULL]. (The form Cj^2L^{-2j} of the first version of these notes is 0 at $j = 0$ and is corrected here.) The identification of σ-Taylor data with the data of (6.1) is cited from `U3_VERIFY` §7, not re-shown. (N) (clean rate L^{-2j}) is not needed and is judged plausibly false.

- Value of the limit. Given convergence, Theorem K or Theorem $\mathrm{B_{PW}}$ gives $\beta_\infty = c \log L$ at once: $S_k/k \to \beta_\infty$ by Cesàro and $S_k/k = c \log L + O(1/k)$ [PROVED+VERIFIED AS AN IMPLICATION: `THEOREM2_PROOF_VERIFY` POINT 4; `BETA_FP_PW` THEOREM 3.2(III)]. Older routes to the value: `BETA_FP_H` Theorem 3.1′ with (H) restated at the constant field, which given (W) reduces to Lemma 5.1 (1.2) alone (`BETA_FP_H_VERIFY` items (1)–(4)), plus (PI), (F), (M) [CONDITIONAL; (M) ARGUED]; and `BETA_FP_PW` Theorem 3.1 (limit form) given (U), ($C\theta$), LW (5.7) [PROVED, UNVERIFIED]. The original (H) with IR weights (`BETA_FP_LEMMA43` (4.2)) was graded UNSUPPORTED and is not used.

- Rate of $\mathcal{C}(\Lambda)$: $O(\Lambda^{-2} \log \Lambda)$ given Lemma 5.1 with rates, via `U3_AW` §3.4(b) slaving [ARGUED, `BETA_FP_H_VERIFY` §5].

3.9.2 Early steps and j_*

Under Prop. 3.6 and $\beta_\infty = c \log L$, $\beta_{j+1}(0) \ge (c \log L)/2$ for $j \ge j_* := \min\{J : C(1+j)^2 L^{-2j} \le (c \log L)/2 \text{ for } j \ge J\}$. Heuristic size [ARGUED]: $C \sim b^3$ gives $j_* \approx 110$ ($L = 3$), ≈ 70 ($L = 13$); $C \sim b^6$ (one-step inverses inside C_{data}) gives ≈ 218, ≈ 138 (`THEOREM2_PROOF_VERIFY` Point 5(e)). With the measured $\|\mathfrak{B}^{-1}\| \approx 195$: $j_* \approx 10$ ($L = 3$), ≈ 4 ($L = 13$). Certifying $\beta_1(0), \dots, \beta_{j_*}(0) > 0$ needs interval arithmetic at odd $L \ge 13$ (blocks $\approx 3.4 \cdot 10^5$ rows per momentum by dense methods; the umklapp code reduces this to 12×12 blocks at L^4 momenta). It is needed only for (L2), not for Theorem 1.

4 The finite-torus existence theorem

This section proves Theorem B. It takes the trajectory of Section 3 as the input block (B-T2). Section 4.1 fixes what "exists" means here. Section 4.2 states the uniform bound (H-U) that the blocks must deliver, and Sections 4.3–4.4 state the theorems. Section 4.5 proves them modulo the named blocks, and Section 4.6 gives the dependency table, which is the single status table of the paper. Section 6 then collects the numerical support for the hypotheses.

Source: `ASSEMBLY.md` (2026-10-27), verified in `ASSEMBLY_VERIFY.md`. The statements carry the verifier's corrections (Section 4.8). The theorem is a reduction to named blocks. It is not unconditional. The notation is that of Sections 2.1 and 2.5.

4.1 What "exists" means here

Jaffe–Witten footnote 2 [15] [VERIFIED QUOTE] (`reports/clay_ym.txt` line 619; anchor: the sentence "Such a result presumably would enable the study of the limit as $T^4 \to \mathbb{R}^4$", line 582): "We specifically exclude weak-existence (compactness) as the solution to the existence part of the Millennium problem, unless one also uses other techniques to establish properties of the limit (such as the existence of a mass gap and the axioms)."

Reading used here. "Exists" means: (i) the family $\{S_{n,\varepsilon}\}$ is relatively compact (uniform bounds); (ii) at least one limit point exists along a sequence $\varepsilon_k \to 0$; (iii) every limit point has the properties (OS0)–(OS3) below, proved on the limit, not assumed. No uniqueness is claimed. Full-sequence convergence needs E13 (comparison of two trajectories; `SOURCES.md` (E14-d), D6), which is not in this paper.

This reading is weaker than the footnote's parenthesis. The theorem below gives no gap and only the torus versions of four axioms. It is NOT an admissible Clay solution. It is the finite-torus step of Jaffe–Witten §6.5 [VERIFIED QUOTE]: "One must then verify the existence of limits of appropriate expectations of gauge-invariant observables as the lattice spacing tends to zero and as the volume tends to infinity" (only the first limit is done here).

The OS axioms are not stated in `reports/clay_ym.txt`. It cites them as OS (CMP 31 (1973), CMP 42 (1975)) [19, 20] and requires "axiomatic properties at least as strong as those cited" (lines 279–280) [VERIFIED QUOTE]. The forms used follow the template's application of OS II (`reports/o3_continuum.txt` lines 4987–5003) [VERIFIED QUOTE]: "The linear growth condition requires a bound $|S_n(F)| \le \sigma_n |F|_{s_n}$ with fixed s and $\sigma_n \le A(n!)^D$ for fixed A, D", and restriction to Schwartz tests flat on the coincidence diagonals gives the test class of that reference. OS II itself was not re-read [CITED].

4.2 Hypothesis delivered by the blocks

Definition 4.1 ((H-U) uniform bound)**.** There are $s \in \mathbb{N}$ and $A, D < \infty$ such that for all $K \ge K_0$, all $n \ge 1$, and all $F \in \mathcal{D}_n$:

$$|S_{n,\varepsilon}(F)| \le \sigma_n \|F\|_{sn}, \qquad \sigma_n := A(n!)^D, \quad \varepsilon = L^{-K}.$$

The derivative order sn grows linearly in n. The OS linear growth condition admits this. (H-U) is the output of blocks (B-T2) + (B-S) + the bridge (B-N1) of Section 4.5.

4.3 Theorem (finite-torus existence, compactness form)

Theorem 4.2 (finite-torus existence, compactness form [PROVED MODULO NAMED BLOCKS])**.** *Let $G = SU(2)$, $d = 4$, L odd and large, $\ell \in 2\mathbb{N}$ fixed. Let $\gamma > 0$ be Balaban's small constant and $g \in]0, \gamma]$ small. For $K \ge K_0$ let $\varepsilon = L^{-K}$ and $g_0 = g_0(\varepsilon, g)$ be a shooting solution (Lemma 3.4) whose effective couplings satisfy $g_k \in]0, \gamma]$ for $0 \le k \le K$ and $g_K = g$. Assume (H-U). When (H-U) is taken from the blocks (Section 4.2) rather than assumed, the chain also assumes the $\kappa \to \kappa/2$ decay recovery of D2 [R, B22] (not stated, `SOURCES_D2_VERIFY.md` (5)), which does not follow from the other hypotheses. Then:*

(a) Compactness. Every sequence $K_m \to \infty$ has a subsequence K_{m_i} such that, with $\varepsilon_i = L^{-K_{m_i}}$, the limit $S_n(F) := \lim_i S_{n,\varepsilon_i}(F)$ exists for every $n \ge 1$ and every $F \in \mathcal{D}_n$.

(b) For every limit family $(S_n)_{n \ge 1}$ obtained in (a):

- *(OS0) Regularity. S_n is linear on $\mathcal{D}_n$, and $|S_n(F)| \le A(n!)^D \|F\|_{sn}$. In particular S_n restricted to $C_c^\infty(T^{4n} \setminus \bigcup_{i<j}\{x_i = x_j\})$ is a distribution of order $\le sn$.*
- *(OS1) Invariance. $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for every $a \in T_\ell^4$ (all continuous translations), and $S_n(F \circ R^{\times n}) = S_n(F)$ for every R in the hypercubic group B_4 (signed permutations of the axes, acting about a lattice site).*

- *(OS2) Reflection positivity. Let* $\theta(x_0,\vec{x}) = (-x_0,\vec{x})$ *on* T^4_ℓ *and* $T_+ := \{x : x_0 \in]0,\ell/2[\}$. *For every finite family* $F_0 \in \mathbb{C}$, $F_n \in \mathcal{D}_n$ *with* $\operatorname{supp} F_n \subset T^n_+$ *(*$1 \le n \le N$*):*

$$\sum_{n,m=0}^{N} S_{n+m}(\Theta F_n \otimes F_m) \ge 0, \quad (\Theta F_n)(x_1,\dots,x_n) := \overline{F_n(\theta x_n,\dots,\theta x_1)}, \quad S_0 := 1.$$

- *(OS3) Symmetry and reality.* $S_n(F\circ\pi) = S_n(F)$ *for every permutation* π *of the* n *arguments, and* $S_n(F) \in \mathbb{R}$ *for real* F.

(c) Nontriviality is a separate statement, Theorem 4.3.

The sourced blocks use the unit-scale condition (U), $\rho_1 \le c_* r_0/M^a$ at $\lambda_K = 1$ (r_0 the Cauchy margin of the constant-source family, `SOURCES_D2C.md` (3.5)), as a constant choice (C23), made once after r_0 and before K; B-N1 meets it by the rescaling $\lambda_i = 2N_U(f_i)$ at a C^n cost, s and D unchanged (`SOURCES_UN1.md`, `SOURCES_UN1_VERIFY.md`). This (U) is not the Route B hypothesis (U). *What is claimed about measures.* No probability measure is claimed. The limit is a family of Schwinger functions on $\mathcal{D}_n$. A measure on distributions would need finite moments of smeared fields, and $\operatorname{Var}\sum_p f(x_p)\varphi_p \sim \varepsilon^{-4}$ (Section 2.5). (OS0)–(OS3) on $\mathcal{D}_n$ are the inputs of OS reconstruction.

4.4 Theorem (nontriviality), conditional on (B-NT)

Theorem 4.3 (nontriviality [CONDITIONAL ON (NT) OF SECTION 4.5.9])**.** *Under the hypotheses of Theorem 4.2 and (NT), every limit family satisfies: (i)* $S_2 \ne 0$*: there are* $f_1, f_2 \in C^\infty(T^4)$ *with disjoint supports and* $S_2(f_1\otimes f_2) \ne 0$. *(ii)* S_2 *is not the two-point function of a scale-invariant (in particular free Maxwell) composite:* $d^8 S_2(x,y)$, $|x-y| = d$, *is not asymptotically constant as* $d \to 0$.

Part (ii) is stated here in the form the proof supports (two-sided envelope only); see Section 4.8, item 3. The source stated (ii) as "behaves like $c\,g(d)^4$ with $g(d)^2 \sim (\beta\log(1/d))^{-1}$", which needs $\beta_j \to \beta_\infty$ and is not proved.

4.5 Proof modulo blocks

Structure. (B-T2) gives the trajectory. (B-S) gives sourced bounds uniform in ε. (B-N1) turns them into (H-U). (B-C) gives (a). (B-REG), (B-TI), (B-RP), and (OS3) give (b). (B-NT) gives Theorem 4.3. Only (B-C), (B-REG), (B-TI), (B-RP), and (OS3) are proved in this section; they use nothing beyond (H-U) and exact lattice symmetries.

4.5.1 (B-T2) Theorem 2: the running coupling

Needed. The hypothesis of Balaban's Theorem 1 (CMP 122 II p. 355), $g_k \in]0,\gamma]$ for all k, quoted in Section 1.2 [VERIFIED QUOTE]. It was to be discharged by Theorem 2 of CMP 109 p. 259 [8]: for small g there is $g_0(\varepsilon,g)$ with all $g_k \in]0,\gamma]$, $g_K = g$, and (0.31): $1/g^2 + \beta\log(L^k\varepsilon)^{-1} \le 1/g_k^2 \le 1/g^2 + \beta'\log(L^k\varepsilon)^{-1}$. Balaban: "has not been published yet" [VERIFIED QUOTE].

Status from Section 3 (and §10 of `THEOREM2_PROOF_v4.md`). The weak form, Theorem 2w, is [PROVED+VERIFIED, CONDITIONAL] by Route K (Theorem K + Lemma D″) on (H2) and the fixed-level inputs (PI), (F), (M_j), V1, (G2), plus L5 and item 7′ ((H0) at odd $L > 11$, or (G-gain) with the running-maximum lift, verified (`SOURCES_GGAIN_NONMONO_VERIFY`)); item 9 (K-independence) is closed. It suffices for Theorem 1 by the cross-scale sweep [PROVED, UNVERIFIED]. Literal (0.31) is not obtained (it needs eventual positivity of $\beta_j(0)$). Route B (Theorem B of `BETA_FP_PW`) is

a formally distinct second route; its logic is [PROVED+VERIFIED, CONDITIONAL], its inputs (U), (DF), (T1) are [PROVED, UNVERIFIED] and rest on (H2). The reduction pieces are Lemma D′, Lemma D″, and Lemmas D, E (the one-argument forms in Lean).

Conclusion. At $L = 13$, (H2) is certified (Theorem 6.1); (B-T2) remains conditional on the fixed-level inputs, L5, and item 7′; see the dependency table (Section 4.6).

4.5.2 (B-S) Sourced bounds E14-a..g

Needed. A version of Balaban's Theorem 1 with the (C1) source, i.e. with the local bare coupling $w_p = 1/g_0^2 - f(x_p)$, f real, $|\partial^m f| \le r$ for $m \le 5$ (class F_r), uniformly in ε. Output form (`SOURCES.md` (N1)): $\log Z_\varepsilon[f] - W_\varepsilon[f]$ is C^∞ in source amplitudes z with $|\partial_z^\alpha(\log Z_\varepsilon[f] - W_\varepsilon[f])| \le C^{|\alpha|}\alpha!\, r^{-|\alpha|}$ for $f = \sum_i z_i f_i$ with disjoint supports. $W_\varepsilon[f]$ is a local field-independent contact functional (allowed to diverge; it contributes zero to every mixed derivative of order ≥ 2 for disjoint supports, `SOURCES_VERIFY` item 5(a)).

Estimates needed (`SOURCES.md` §2.5): (E14-a) sourced small-field step; (E14-b) sourced large-field step and R operation; (E14-c) derivative bounds in f. Not needed: (E14-d) fixed-torus convergence (= E13); (E14-g) tensor sources. (E14-e) RP and symmetry transfer is done in this section (Sections 4.5.6, 4.5.7). (E14-f) nontriviality at a fixed layer is needed only for Theorem 4.3. The current status of each sub-block is in the dependency table (Section 4.6); (B-S) remains a [HYPOTHESIS].

4.5.3 (B-N1) Bridge: source-form bounds ⇒ (H-U) [OPEN]

`SOURCES_VERIFY` item 5(b) [VERIFIED QUOTE]: the source form "does not give $|S_n(F)| \le C^n n! \|F\|_{s,n}$ for F flat on diagonals with a fixed s … Mark N1 for general F as a gap." This paper does not close the gap. It gives a route [DERIVED SKETCH], which shows that the gap is removed by the OS index growth sn.

1. Moments from cumulants. $\langle \prod_i \phi(f_i) \rangle = \sum_{\text{partitions}} \prod_{\text{blocks}} \kappa(\cdot)$. Singleton cumulants vanish (φ is centred). For disjoint supports the source form bounds every cumulant by $C^{|B|}|B|!\, r^{-|B|}$, so the moment by $C^n n!\, r^{-n}$.

2. Scale-δ tests. A bump of side $\delta \le 1$ and amplitude a has $|\partial^5 f| \sim a\delta^{-5}$; it lies in F_r if $a \le cr\delta^5$. So n disjoint unit-amplitude scale-δ bumps have $|\text{moment}| \le C^n n!\, \delta^{-5n}$. This is lossy but polynomial in δ^{-1}.

3. Whitney decomposition. Split T^{4n} by a smooth dyadic partition in $\rho(x) := \min_{i<j} |x_i - x_j| \in [\delta, 2\delta]$, $\delta = 2^{-m} \ge \varepsilon$. On each piece cover by cubes of side $\delta/100$; cube tuples are pairwise disjoint. Expand $F\cdot$(cutoffs) in a Fourier series on each cube tuple; the coefficients decay faster than any power; each term is a product of disjointly supported scale-δ bumps. Flatness gives $|\partial^\beta F(x)| \le C_k \|F\|_{k+|\beta|} \rho(x)^k / k!$. The count is $\le C^n \delta^{-4n}$ cube tuples. Derivatives of the cutoffs cost at most $(Cn^2/\delta)^M$; with $M = O(n)$ this is $\le C^n (n!)^c \delta^{-M}$.

4. Bound per dyadic scale: $\le C^n (n!)^{D'} \delta^{k-9n-M} \|F\|_{k+M}$. Choose $k = sn - M$ with s large enough ($s \ge 20$ suffices on this count) so the δ-exponent is $\ge n$; the sum over dyadic $\delta \in [\varepsilon, 1]$ converges uniformly in ε.

5. Sub-lattice region $\rho(x) \lesssim \varepsilon$: $\le C^n \varepsilon^{-4n}$ plaquette tuples, each moment $\le 2^n$, and $|F| \le \|F\|_k \varepsilon^k$; this is $\le C^n \|F\|_{sn}$ for $k = sn \ge 4n$.

Result (if the steps hold): (H-U) with some fixed s and D. To check: uniformity of the source bound in the position of the bumps and in ε; the C^∞ real-source form of (E14-c) with n-th derivatives for all n; the cutoff-derivative count in step 3. The verifier lists further gaps: complex Fourier modes against real sources (a factor 2^n), the extra derivatives of Fourier summation (absorbed in s), and the smooth min-distance partition. Later work (`SOURCES_D4D5`, graded in `SOURCES_D4D5_VERIFY`) confirms "(S_{src}) $\Rightarrow$ (H-U) with $s = 6$, $D > 1$" given (FX) (see Section 4.6). `ASSEMBLY_N1.md` closes the three gaps given (S) (confirmed in `SOURCES_HHIST_N1_VERIFY.md`) and, since 2026-10-28, uses the verified form: C^1 norm $\max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$, order $(n!)^{1+\eta}$, $s = 6$, $D > 1$ ($D > 2$ under the fallback order $2 + \eta$); the ∂^5, δ^{-5n}, and $s \geq 20$ counts in the sketch above are superseded.

4.5.4 (B-C) Compactness [PROVED HERE, GIVEN (H-U)]

Proposition 4.4. *Under (H-U), every sequence $\varepsilon_m \to 0$ has a subsequence along which $S_{n,\varepsilon}(F)$ converges for every n and every $F \in \mathcal{D}_n$. The limit S_n is linear and satisfies $|S_n(F)| \leq \sigma_n \|F\|_{sn}$.*

Proof. 1. Separability. $C^{sn}(T^{4n})$ with $\|\cdot\|_{sn}$ is separable (trigonometric polynomials with rational coefficients are dense; Fejér means converge in C^k). A subset of a separable metric space is separable. So $\mathcal{D}_n$ has a countable subset Q_n dense for $\|\cdot\|_{sn}$. 2. Diagonal extraction. $Q := \bigcup_n Q_n \times \{n\}$ is countable; enumerate it as $(F^{(1)}, n_1), (F^{(2)}, n_2), \ldots$. Each sequence $S_{n_t,\varepsilon_m}(F^{(t)})$ is bounded by $\sigma_{n_t}\|F^{(t)}\|_{sn_t}$ (H-U), so it has a convergent subsequence. Extract successively and take the diagonal subsequence ε_i. 3. Extension. Let $F \in \mathcal{D}_n$, $\eta > 0$, and $G \in Q_n$ with $\|F - G\|_{sn} \leq \eta$. By (H-U) and linearity, $|S_{n,\varepsilon_i}(F) - S_{n,\varepsilon_j}(F)| \leq 2\sigma_n\eta + |S_{n,\varepsilon_i}(G) - S_{n,\varepsilon_j}(G)|$. The last term $\to 0$. So the sequence is Cauchy; call the limit $S_n(F)$. 4. Linearity passes to pointwise limits, and $|S_n(F)| = \lim |S_{n,\varepsilon_i}(F)| \leq \sigma_n\|F\|_{sn}$. □

Remarks. (i) The same subsequence serves all n; this matters for (OS2). (ii) Nothing here needs E13. (iii) The bounded normalizations B_K and c_ε can be included in the extraction.

4.5.5 (B-REG) Regularity (OS0) [PROVED HERE, GIVEN (H-U)]

By step 4 of the proof of Proposition 4.4 the limit satisfies $|S_n(F)| \leq \sigma_n\|F\|_{sn}$, $\sigma_n = A(n!)^D$. On the compact torus the Schwartz seminorm is replaced by $\|F\|_p$ (the weights control decay at infinity on $\mathbb{R}^{dn}$; the torus has none). This is the quoted OS II form. Distribution property: for $F \in C_c^\infty$ of the off-diagonal open set, $F \in \mathcal{D}_n$ and the bound is continuity in the C^{sn} topology, so S_n is a distribution of order $\leq sn$ there.

Caveat. OS II is formulated on $\mathbb{R}^{dn}$. On T^{4n} the axioms are used in their torus form (periodic test functions). This is a change of setting, not a consequence of OS II.

4.5.6 (B-TI) Translation and hypercubic invariance (OS1) [PROVED HERE, GIVEN (H-U)]

Lattice, exact. Put the origin at a lattice site.

- Translations $a \in \varepsilon\mathbb{Z}^4$. The map $U \mapsto U^a$, $U^a(b) := U(b - a)$, permutes the links, preserves the product Haar measure and the Wilson action, and $P_p(U^a) = P_{p-a}(U)$. So $\langle \prod_i \varphi_{p_i+a} \rangle_\varepsilon = \langle \prod_i \varphi_{p_i} \rangle_\varepsilon$ and $S_{n,\varepsilon}(F \circ \tau_a^{\times n}) = S_{n,\varepsilon}(F)$. The coupling $g_0(\varepsilon)$ is one number, so nothing in μ_ε depends on position.

- Hypercubic $R \in B_4$ (signed permutation matrices, $|B_4| = 384$). R maps sites to sites, links to links (possibly reversed), plaquettes to plaquettes, and $x_{Rp} = Rx_p$. Define $(RU)(b) := U(R^{-1}b)$ with $U(-b) = U(b)^{-1}$. Haar measure is invariant under $U \mapsto U^{-1}$ and $\mathrm{Re}\,\mathrm{tr}\, W^{-1} =$

$\operatorname{Re}\operatorname{tr} W$ for $W \in SU(2)$, so A and the measure are invariant and $P_{Rp}(RU) = P_p(U)$. Hence $S_{n,\varepsilon}(F \circ R^{\times n}) = S_{n,\varepsilon}(F)$.

- These identities hold at every ε. $F \circ \tau_a^{\times n}$ and $F \circ R^{\times n}$ are again in $\mathcal{D}_n$. They pass to the limit along the subsequence of Proposition 4.4: $S_n(F \circ R^{\times n}) = S_n(F)$, and $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for $a \in \varepsilon_i \mathbb{Z}^4$ for all large i.

Continuous translations. Let $a \in T_\ell^4$. Pick $a_i \in \varepsilon_i \mathbb{Z}^4$ with $|a - a_i| \le \varepsilon_i$ (componentwise rounding gives $\le \varepsilon_i \sqrt{4}/2 = \varepsilon_i$). Write $F_a := F \circ \tau_a^{\times n}$. The mean value theorem along the diagonal direction gives $\|F_a - F_{a_i}\|_{sn} \le 4n|a - a_i|\,\|F\|_{sn+1}$. Then

$$|S_n(F_a) - S_n(F)| \le |S_n(F_a) - S_{n,\varepsilon_i}(F_a)| + |S_{n,\varepsilon_i}(F_a) - S_{n,\varepsilon_i}(F_{a_i})| + |S_{n,\varepsilon_i}(F_{a_i}) - S_{n,\varepsilon_i}(F)| + |S_{n,\varepsilon_i}(F) - S_n(F)|.$$

The first and last terms $\to 0$ by Proposition 4.4. The third is 0 by lattice invariance. The second is $\le \sigma_n \cdot 4n\,\varepsilon_i \|F\|_{sn+1}$ by (H-U) and linearity. Let $i \to \infty$. F is smooth, so $\|F\|_{sn+1} < \infty$. Rotations outside B_4 are not available (Section 4.7).

4.5.7 (B-RP) Reflection positivity (OS2)

Lattice theorem [CITED]: Osterwalder–Seiler [21], Ann. Phys. 110 (1978) 440–471. The paper is not in the project folder; no verbatim quote. Content as usually stated: for the Wilson action with compact gauge group and any $\beta \ge 0$, and θ the reflection in a lattice hyperplane that bisects the links orthogonal to it, $\langle \Theta(\Phi)\Phi \rangle \ge 0$ for every function Φ of the link variables in the positive half-lattice, $(\Theta\Phi)(U) := \overline{\Phi(\theta U)}$. Jaffe–Witten [15] line 546 [VERIFIED QUOTE]: "Reflection positivity holds for the Wilson approximation [36], a major advantage". The template uses the same input (`o3_continuum.txt` line 4902) [VERIFIED QUOTE]: "At every cutoff the nearest-neighbour measure is reflection positive across microscopic site and link seams." Proof idea (standard): fix the crossing links to 1 by a gauge transformation; the action splits as $A_+ + \theta A_+ + \sum_{\text{crossing}} \operatorname{Re}\operatorname{tr}(X_p \theta(X_p)^*)$, and $\exp(\beta \operatorname{Re}\operatorname{tr}(XY^*))$ is a positive-type kernel (positive combination of characters), which gives a sum of squares. On the periodic lattice there are two fixed planes, $x_0 = \varepsilon/2$ and $x_0 = \varepsilon/2 + \ell/2$; both are link planes because N_ε is even (ℓ even, Section 2.5). The periodic two-seam version is standard [CITED]; its text was not checked. For site-plane reflections see Menotti–Pelissetto [17] (Section 4.8, item 1).

Passage to the limit [PROVED HERE, GIVEN (H-U) AND THE LATTICE THEOREM]. Setup. $\theta_\varepsilon(x) := \theta(x) + \varepsilon e_0 = \theta(x - \varepsilon e_0)$, fixed planes $x_0 = \varepsilon/2,\ \varepsilon/2 + \ell/2$. The positive half Λ_+ consists of links with both ends in $\{x_0 \in \{\varepsilon, 2\varepsilon, \dots, \ell/2\}\}$. Fix $F_0 \in \mathbb{C}$ and $F_n \in \mathcal{D}_n$, $\operatorname{supp} F_n \subset T_+^n$, $1 \le n \le N$. Supports are compact in the open set, so there is $\delta > 0$ with $x_0 \in [\delta, \ell/2 - \delta]$ on all supports.

Step 1 (lattice form). For $\varepsilon < \delta/2$ every plaquette with centre in a support has all four links in Λ_+. Define $\Phi := F_0 + \sum_{m=1}^N \sum_{p_1 \dots p_m} F_m(x_{p_1}, \dots, x_{p_m}) \prod_i \varphi_{p_i}$. Φ depends only on Λ_+ links. Since φ_p is real and $P_p(\theta_\varepsilon U) = P_{\theta_\varepsilon p}(U)$, re-indexing gives $\Theta\Phi = \bar{F}_0 + \sum_n \sum_q \overline{F_n}(\theta_\varepsilon x_{q_1}, \dots, \theta_\varepsilon x_{q_n}) \prod \varphi_{q_i}$. Hence

$$0 \le \langle \Theta\Phi\, \Phi \rangle_\varepsilon = \sum_{n,m=0}^{N} S_{n+m,\varepsilon}(\Theta_\varepsilon F_n \otimes F_m), \quad \Theta_\varepsilon F_n := (\Theta F_n) \circ \tau_{-\varepsilon e_0}^{\times n},\ S_{0,\varepsilon} := 1.$$

(Argument order inside ΘF_n is immaterial by (OS3).) Each $\Theta_\varepsilon F_n \otimes F_m$ lies in $\mathcal{D}_{n+m}$: its two factors have supports at positive distance and each factor is flat on its own diagonals.

Step 2 (seam shift). $\|\Theta_\varepsilon F_n \otimes F_m - \Theta F_n \otimes F_m\|_{s(n+m)} \le 2^{s(n+m)} \cdot 4n\,\varepsilon\, \|F_n\|_{s(n+m)+1} \|F_m\|_{s(n+m)}$. By (H-U) the difference of the $S_{n+m,\varepsilon}$ values is $\le \sigma_{n+m}$ times this, which $\to 0$.

Step 3 (limit). Along the subsequence of Proposition 4.4, $S_{n+m,\varepsilon_i}(\Theta F_n \otimes F_m) \to S_{n+m}(\Theta F_n \otimes F_m)$ for each of the finitely many (n,m). So $\sum_{n,m} S_{n+m}(\Theta F_n \otimes F_m) = \lim_i \sum_{n,m} S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m) \geq 0$.

Lemma 4.5 (positivity of a finite form survives pointwise limits [PROVED HERE])**.** *Let $a^{(i)} \in \mathbb{C}^{(N+1)\times(N+1)}$ with $\sum_{n,m} \bar{c}_n c_m a^{(i)}_{nm} \geq 0$ for all c and all i, and $a^{(i)}_{nm} \to a_{nm}$ for every (n,m). Then $\sum \bar{c}_n c_m a_{nm} \geq 0$ for all c.*

Proof. For fixed c the left side is a finite linear combination of the entries, so it converges to $\sum \bar{c}_n c_m a_{nm}$; $[0,\infty)$ is closed. □

Step 3 is this lemma with $a^{(i)}_{nm} := S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m)$ (replace F_n by $c_n F_n$), plus Step 2.

4.5.8 (OS3) Symmetry and reality [PROVED HERE]

$S_{n,\varepsilon}(F \circ \pi) = S_{n,\varepsilon}(F)$: the φ_p are commuting real random variables; re-index the sum. For real F, $S_{n,\varepsilon}(F) \in \mathbb{R}$. Both identities are linear and pass to the limit. $S_0 = 1$ by definition.

4.5.9 (B-NT) Nontriviality: what is needed

- A two-point function alone never excludes a Gaussian: every positive-definite S_2 is the covariance of some Gaussian generalized field.
- The plaquette density is a composite ($\approx \frac{1}{4} \operatorname{tr} F^2 \times \varepsilon^4$). Even in the free theory its connected $n \geq 3$ functions are nonzero (Wick squares). So $S^c_n \neq 0$ for some $n \geq 3$ does not prove interaction.
- The meaningful test excludes (a) the zero theory and (b) the free Maxwell composite. The free composite has, at short distance, the scale-invariant form $c_0 d^{-8}$. Asymptotic freedom predicts $c\, g(d)^4 d^{-8}$ with a running coupling.

Free-theory prediction along this sequence. At tree level with the bare coupling, $\langle \varphi_p \varphi_{p'} \rangle \approx c\, \varepsilon^8 g_0^4 |x-y|^{-8}$. Since $g_0(\varepsilon)^2 \leq (1/g^2 + \beta \log \varepsilon^{-1})^{-1} \to 0$ by (0.31), the bare-coupling prediction for S_2 at fixed separation tends to 0. The claim is that the true S_2 does not.

Statement (NT) needed [HYPOTHESIS]; owned by (E14-f), a corollary of (E14-a) at a fixed layer. There are $c > 0$ and C such that for all K, all $f_1, f_2 \in C^\infty(T^4)$ with disjoint supports at distance $\geq d$, $d \geq L\varepsilon$, and $k(d)$ the layer with $L^k \varepsilon \leq d < L^{k+1}\varepsilon$:

$$S_{2,\varepsilon}(f_1 \otimes f_2) = c \iint f_1(x) f_2(y)\, g^4_{k(|x-y|)} |x-y|^{-8}\, dx\, dy\, (1+R), \quad |R| \leq C(g^2_{k(d)} + (d/\ell)^2).$$

c is the one-loop (tree-level in the composite) coefficient: $c \propto \sum (\text{free field-strength covariance})^2$, a sum of squares, so $c > 0$ [DERIVED, STANDARD]; its value must be computed in Balaban's normalization.

Derivation of Theorem 4.3 from (NT) and (B-T2) [PROVED HERE, GIVEN (NT)]. (i) Take $f_1, f_2 \geq 0$, not identically 0, with supports at distance $\geq d_0$. Corrected condition (Section 4.8, item 3): the support diameters are $\ll \ell$, so that $g^2 + (|x-y|/\ell)^2$ is small on the whole support pair and $|R| \leq \frac{1}{2}$. By (0.31) at layer $k(d)$ with $d \in [d_0, \ell]$: $1/g_k^2 \leq 1/g^2 + \beta' \log(L^k\varepsilon)^{-1} \leq 1/g^2 + \beta' \log(L/d_0)$, uniformly in ε. So $S_{2,\varepsilon}(f_1 \otimes f_2) \geq \frac{1}{2} c (1/g^2 + \beta' \log(L/d_0))^{-2} \iint f_1 f_2 |x-y|^{-8} > 0$ uniformly in ε, and the limit inherits the bound. (ii) For bumps concentrated at separation $d \to 0$ (with d a fixed multiple of their width), $d^8 S_2 \leq (1 + C g^2_{k(d)})\, c\, (1/g^2 + \beta \log(1/d))^{-2} \to 0$ as $d \to 0$ by the

lower bound in (0.31), while $d^8 S_2 > 0$ by (i). A scale-invariant composite would give a nonzero constant limit. So the limit is neither zero nor the free composite. Only the two-sided bound (0.31) is used. Caveat. (ii) needs (NT) uniformly down to $d \geq L\varepsilon$ with the remainder controlled by $g^2_{k(d)}$, i.e. the fixed-layer expansion at every layer. If (E14-f) is proved only at a fixed layer j, (i) holds, and (ii) holds only in the weaker form "$d^8 S_2(d)$ is bounded between the two (0.31) envelopes for $d \in [L^{-j}, 1]$".

4.6 Dependency table

The table gives the status of every block of Theorem B and of every named hypothesis of Theorem A. It follows the state recorded in `PLAN.md` (through the entry `SOURCES_S2_G1_V7_VERIFY.md`, 2026-10-28) and §10 of `THEOREM2_PROOF_v4.md`. Global conditions carried by every row: (GC1) Balaban's published proofs CMP 95–122 (1984–89) are correct as published; nothing in this paper re-checks them line by line. (GC2) Balaban's inductive assumptions (CMP 109 pp. 263–264) hold with the stated uniformity. (GC3) L odd, $L > 11$, M large, γ small as Balaban requires; ℓ even (sufficient, Section 2.5). Rows "item n" are the named hypotheses of Section 3; n is their item in Section 7. Their file columns use the short names of `THEOREM2_PROOF_v2_VERIFY` §K (KAPPA = `KAPPA_BOUNDED`, H_VERIFY = `BETA_FP_H_VERIFY`, PW_VERIFY = `BETA_FP_PW_VERIFY`, v1 verifier = `THEOREM2_PROOF_VERIFY`).

Block	Content	Current tag	Files	Hypotheses carried
B-T2	Theorem 2w (Theorem 3.1): trajectory $g_0(\varepsilon, g)$, (b), (c′), (S)	[PROVED+VERIFIED, CONDITIONAL] (Route K = Theorem K + Lemma D″). Literal (0.31) not obtained	`THEOREM2_PROOF_v4.md`, `KAPPA_BOUNDED.md`	(H2), (PI), (F), (M_j), V1, (G2), L5, item 7′ (from (H0) or (G-gain), verified (`SOURCES_GGAIN_NONMONO_VERIFY`))
B-T2.a	Lemmas D, D′, D″, E (shooting)	[PROVED+VERIFIED; LEAN] (D′, D, E, D″ one-argument form; history form open)	`LEMMA_DPRIME.md`, `lean/`	continuity of β_{k+1} in history
B-T2.b	Sweep: (c′)+(S) ⇒ Balaban's Theorem 1 inputs (Section 3.4.3; item 10)	[PROVED, UNVERIFIED]; 4 of 19 rows verified (LEMMA_DPRIME_VERIFY Steps 4–5, rows 11, 12, 16, 18)	`LEMMA_DPRIME_SWEEP.md`	Balaban readings (unchecked list)
B-T2.1	U3 Thm 4.1 (Gaussian RG, rate L^{-2j}) + O5	[PROVED+VERIFIED]	`U3_GAUSSIAN_RG.md`, `U3_REPAIRS.md`	trivial background

Block	Content	Current tag	Files	Hypotheses carried
B-T2.2	U3 Lemma 5.1; (A), (W)	neutral sector [PROVED+VERIFIED]. (W) numerically exact; (A) confirmed qualitatively and quantitatively (`U3_AW_VERIFY.md`); (HS) [REFUTED] at $L = 3$. Not on the path of Theorem 2w after (H2); needed for the convergence route only	`U3_LEMMA51.md`, `U3_AW.md`	(A) in full, (N) (plausibly false)
B-T2.3	O6 bordered-matrix invertibility	one-step [PROVED+VERIFIED]; composed constraint reduced to one-step O6 + (A)	`U3_REPAIRS.md`	—
B-T2.4	$c = 11N^2/(12\pi^2)$, anchor-free identity Prop. 2.1, telescoping	[PROVED+VERIFIED]. $\beta_\infty = c \log L$ itself replaced by window sums (Theorem K, Route B)	`BETA_FP_LEMMA43.md`, `BETA_FP_LEMMA43_VERIFY.md`	(F) [CITED, SOURCE CHECKED]
B-T2.5	κ_k bounded (Theorem K)	[PROVED+VERIFIED, CONDITIONAL ON (B2)]; (B2) [PROVED+VERIFIED, CONDITIONAL ON (H2)]	`KAPPA_BOUNDED.md`, `KAPPA_BOUNDED_VERIFY.md`, `B2_TWOSTEP.md`, `B2_TWOSTEP_VERIFY.md`	(H2), (PI), (F), (M_j), V1, (G2)
B-T2.H2	(H2) uniform two-step inequality $\rho_2 < L$ (Section 3.7.3; item H)	[CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)] at $L = 13$: $\rho_2 \le \sup \rho_1^2 \le 10 < 13$ (Theorem 6.1); residue: reading of (0.4), gauge choice, Prop. A at $M \ge 27$, Lemma G beyond point checks, `bb_arb.py` audited only. At $L = 3, 5, 9$ [NUMERICAL]; point enclosure $[1.9993, 2.1273] < 3$ at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED]	`H1_ANALYTIC.md`, `H1_ANALYTIC_VERIFY.md`, `H1_ANALYTIC_VERIFY2.md`, `H2_CERTIFY_P5_VERIFY.md`, `H2_LARGE_L_VERIFY.md`, `H2_CERTIFY_VERIFY.md`; B2_TWOSTEP §2	— (box sweep at $L = 13$ running as a cross-check)
B-T2.H1	one-step route at $L = 13$: $\sup \rho_1^2 < L \Rightarrow$ (H2)	[CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]: $\sup_{j,K} \rho_1^2 \le 9.9995 < 13$ by the majorant B_{13} and `bb_arb.py`; factorization [PROVED+VERIFIED; LEAN] (`rho2_le`); majorant core [LEAN] (`Tower.lean`)	`H1_ANALYTIC.md`, `H1_ANALYTIC_VERIFY.md`, `H1_ANALYTIC_VERIFY2.md`, `H2_CERTIFY_VERIFY.md`	—

Block	Content	Current tag	Files	Hypotheses carried
B-T2.B	Route B (Theorem B_{PW}; item B8)	logic [PROVED+VERIFIED, CONDITIONAL] (tag decision, `PLAN.md`; PW_VERIFY §6 items 1–4); inputs (U), (DF), (T1) [PROVED, UNVERIFIED]	`BETA_FP_PW.md`, `BETA_FP_PW_VERIFY.md`, `BETA_FP_U.md`	(H2), $(B\theta)$, A6, LW (5.7), (H0)
item 1	(B2) / (B2w) (Section 3.7)	[VERIFIED REDUCTION] to (H2); holds at $L = 13$ (Theorem 6.1); open at other L	KAPPA §2, Rem. 2.3; KAPPA_VERIFY §7–§9	(H2)
item 1	(H2) $\Rightarrow$ $(C1\sigma)$ $\Rightarrow$ (B2) (Theorems 3.10, 3.11)	proved+verified, conditional on (H2)	B2_TWOSTEP §2–§3; B2_TWOSTEP_VERIFY grades 1, 3, 4	(H2)
item 1	$(C1\sigma)$ + (h) $\Rightarrow$ (B2)	implication verified; $(C1\sigma)$ $\Leftarrow$ (H2); (h) numerical $L = 3, 5$	KAPPA Prop. 4.1; KAPPA_VERIFY §7	—
item 1	(H″) $\Rightarrow$ (A) $\Rightarrow$ (B2)	implication argued; not needed after (H2)	U3_AW §4.7; no verifier (U3_AW_VERIFY grade 5 covers (H′) only)	—
—	(HS) (Section 3.7.2)	refuted $L = 3$	U3_AW (H′); U3_AW_VERIFY §6	—
item H	$\rho_1^2 < L \Rightarrow$ (H2) (one step, $L \geq 5$)	arithmetic; $\rho_1^2/L = 0.771$ at $L = 5$, numerical	H2_LARGE_L §3; H2_LARGE_L_VERIFY grade 4 (measurement)	—
item H	certification tail j_0	derivation confirmed; $j_0 \approx 8$ ($L = 3$), ≈ 6 ($L = 5$), corrected	H2_LARGE_L §4.1; H2_LARGE_L_VERIFY grade 6	—
item 2	(PI)	argued	BETA_FP_H §3.1; H_VERIFY §5	—
item 3	(F)	cited, source checked	BETA_FP_LEMMA43; H_VERIFY §4	—
item 4	(M_j): (S1)–(S3), Prop. 4.4	(S1), (S3), $(S2)|_{t=0}$, (S2c) proved at fixed j given (I3); (M_j) [VERIFIED REDUCTION] to (I3), (G'_A), Prop. 4.4 region 1 (argued)	BETA_FP_M §4; `SOURCES_T2_S123.md`, `SOURCES_T2_S123_VERIFY.md`; H_VERIFY §5 refuted the old step	(I3), (G'_A)
item 5	V1	open	R2; U3_VERIFY C4 (order p^2)	—
item 6	(G2)	formal	BETA_FP_LEMMA43	—

Block	Content	Current tag	Files	Hypotheses carried
item 7′	$\lvert\beta_{k+1}(h,g) - \beta_{k+1}(0)\rvert \le C_\beta g$ on every past (was (H0), uniform in j and history)	from (H0) [BALABAN], odd $L > 11$, or from (G-gain) with the running-maximum lift [PROVED+VERIFIED]	CMP 109 (1.22); `SOURCES_GGAIN_NONMONO.md`, `SOURCES_GGAIN_NONMONO_VERIFY.md`; KUNIF_VERIFY item 3	(G-gain) inputs if taken from it
item 8	L5 history form and domain (Section 3.4.5)	open	LEMMA_DPRIME Rem. 3; LEMMAS_DE_VERIFY §2.1; v2_VERIFY §F	—
item 9	$\beta_j(g)$ independent of K	closed as a reduction [VERIFIED]: volume statement, β_j whole-lattice by definition, D′/D″ at fixed K; residual volume question = ledger B8, an input of Theorem 1	`SOURCES_T2_KUNIF.md`, `SOURCES_T2_KUNIF_VERIFY.md`	(K1), (K2) = item 7′, (K3)
item 10	BALABAN_CHECK item 11 readings	verified with corrections (former ledger A12, A20, A22); residue: A11 not located, A23 (T-step rate loss)	BALABAN_EXCERPTS; `SOURCES_BOOKKEEPING.md`	A11, A23
item 11	odd L; $L > 11$ in Balaban	input	CMP 109 pp. 251–252; v1 verifier Step 6	—
item 12	SU(2)	scope	—	—
item 13	Lean history form	open	—	—
B1, B2′	Route B: (H2) + (T1) ⇒ (U) (Theorem 2.4 of BETA_FP_U)	proved, unverified	BETA_FP_U §2.4; no verifier	(H2), (T1)
B1, B2′	Route B: (U)	hypothesis; ⇐ (T1) + (H2) (BETA_FP_U); proved, unverified	BETA_FP_PW §1.2; PW_VERIFY §3	(T1), (H2)
B1	Route B: (T1)	argued	BETA_FP_U §2.3; no verifier	—
B3	Route B: $(B\theta) = \gamma_{rec} + \mathfrak{d}_F$	γ_{rec} bounded given Thm 2.4 (proved, unverified); (DF) argued	BETA_FP_PW §3.3; BETA_FP_U §3; PW_VERIFY §6 (logic of Theorem B_{PW}); none for BETA_FP_U	(DF)
B4	Route B: LW (5.7)	cited	cited; PW_VERIFY §3 (regime correct)	—
B5	Route B: A6 (β identification, uniform in k)	argued	SOURCES_D1_VERIFY 8; PW_VERIFY §3; BETA_FP_U §4.2	—
B7	Route B: $\theta_* < 1$	trivial condition	PW_VERIFY §3	—

Block	Content	Current tag	Files	Hypotheses carried
B-T2.6	certified early steps $\beta_j(0) > 0$	needed only for literal (0.31) at every K, not for Theorem 1; [NUMERICAL], no certificate; $\beta_1(0)$ computed to $L = 11$	`numerics/NUMERICS.md`, `UMK_BETA1.md`	$L > 11$ is Balaban's range
B-S	sourced Theorem 1 (E14-a,b,c)	[OPEN]	`SOURCES.md` §2.5	B-T2 for the sourced trajectory
B-S.D1	one sourced step, Gaussian level	[PROVED+VERIFIED] (with corrections)	`SOURCES_D1.md`, `SOURCES_D1_VERIFY.md`	—
B-S.D2	sourced small-field step (E14-a)	(a), (b), (d), (f) confirmed; (c) rejected; (e) reduced	`SOURCES_D2.md`, `SOURCES_D2_VERIFY.md`	analyticity in input activities; $\kappa \to \kappa/2$ [R, verified, B22; uses A23]; (U) [C23, constant choice]
B-S.D2C	redo of (c): frozen source per domain, Lemma A, $h \equiv 0$	Lemma A and recursion confirmed; (I) corrected (exact only with adapted gauge-fixing weight)	`SOURCES_D2C.md`, `SOURCES_D2C_VERIFY.md`	domain hypothesis (D); (Hβ)
B-S.Hβ	two-loop bound $\sup \sum_j \lvert B'_j \rvert \le \frac{1}{2}$	[PROVED+VERIFIED, CONDITIONAL] on (I), (H-hist)	`SOURCES_HBETA.md`, `SOURCES_HBETA_VERIFY.md`	(I), (H-hist)
B-S.hist	(H-hist) history sensitivity	[VERIFIED REDUCTION] on (W4), Facts A/B, (I)/L3 and (G-gain) in both slots (`SOURCES_SM.md`; verifier: items 1–4 confirmed, 5–6 corrected); the (G-gain) chain continues in row B-S.Ggain (G3, (Q)); with the large-field half (`SOURCES_GGAIN_LF_VERIFY`) the real uniform (H-hist) is a [VERIFIED REDUCTION] in both halves (ledger B13); Lemma HH: `SOURCES_HHIST.md`, `SOURCES_HHIST_N1_VERIFY.md`	`SOURCES_SM.md`, `SOURCES_SM_VERIFY.md`	(G-gain), small and large field

Block	Content	Current tag	Files	Hypotheses carried
B-S.Ggain	(G-gain): $g_k\|B\|$ gain in the small-field step	pointwise Cauchy and uniform C_β on $]0,\gamma]$ [PROVED+VERIFIED] (small-field, incl. vacuum subtraction); Lemma G4 off-chain (SOURCES_G4.md); χ_k surface term = Lemma G3, proved given (Q) [PROVED+VERIFIED, CONDITIONAL] (SOURCES_G3_VERIFY); (Q) reduced to CMP 99 (3.132), (E_U), and the localization lemma LOC [VERIFIED REDUCTION] (SOURCES_Q_VERIFY, SOURCES_CDC_VERIFY, SOURCES_EU_VERIFY, SOURCES_LOC_VERIFY); large field [VERIFIED REDUCTION] (one Cauchy step on the large-field disc, SOURCES_GGAIN_LF_VERIFY); non-monotone pasts: running-maximum lift (SOURCES_GGAIN_NONMONO.md, verified (SOURCES_GGAIN_NONMONO_VERIFY))	SOURCES_GGAIN.md, SOURCES_GGAIN_VERIFY.md, SOURCES_GGAIN_LF.md	(Q)/c_-, (M_r), CMP 102 (189), (M-B) bookkeeping; not at $g = 0$
B-S.G3	Lemma G3: χ_k surface term, uniform in g	[PROVED+VERIFIED, CONDITIONAL] on (Q) (Route 2 surface estimate; the B/B′ cut closed in SOURCES_Q)	SOURCES_G3.md, SOURCES_G3_VERIFY.md	(Q)
B-S.Q	(Q): covariance lower bound $c_- > 0$	[VERIFIED REDUCTION] to CMP 99 (3.132) (printed; ledger A1, kernel-decay form now a verified reduction via Lemma CT to CMP 99 Thms 3.1, 3.3, 3.12, SOURCES_S2_FORM_VERIFY), (E_U), and LOC; Woodbury identity confirmed (CDC); free-case constants [NUMERICAL] (numerics/kappa_d.py)	SOURCES_Q.md, SOURCES_Q_VERIFY.md, SOURCES_CDC.md, SOURCES_CDC_VERIFY.md, numerics/KAPPA_D.md	A1 residue (torus-side hypothesis, conventions, reading $a = 1$; G_1 clause closed, SOURCES_S2_G1_V7_VERIFY); two [I] bridges

Block	Content	Current tag	Files	Hypotheses carried
B-S.EU	(E_U): uniform block extension	$K_0(4) = 195.19$ ($N = \infty$) [NUMERICAL, REPRODUCED]; $K \le K_0(1 + c_{2/3}(M\alpha_0)^{2/3} + c_1 M\alpha_0)$ [DERIVED SKETCH]; (E_U) [VERIFIED REDUCTION] to kernel-moment bounds Bałaban does not print	`SOURCES_EU.md`, `SOURCES_EU_VERIFY.md`, `numerics/k0_d.py`	k-uniform kernel moments; unprinted C_1'
B-S.LOC	LOC: localized covariance $C^{(k)}(Z_0, 0)$	[PROVED+VERIFIED, CONDITIONAL] (short lemma; symmetry (S) not needed)	`SOURCES_LOC.md`, `SOURCES_LOC_VERIFY.md`	R_2 bound (2.16) (ledger A2, verified reduction with the V7 repair to (W^c) in the exponential gauge + A7 at real U + two [I]); γ_0 for general U (ledger A3, full proof verified, $\gamma_0 \ge m_L/(48d^2L^d)$); α_0, α_1 small
B-S.thr	(H-thr) threshold shell sum	Lemma T and trajectory sum [PROVED+VERIFIED], conditional on (H-hist2); (H-hist2) [VERIFIED REDUCTION] to the real uniform (H-hist), no new input (`SOURCES_HHIST2`, `SOURCES_HHIST2_VERIFY`)	`SOURCES_HTHR.md`, `SOURCES_HTHR2.md`, `SOURCES_HTHR2_VERIFY.md`	(H-hist2), M3 weights, frozen integers (L3)
B-S.hist2	(H-hist2): mixed second history derivatives	[VERIFIED REDUCTION] to the real uniform (H-hist), no new input (Route 2(a); decay $\lambda_C^{m/2}$; the product form λ^m [REFUTED])	`SOURCES_HHIST2.md`, `SOURCES_HHIST2_VERIFY.md`	real uniform (H-hist)
B-S.D3	sourced large-field step + R operation (E14-b)	draft verified with corrections (`SOURCES_D3_VERIFY`, `SOURCES_D3R_VERIFY`, `SOURCES_D3W8_VERIFY`; route (a), Lemma F2′ confirmed given (W4), (Eps), F1, (D)). (FX) is a genuine new hypothesis (`SOURCES_D4D5_VERIFY.md`); its state is in row B-S.FX	`SOURCES_D3.md`, `SOURCES_FX.md`, `SOURCES_FX_VERIFY.md`	(FX), (FX2); (U) [C23, constant choice]

Block	Content	Current tag	Files	Hypotheses carried
B-S.FX	O1(b), O2, O3, O4, R2	O1(b) [PROVED+VERIFIED] (Lemma [P], no additive constant); O2 [PROVED+VERIFIED] (the R-denominator is the bare Wilson integral (1.100)); O3 [PROVED+VERIFIED, CONDITIONAL ON N3] (N3 = the open HBETA strip bound $\|\delta w\| \le \frac{3}{2}\tau$); R2 ($\kappa_{fin} \ge \kappa_{final}$) [PROVED+VERIFIED] as a reduction, in volume form, given $R_{K+1} \ge 1$ and C_c independent of g; O4 (unbounded components in (1.103)) avoided by keeping Balaban's main form (1.98), whose fallback was Gevrey $2+\eta$. N3 reduced to (H-hol) + real (H-hist) + (H2-hist), condition (a) settled [VERIFIED REDUCTION] (`SOURCES_HBETA_N3_VERIFY`, `SOURCES_N3A_VERIFY`); (H-hol) small field [VERIFIED REDUCTION] (`SOURCES_HHOL_VERIFY`); large field [VERIFIED REDUCTION] to (M-A), (M-B), (M-C), (H-hist2) (`SOURCES_HHOL_LF_VERIFY`); (M-B) [PROVED+VERIFIED] with $p_0 \ge 17/(\gamma_0 A_1^2)$, $\theta_W + \tau_C \le 0.2$; (M-A) reduced to (H−), (M-A′) closed, (H−) rests on CMP 102 (189) [VERIFIED REDUCTION] (`SOURCES_MA_VERIFY`, `SOURCES_MA2_VERIFY`); (M-C) [VERIFIED REDUCTION] to printed CMP 116 §2 conditions (`SOURCES_MC_VERIFY`); $R_{K+1} \ge L$ and C_c free of g are verified readings (`SOURCES_BOOKKEEPING.md`)	`SOURCES_FX_O1.md`, `SOURCES_FX_O1_VERIFY.md`, `SOURCES_FX_O23.md`, `SOURCES_FX_O23_VERIFY.md`, `SOURCES_FX_R2.md`, `SOURCES_FX_R2_VERIFY.md`	residual inputs of (M-C); CMP 102 (189) (ledger A4, verified reduction to A1)

Block	Content	Current tag	Files	Hypotheses carried
B-S.N3	N3: HBETA strip bound $\lvert\delta w\rvert \le \frac{3}{2}\tau$	[VERIFIED REDUCTION] to (H-hol) + real (H-hist) + (H2-hist)	`SOURCES_HBETA_N3.md`, `SOURCES_HBETA_N3_VERIFY.md`	(H-hol), (H-hist), (H2-hist)
B-S.N3A	condition (a) of Lemma N3	settled [PROVED+VERIFIED] (slack $> 5\times$; item 1 corrected)	`SOURCES_N3A.md`, `SOURCES_N3A_VERIFY.md`	constant order $L \to$ Bałaban's constants $\to \gamma \to g \le \gamma/2 \to r_0 \to K$
B-S.hol	(H-hol), small field: β_j holomorphic on $\lvert g - x\rvert < \rho_0 x$	[VERIFIED REDUCTION] to Bałaban's conditions with modified constants	`SOURCES_HHOL.md`, `SOURCES_HHOL_VERIFY.md`	(C1′)–(C4′), (C4″), (2.24′)
B-S.holLF	(H-hol), large field (frozen thresholds at x^*)	[VERIFIED REDUCTION] to (M-A), (M-B), (M-C), (H-hist2); (M-C) itself a verified reduction	`SOURCES_HHOL_LF.md`, `SOURCES_HHOL_LF_VERIFY.md`	(M-A)–(M-D)
M-A	equipartition $\langle A - A_{\min}\rangle_u \le (n_i/u)(1+\eta_x)$	[PROVED+VERIFIED] given (H−); (M-A′) closed [PROVED+VERIFIED]; (H−) [VERIFIED REDUCTION] to CMP 102 (189), an existing trust item	`SOURCES_MA.md`, `SOURCES_MA_VERIFY.md`, `SOURCES_MA2.md`, `SOURCES_MA2_VERIFY.md`	CMP 102 (189); (C1), (C2)
M-B	(1.80)–(1.98) bookkeeping, changed constants	[PROVED+VERIFIED] with $p_0 \ge 17/(\gamma_0 A_1^2)$, $\theta_W + \tau_C \le 0.2$	`SOURCES_MB.md`, `SOURCES_MB_VERIFY.md`	(M-A), (M-C)
M-C	complex small-field step	[VERIFIED REDUCTION] to printed CMP 116 §2 (B1–B23 + 4), Lemma 2, (1.36), p. 12 (2.5)–(2.6); shrinks $\varepsilon_1/(4\lambda_V)$, $\gamma/(8\lambda_V)$; (I-X) resolved	`SOURCES_MC.md`, `SOURCES_MC_VERIFY.md`	(I-ε), (I-V0) residual, (M1)
M-D	frozen map vs Bałaban's map	not a convention: needs the frozen map's derivative at x^*, which is (H-hist) + (H-thr) [VERIFIED REDUCTION]	`SOURCES_HHOL_LF_VERIFY.md`	(H-hist), (H-hist2)
B-S.D4	real-source derivative bounds (E14-c)	D4.1, D4.3, D4.4 and Lemma C (order $1+\eta$, given (FX)) [PROVED+VERIFIED]; (S_{src}) $\Rightarrow$ (H-U), $s = 6$, $D > 1$, confirmed given (FX)	`SOURCES_D4D5.md`, `SOURCES_D4D5_VERIFY.md`	(FX), (FX2), S1 at all scales

Block	Content	Current tag	Files	Hypotheses carried
B-N1	source form ⇒ (H-U) on $\mathcal{D}_n$	derived given the sourced Gevrey bound (S) (`ASSEMBLY_N1.md`; confirmed, `SOURCES_HHIST_N1_VERIFY.md`); (S_{src}) ⇒ (H-U), $s = 6$, $D > 1$, confirmed given (FX); uniformity of (S) in support size and ε [VERIFIED REDUCTION] (h-independent; `SOURCES_B21_S_VERIFY`); ASSEMBLY_N1 aligned 2026-10-28 with the C^1 norm, $(n!)^{1+\eta}$, $s = 6$, $D > 1$ ($D > 2$ if the coarsest-scale (FX2) radius fails); open: the (E14-c) clause	this paper, `SOURCES_D4D5.md`, `SOURCES_B21_S.md`	(FX) main form, coarsest-scale (FX2) radius, S1 at all scales, Lemma C
B-C	compactness (diagonal extraction)	[PROVED HERE]; verifier: correct	Proposition 4.4	(H-U)
B-REG	(OS0)	[PROVED HERE]; verifier: correct	Section 4.5.5	(H-U); torus form of OS II
B-TI	(OS1) all of T^4 and B_4	[PROVED HERE]; verifier: correct	Section 4.5.6	(H-U); origin at a site
B-RP	(OS2)	lattice: [CITED]; passage: [PROVED HERE]; verifier: correct	Section 4.5.7	ℓ even (sufficient); (H-U)
OS3	permutation symmetry, reality	[PROVED HERE]	Section 4.5.8	—
B-NT	nontriviality (Theorem 4.3)	[OPEN]: needs (NT) = (E14-f); derivation from (NT) [PROVED HERE], corrected statement	Section 4.5.9	(NT) at every layer for (ii); $c > 0$ in Balaban's normalization
E13	full-sequence convergence	not needed here; [OPEN]	`SOURCES.md` (E14-d), D6	—

Critical path to an unconditional (modulo GC1–GC3) Theorem 4.2: (H2) is certified at $L = 13$ (Theorem 6.1); B-T2 still needs (PI), (F), (M_j), V1, (G2), L5, and item 7′, and the step to Theorem 1 needs the sweep (B10, A11) and the finite-volume identification B8; then B-S: the open inputs of the real uniform (H-hist) ((Q)/c_-, (M_r), CMP 102 (189), (M-B) bookkeeping), the residue under A1–A4 ((W_G) with O3, O4; (W^c) and A7; R-Δ2; A18; the torus-side hypothesis), M2′, M4, M5 of (H-thr), the Λ_j bookkeeping of B20, the residual inputs of (M-C), and the (E14-c) clause of B-N1. The blocks proved in this section are short and are not on the critical path.

4.7 What the theorem does not give

Each item needs a separate argument that this paper does not give.

- Rotation invariance. Only B_4 and the torus translations are proved. Full $O(4)$ is not a symmetry of T^4_ℓ. `LEMMAS.md` ledger item 5 [VERIFIED QUOTE]: on the torus the inclined

comparison "gives only the tilted-axis reflection symmetry $R_{2\theta}s$, never a rotation"; the template reaches rotations only after the torus $\to$ plane limit, which it controls with its mass gap.

- Infinite volume. ℓ is fixed. Jaffe–Witten tie the limit $T^4 \to \mathbb{R}^4$ to a volume-uniform gap [VERIFIED QUOTE]: "New ideas are needed to prove the existence of a mass gap that is uniform in the volume of space-time. Such a result presumably would enable the study of the limit as $T^4 \to \mathbb{R}^4$." No public mechanism without a gap is known to us (`LEMMAS.md` ledger item 8).

- Mass gap. Not addressed. On a finite torus a "gap" of the transfer matrix is not the Clay mass gap.

- Uniqueness. Different subsequences may give different Schwinger functions. Uniqueness needs E13, which Balaban does not have. Also not given: uniqueness of the vacuum after reconstruction (no clustering on a torus).

- Vacuum state. Time is periodic, so OS reconstruction from torus reflection positivity gives a transfer matrix on T^3 and a thermal state at inverse temperature ℓ, not a vacuum state (Section 4.8, item 4).

- Also not given: clustering (OS4); a probability measure for the density field; Wilson-loop observables (class C3); the stress tensor and the operator product expansion (class C2 and beyond); independence of the limit from the block-averaging choices of Balaban's RG (only the lattice measure enters $S_{n,\varepsilon}$, but $g_0(\varepsilon)$ is chosen through Balaban's flow).

- Status of the whole statement. Even with every block closed, the theorem is conditional on Balaban's published 1984–89 proofs (GC1). Under Jaffe–Witten footnote 2 it is a compactness ("weak-existence") result on a torus with the torus forms of OS0–OS3; it is not a solution of the Millennium problem.

4.8 Corrections after verification

The verifier confirmed the theorem statement, all in-chapter proofs (B-C, B-REG, B-TI, B_4, the RP passage, OS3, the finite-form lemma), the quotes of Jaffe–Witten, and the open status of B-N1. Four items were corrected (`ASSEMBLY_VERIFY.md`). The statements above carry the corrected wording; the original claims are listed here.

1. *ℓ even is sufficient, not necessary* (verifier §5, grade 3). The parity count is right: L odd gives $N_\varepsilon = \ell L^K$ even iff ℓ even; for odd N_ε every time reflection has one site seam and one link seam. Osterwalder–Seiler (1978) [21] treat link-plane (between-site) reflections. Site-plane reflections were added by Menotti–Pelissetto, CMP 113 (1987) 369–373 [17] [VERIFIED QUOTE] (abstract, per the verifier): "We extend the proof by Osterwalder and Seiler of reflection positivity of lattice gauge theories with fermions to the case in which the reflection is performed with respect to planes containing sites." and "Our proof applies to all observables and relies on the particular structure of the Wilson action." The mixed case (one site seam, one link seam on a periodic lattice) is very likely routine, since each mechanism is local to its seam, but no citation states it. So odd ℓ is not excluded; the hypothesis ℓ even only avoids the mixed case. The source's phrase "plausible but not cited" stands. The periodic two-seam link version attributed to Osterwalder–Seiler is [CITED], not [VERIFIED QUOTE]. Two minor points from the verifier: the support distance after the seam shift is $\geq 2\delta - \varepsilon$ (the source's $2\delta - 2\varepsilon$ is conservative); the factor $2^{s(n+m)}$ in Step 2 is unneeded but harmless.

2. *Reference title.* The Jaffe–Witten reference list gives the title "Gauge theories on the lattice"; the bibliography entry follows the Annals of Physics title.

3. *Nontriviality needs the two-sided envelope only and support diameters $\ll \ell$* (verifier §6, grade 6). In Theorem 4.3(i), $|R| \le \frac{1}{2}$ needs $g^2 + (|x-y|/\ell)^2$ small on the whole support pair, so the support diameters must be $\ll \ell$, not only the minimum distance d_0. In Theorem 4.3(ii) the source stated the asymptotic $d^8 S_2 \sim c\, g(d)^4$ with $g(d)^2 \sim (\beta \log(1/d))^{-1}$. The proof uses only the two-sided bound (0.31); the asymptotic needs $\beta_j \to \beta_\infty$ and is not proved. The statement of Theorem 4.3(ii) above is the supported one. The overclaim was in the statement, not in the proof.

4. *Periodic time gives a thermal state* (verifier §6, grade 8). OS reconstruction from torus reflection positivity gives a transfer matrix on T^3 and a thermal state at inverse temperature ℓ, not a vacuum. This remark is added to the list of non-results (Section 4.7).

5 The sources block

This section gives one paragraph for each project file `SOURCES_*.md` that has a verifier file. Each paragraph states the result, describes the method in two sentences, gives the verifier's grade in the tags of Section 1.4, and names the inputs that stay open. Nothing here goes beyond the project ledgers: the verifier files, `TRUST_LEDGER.md`, the dependency table (Section 4.6), and `PLAN.md`. Where a verifier corrected a file, the corrected statement is given. The grouping follows §4 of `TRUST_LEDGER.md`: first the files that Theorem 2w (Target 1) uses, then the files of the finite-torus theorem (Target 2) in the order of its dependency chain (steps 2–9 of §4.2 there). Ledger labels An, Bn, Cn, Xn are those of `TRUST_LEDGER.md`. `SOURCES_A1_KERNEL.md` and `SOURCES_A4_D189.md` share the verifier `SOURCES_A1_A4_VERIFY.md`, and `SOURCES_HHIST.md` is checked in `SOURCES_HHIST_N1_VERIFY.md`. Three files have no verifier and are not listed: `SOURCES_BOOKKEEPING.md`, `SOURCES_G4.md`, `SOURCES_W4.md`. Files verified after this section was assembled have no paragraph of their own: `SOURCES_WC_EXPGAUGE.md` and `SOURCES_WC_RESIDUE.md` (cited under A1w and A2), and `SOURCES_T2_V1.md`, `SOURCES_T2_PI.md`, `SOURCES_T2_A1WO3.md`, `SOURCES_T2_A1WO4.md`, `SOURCES_T2_D189.md`, `SOURCES_T2_R2REM.md`; their state is in `TRUST_LEDGER.md` and `PLAN.md`.

5.1 Target 1: inputs of Theorem 2w

`SOURCES_GGAIN_NONMONO.md`: item 7′, the running-maximum lift. For every step k, every past $h \in]0,\gamma]^k$, monotone or not, and every $g \in [0,\gamma]$, $|\beta_{k+1}(h,g) - \beta_{k+1}(0)| \le C_\beta g$ with $C_\beta = C_\Pi S_0$, the constant of (G-gain), uniform in k, K, and the past. This removes the monotone-history restriction of (G-gain), which blocked item 7′ on the clamped, non-monotone pasts that the shooting evaluates. The only use of monotonicity is Step 2 of Lemma F2′, where $\max_{j\le k} 4g_j^{\kappa_0} = 4g_k^{\kappa_0}$ compares bound sizes; with the running maximum $G_k = \max_{i\le k} g_i$ the induction $a_j \le 2G_j^{\kappa_0}$ closes for every past under the side condition $4\gamma^{\kappa_0} \le E_0$, which is of Bałaban's existing type. The value at zero own coupling, (K1), is history-free because the exponent vanishes at $g_k = 0$ (CMP 109 p. 268) for any input, and the endpoint $g \to 0$ is a dominated-convergence step that is not written line by line. Verifier: `SOURCES_GGAIN_NONMONO_VERIFY.md`, the lift [PROVED+VERIFIED], item 7′ [PROVED+VERIFIED, CONDITIONAL ON (H0) (LEDGER A5) OR ON THE (G-GAIN) INPUTS (LEDGER B13)]; all four items confirmed with small corrections: $C'_\beta = C_\beta$ holds because $E_0 + 4\gamma^{\kappa_0} \le 2E_0$, the input radius that (G-gain) already covers, and the possible dependence of the newest R-input on g (caveat c1) is open on the literal reading of β and void on the pure-part reading. Open

inputs: (H0) (ledger A5), or the (G-gain) inputs of ledger B13 ((Q)/c_-, (M_r), CMP 102 (189), the (M-B) bookkeeping); L5, the evaluation of β off Bałaban's trajectory (item 8, ledger B7); caveat c4, CMP 119 (2.31) and the off-trajectory small-field induction of `SOURCES_SM.md` §5.1 not swept for coupling-order uses.

`SOURCES_T2_KUNIF.md`**: item 9,** K**-independence of** $\beta_j(g)$**.** Item 9 is a volume statement, not a history identity, and it closes as a reduction: CMP 119 (2.24), p. 253, and p. 279 define β_j as the whole-lattice coefficient of CMP 109 (whose Π in (4.37) is a sum over $\mathbb{Z}^4$), so the β_j of the recursion is K-free by definition. The shooting needs only (K1) $\beta_j(h,0) = \beta_j(0)$ for every admissible past, (K2) $|\partial\beta_j/\partial g_{j-1}| \le C_\beta$, and (K3) the window bound $\sum_{i=k+1}^{n} \beta_i(0) \in [b(n-k) - 2R,\, b(n-k) + 2R]$ of Theorem K; with $\gamma \le b/(4C_\beta)$ these give $T(m,n) \in [(3b/4)(n-m) - 2R,\, (5b/4)(n-m) + 2R]$ on every admissible history. Lemmas D′ and D″ run at one fixed K and never compare two trajectories or two values of K, so only their constants must be free of K. The bound $|\beta_{k+1}(h_k) - \beta_{k+1}(0)| \le C_\beta g_k$ follows from (K1) and the mean-value theorem in the own slot, and it gives the window bracket pointwise on each clamped history of the intermediate-value argument. Verifier: `SOURCES_T2_KUNIF_VERIFY.md`, [VERIFIED REDUCTION]; items 1, 2, 4 confirmed, item 3 corrected: (G-gain) gave C_β only on monotone slot histories, so (K2) via (G-gain) needs item 7′ (now lifted by `SOURCES_GGAIN_NONMONO.md`); the CMP 119 page for the definition of β_j is p. 253, not p. 249. Open inputs: item 7′ for (K2) (from (H0), ledger A5, or from (G-gain), ledger B13); L5, the domain and continuity of β off Bałaban's trajectory (item 8, ledger B7); the inputs of Theorem K for (K3); the finite-volume identification of the step (ledger B8, with A15), an input of Theorem 1, not of the shooting.

`SOURCES_T2_S123.md`**: (**M_j**), ledger B4.** At fixed $j \ge 1$, odd $L \ge 3$, and small flux f, the lattice spectral inputs hold: (S1) $\mathcal{K}_t(f) \ge -C_1|f|$ for $t \in [0,1]$, (S3) $\pm V \le C_3[(\mathcal{K}_t + \mu)^2 + f^2]$ in the subdivided form of `KAPPA_BOUNDED.md` §1.3, (S2) at $t = 0$, and the coarse comparison (S2c) $N_0(\lambda - \eta_1) \le N_t(\lambda) \le N_0(\lambda + \eta_1)$ with $\eta_1 = O(f^2)$. Corollary 7.1 then reduces (M_j), the limit of $[\Gamma_c(\Lambda_j; f) - \Gamma_c(1; f) - (f = 0 \text{ value})]/(n_2 f^2)$ to d_j, to the near-zero input (Z_j), itself reduced to the 2D gap (G'_A), and to region 1 of Prop. 4.4 of `BETA_FP_M.md`. The proofs work in the twisted algebra of operators that commute with the magnetic translations, with Peierls products, Moyal expansions with kernel-norm remainders, a Nenciu-type inverse, and operator bounds from the vanishing order of symbols. At $t = 0$ a fibre decomposition over $k_\perp$ separates the operator, and a gauge-copy lemma (Lemma 5.8) shows that the near-zero density of states needs only (G'_A): the transverse charged 2D SU(2) Wilson Hessian at constant flux has no spectrum in $(-c|f|, c|f|)$, and numerically its gap is about $|ef|$. Verifier: `SOURCES_T2_S123_VERIFY.md`, (M_j) [VERIFIED REDUCTION], with (S1), (S3), (S2)$|_{t=0}$, (S2c) and region 2 of Prop. 4.4 [PROVED+VERIFIED AT FIXED j, CONDITIONAL ON (I3)]; 25 claims confirmed, 2 corrected (in Lemma 1.4 ∇^β is one twisted convolution; the explicit L-power of C_j in Prop. 3.2 is convention-dependent and is not quoted), and the input list of (M_j) gains the neutral-sector differentiation under the integral, an inference. Open inputs: (I3), the entrywise analyticity in f of Bałaban's linearized averaging at each of the j steps (argued); (G'_A) (open, independent of j and L, numerically confirmed at six rational fluxes with margin about $|ef|$); region 1 of Prop. 4.4 of `BETA_FP_M.md` (argued); all under ledger B4.

5.2 Target 2, step 2: the sourced Theorem 1 (ledger B11)

`SOURCES.md`**: the E14 reduction of (B-S).** The plaquette-density source is a local bare coupling, $\rho_0^f = \exp(-\sum_p (g_0^{-2} - f_p) P_p(U))$, so the Schwinger functions on test functions flat on the diagonals reduce to a sourced version of Bałaban's Theorem 1 with a pointwise coupling flow (S-A), sourced

regular terms (S-B), a local contact functional $W_k[f]$ (S-C), a local β-function (S-D), and the normalization (S-E). The new estimates are (E14-a) the sourced small-field step, (E14-b) the sourced large-field step and R operation, and (E14-c) the derivative bounds in f; (E14-d) = E13 is needed only for full-sequence convergence, and (E14-f) only for nontriviality. A dimension count shows that the source of the dimension-4 density is marginal at every degree, that its field-independent part is relevant and is removed as local contact terms, and that the first field-dependent gradient term has dimension 6. One step is written out at the Gaussian level: with $K_f = (\Delta^{(1)})^{-1/2}\Delta^{(f)}(\Delta^{(1)})^{-1/2}$ and $\|K_f\| \le r$ the source enters through $-\frac{1}{2}\log\det(1-g_0^2K_f)$, and the linear response per step is $c_j = 1 + \frac{1}{2}g_{j-1}^3\beta_j'(g_{j-1})$. Verifier: `SOURCES_VERIFY.md`, [VERIFIED REDUCTION] (the structure confirmed; items 5, 6, 8 corrected, item 10 confirmed with corrections); the printed CMP 119 (2.24) is $1/g_{j-1}^2(x) = 1/g_j^2(x)+\beta_j(g_{j-1})\varphi_j(x)$, and by (2.25) Bałaban evaluates β_j at the constant coupling, so (S-A) and (S-D) are new; derivatives of $\log Z$ are cumulants; $\prod_j c_j$ is uniformly bounded, not shown convergent, and assumes a one-step recursion; the uniform bound for general F flat on the diagonals is not derived from the source form. Open inputs: (B-S) as a whole, (E14-a, b, c) (ledger B11, [OPEN]), with its parts B12–B20 and B22; the (E14-c) clause of the bridge (B-N1) (ledger B21); (NT) = (E14-f) for Theorem 4.3; E13 = (E14-d), not needed for Theorem B.

`SOURCES_D1.md`: one sourced step at one loop (D1). At one loop the plaquette-density source, a local bare coupling $w_p = 1/g_0^2 - f_p$, gives the multiplicative source renormalization $c_1 = 1$ exactly, for every background; the first correction is $c_1 - 1 = \frac{1}{2}g^3\beta_1'(g)$, which is $\beta_1^{(2)}g_0^4 + O(g_0^6)$ if β_1 is even and $O(g_0^3)$ with a C^1 bound only. The field-independent terms are the tadpole $\frac{1}{2}(9L^4-9)g_0^2$ per coarse cell ($\frac{3}{4}(1-L^{-4})g_0^2$ per plaquette) and a contact term $C(q)$ that is analytic at $q = 0^+$, and the first field-dependent gradient terms have dimension 6. The source enters the step only through the Hessian $\Delta_f = \sum_p w_p H_p$ and a current term $\langle J_f, B\rangle$ at Bałaban's unweighted minimizer, while gauge fixing, ghosts, and the constraint carry no f; for constant f the quadratic form scales by $1 - g_0^2 f_0$, so the determinant changes by a field-independent constant over $9L^4 - 9$ modes per cell. Per-momentum formulas for the tadpole, $C(q)$, and the mixed derivative are evaluated by `numerics/d1_check.py` and checked against real-space finite differences on the periodic 6^4 lattice. Verifier: `SOURCES_D1_VERIFY.md`, [PROVED+VERIFIED] for the algebra (items 1–4, 6, 8–10 confirmed) and [NUMERICAL, REPRODUCED] for the rerun, which matched every digit; corrected: the dimension-6 count holds by index parity, the γ_2 values are unreliable, the data support $|c_1 - 1| \le 5\times10^{-4}$ but not geometric convergence in the grid, and the expansion-point error has the small parameter Lp. Open inputs: none at the Gaussian level; the non-Gaussian sourced step and the pointwise flow belong to D2–D3 and to (B-S), ledger B11.

`SOURCES_D2.md`: sourced small-field step (E14-a). The sourced small-field step is Bałaban's step with extra source vertices of order g_k, g_k^2, g_k^3, all in his $P^{(k)}$ class (each $\le O(g_k)(1+|B|^2)$ on $|B| < \varepsilon_1/g_k$), a propagator for the sourced Hessian with Bałaban's decay and regularity bounds (constant $3c_1$ with $c_1 = O(\delta_0^{-5})c_0$, rate $\delta_0/2$), and a field-independent vacuum term W that factors out. Its step (c), the pointwise flow $1/g_{j+1}^2(x) = 1/g_j^2(x) - B_{j+1}(f(x))$ with $h \equiv 0$, is not established here and is redone in `SOURCES_D2C.md`. The source term is evaluated at Bałaban's unweighted minimizer, so his measure and propagators are used without change, and the sourced propagator follows from a weighted Neumann series in his cube-to-cube operator norms (CMP 116 (1.7)). Real-source derivative bounds are tied to polystrip holomorphy by the interpolation Lemma 7.1, and the remainder bounds are reduced to Bałaban's estimates with a freezing property and a constant-source family. Verifier: `SOURCES_D2_VERIFY.md`, steps (a), (b), (d), (f) [PROVED+VERIFIED], step (e) a [VERIFIED REDUCTION], and step (c) rejected, since its claim that CMP 109 Sects. 3–4 run

verbatim is [REFUTED] ((4.34)–(4.37) use Euclidean symmetry, and the cubes are at scale k, not j); corrected: (a) needs a block-constant $\bar{f}$, the constant is $3c_1$ and not $\times 3$, W factors only in the small-field region, and the polydisc radius is $r/(2n)$. Open inputs: analyticity of the step in its input activities, (W4), and the field-dependent domains Λ_j in W (ledger B20); the recovery $\kappa \to \kappa/2$ (ledger B22, a verified reduction that uses A23); (Hβ), reduced in `SOURCES_HBETA.md`; the scale condition (U) is the constant choice C23, not an open input.

`SOURCES_D2C.md`**: step (c) redone, frozen source per domain (Lemma A).** Each step term $E^{(j)}(X, \cdot\,; f)$, $X \in D_j$, is frozen at $c_X = f(b_X)$, and the freezing error obeys $|D_X| \leq E_0 C M^3 (\rho_1/r)(L^j\eta)^5 \lambda_k e^{-\kappa d/2}$ (Lemma A), irrelevant with exponent 1 and with no growth in k. With the identification (I) of the constant-source increment $B_j(c)$ with Bałaban's β_j on the trajectory from $1/g_0^2 - c$, this gives the pointwise recursion $1/g_{j+1}^2(x) = 1/g_j^2(x) - B_{j+1}(f(x))$ with $h \equiv 0$ and gradient corrections irrelevant with exponent $\min(\alpha, \beta, 1)$. The non-constant source is handled at scale j, per localization domain, as a gauge-invariant difference bounded in an axial gauge on X, where each field costs two powers of $L^j\eta$. Bałaban's CMP 109 Sects. 4–5 are then used only for constant sources, where they apply at bare coupling $1/g_0^2 - c$. Verifier: `SOURCES_D2C_VERIFY.md`, Lemma A and the recursion [PROVED+VERIFIED, CONDITIONAL ON (D) AND (Hβ)]; corrected: the freezing points are the points of X nearest to b_X, (I) is exact only with the adapted gauge-fixing weight $1/u_k(c)$ (otherwise up to $O(e^{-c\varepsilon_1^2/g_k^2})$), $\beta > 0$ in (4.18) must be fixed, and the exact cancellation of the cubic first-moment term of (4.34) is unsupported and not needed. Open inputs: the domain hypothesis (D) (ledger B20; argued in `SOURCES_D2C_VERIFY.md` §2 from the CMP 109 p. 263 remark, which cites CMP 102 Prop. 9, ledger A4); (Hβ) (Lemma H of `SOURCES_HBETA.md`); (I) with large-field thresholds (ledger B12); the adapted gauge-fixing weight changes the construction (ledger C10).

`SOURCES_D3.md`**: sourced large-field step (E14-b).** The sourced large-field step is Bałaban's step with all thresholds frozen at the reference couplings, the local coupling $w_j = w_{j-1} - B_j(f)\varphi_j$, an R operation that resets it on Z to $u_k(f(x))$ (CMP 122 II p. 365), and an adapted local gauge-fixing weight. Real sources keep the positivity of the large-field weights under $3\gamma^2 r \leq \frac{1}{2}$ (revised to $5\gamma^2 r \leq \frac{1}{2}$ in `SOURCES_D3R.md`), Lemma D3.1 gives the local holomorphy radius $c_\tau r/(E_0|Y|)$ of one positive denominator, and the identification (I) is off by a threshold term, the new input (H-thr). The file lists the places (P1)–(P7) where Bałaban uses a constant coupling in the large-field step and states the change at each. Each change is reduced to a quoted Bałaban estimate (CMP 119 (2.23)–(2.31), CMP 122 II (1.73)–(1.75)); a line-by-line rerun of CMP 122 I–II is not done. Verifier: `SOURCES_D3_VERIFY.md`, [VERIFIED REDUCTION] with corrections: the reset term is bounded by Bałaban's sup bound (CMP 122 II p. 366), not cancelled; the t-parameter term (P8) of CMP 119 p. 271 is missing; old R-terms are merged into E_k, so freezing through R is open again; the Gevrey order is $(n!)^{2+\epsilon}$, not $(n!)^2$; and the lower bound on w_k holds relative to Bałaban's local coupling. Open inputs: (H-thr) (ledger B15); the real uniform (H-hist) (ledger B13); strip holomorphy of u_k, now N3 (ledger B18); (W4) and the field-dependent domains Λ_j (ledger B20); (FX), (FX2) for the final step (ledger B19); (U) is the constant choice C23.

`SOURCES_D3R.md`**: revision of D3 ((P3), (P8), freezing through R, Gevrey order).** The sourced reset coefficient differs from Bałaban's by at most $\delta_r = 4r + 2C_h M \rho_1 \lambda_k$, additively and pointwise (Lemma P3), so his sup bound of CMP 122 II p. 366 holds with one changed constant, and the large-field factors survive with $\gamma_0 \to \gamma_0(1-\theta) \geq \gamma_0/2$, $\theta = \gamma^2\delta_r \leq 5\gamma^2 r \leq \frac{1}{2}$. The t-parameter term splits as $d_B + d_S$ with $|d_S| \leq 2r$, freezing through R splits into a pure part (Lemma F1) and

an R-induced part (Lemma F2), and the cumulant bound has Gevrey order $(n!)^{2+\epsilon}$, admissible with $D = 3$. Positivity is kept by a split against Bałaban's local coupling, and the (P8) term is split into a piece with the support of Bałaban's deviation, booked as a boundary term, and the source weight of D2. The regular output is split by an auxiliary parameter into the pure and R-induced parts, and the Gevrey count uses a Lah-number sum over partitions. Verifier: `SOURCES_D3R_VERIFY.md`, [VERIFIED REDUCTION] with corrections: (P3) needs $\rho_1 \le r/(4C_hM)$; in (P8) h_k is booked as irrelevant, and d_S changes the covariance $C^{(k)}$ at every t (relative size $O(\gamma^2 r)$); Lemma F2 is a new claim, not Bałaban's bound for merged R-terms (item W8, taken up in `SOURCES_D3W8.md`); the Gevrey count is confirmed conditional on (B-N1). Open inputs: strip holomorphy of u_k (N3, ledger B18); (H-thr) (ledger B15); the real uniform (H-hist) (ledger B13); (W4) and the covariance change in (P8) (ledger B20, the latter by analogy with A16); (B-N1) (ledger B21).

`SOURCES_D3W8.md`**: covariance change along t and merged R-outputs (Lemma F2′).** Along the t-interpolation the measure is Bałaban's CMP 119 (3.27) with covariance $C_f^{(k)}$, independent of s and t; with $\tau \le 3c_1c_H\gamma^2 r \le \frac{1}{2}$ one has $\|C_f - C_0\|_\delta \le 2c_1\tau$, and the large-field factors survive with $\gamma_0 \to \gamma_0/4$. Lemma F2′: the R-induced part obeys $a_m \le 32\varepsilon' g_m^{\kappa_0}$ with no compounding, and $|E_R^{(m)}(X,U) - E_R^{(m)}(X,1)| \le CM^2 \cdot 32\varepsilon'(L^m\eta)^4 g_m^{\kappa_0} e^{-\kappa' d_m(X)}$ on regular configurations, a marginal term of Bałaban's R-type form, summable in m for $\kappa_0 > 2$. Transported inputs keep their scale label and never pass through the generated map G, so a Schwarz-lemma step on $G - G(0)$ with Bałaban's factor $\varepsilon' = O(\alpha_2^{-1}\varepsilon_1)$ (CMP 109 p. 280) gives $a_{k+1} \le 8\varepsilon' b_k$ per generation. The covariance bounds come from the weighted Neumann series of `SOURCES_D2.md`, and positivity from domination by a rescaled C_0 Gaussian. Verifier: `SOURCES_D3W8_VERIFY.md`, route (a) and Lemma F2′ [PROVED+VERIFIED, CONDITIONAL ON (W4), (EPS), LEMMA F1, (D), AND $32\varepsilon' \le 1$], the t-measure and its bounds corrected: the cross term $g_k^2\langle A, \mathcal{W}_S A_k\rangle$ goes to $B^{(k+1)}$, CMP 116 p. 3 covers C_f only by analogy, the domination must use the marginal region (or Anderson's inequality) to avoid a volume factor, and ε' contains Bałaban's large constant of p. 280. Open inputs: (W4) and (D) (ledger B20; (W4) is reduced to the printed (C1)–(C5) through Theorem SM); (Eps), from Theorem SM and A10; the extension A16 to the fluctuation covariance; the conditions $32\varepsilon' \le 1$, $S \ge 1$, $\kappa_0 > 2$ change the construction (ledger C16); the monotone-history assumption of Lemma F2′ is removed by the running maximum, with side condition $4\gamma^{\kappa_0} \le E_0$ (`SOURCES_GGAIN_NONMONO_VERIFY.md`).

`SOURCES_D4D5.md`**: RP and symmetry transfer, and the cumulant bound ($\mathbf{S}_{\mathrm{src}}$).** The sourced RG output equals the bare integral $Z_\varepsilon[f]$ (Lemma D4.1), reflection positivity and the lattice symmetries are properties of the lattice measure (Lemmas D4.2–D4.4), and given (FX) the cumulants of real sources obey ($\mathrm{S}_{\mathrm{src}}$): $|\kappa_n(f_1,\dots,f_n)| \le (C_S\sigma^{-a})^n (n!)^{1+\eta} \prod_i N(f_i)$ with $N(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$, for $\sigma \in [16\varepsilon, \ell/8]$, supports pairwise at distance $\ge \sigma$, minimal diameter $\le \sigma$, and C_S independent of K. Then ($\mathrm{S}_{\mathrm{src}}$) $\Rightarrow$ (H-U) with $s = 6$, $D > 1$. Each RG operation is an identity of integrals (Haar invariance, δ-function blocking, partitions of unity, convergent expansions, $\int dV\, R\rho = \int dV \rho$), and complex sources follow by the identity theorem. Lemma C takes one Cauchy derivative in each of n separated source amplitudes on a polydisc whose radius does not depend on n, so the sum over polymer sizes gives a single $n!$. Verifier: `SOURCES_D4D5_VERIFY.md`, D4.1, D4.3, D4.4 and Lemma C [PROVED+VERIFIED] (Lemma C given (FX)), ($\mathrm{S}_{\mathrm{src}}$) $\Rightarrow$ (H-U) confirmed, and (FX) confirmed as a new hypothesis not contained in CMP 116–122; corrected: cite CMP 122 I (1.102) for the actual R′ operation, the block geometry is invariant only under translations by $M\mathbb{Z}^4$ or coarser, and the order $1 + \eta$ needs the coarsest-scale (FX2) radius (else an extra factor $(n!)^{2p_R/p_0}$, still OS-admissible). Open inputs: (FX) in Bałaban's main form with the coarsest-scale (FX2) radius (ledger B19; remaining input N3, ledger B18); S1 at all scales, part of (B-S) (ledger

B11); the (E14-c) clause (ledger B21).

`SOURCES_HBETA.md`: **Lemma H, the two-loop bound (Hβ).** Lemma H: given (I), (E), and (H-hist), with $x := C_1 A\gamma/\beta \le \ln(3/2)$ and $3r < g^{-2} - \gamma^{-2}$ ($g = g_K$), every real c with $|c| \le 2r$ keeps all $g_k(c) \in \,]0,\gamma]$ and gives $\sum_{j\le K} |B'_j(c)| \le e^x - 1 \le \frac{1}{2}$ and $\frac{1}{2} \le |v_k(c)| \le \frac{3}{2}$, uniformly in K. With $\beta = \beta_0/2$ the smallness condition reads $\gamma \le \ln(3/2)\,\beta_0/(2C_1A) \approx 0.20\,\beta_0/(C_1A)$. In $u = 1/g^2$ the chain rule gives $\partial\beta_j/\partial u_i = -\frac{1}{2} g_i^3\, \partial\beta_j/\partial g_i$, so a C^1 bound costs $O(g_i^3)$ per step, $\sum_i g_i^3 \le 2\gamma/\beta$ along the trajectory, and a Gronwall product on the variational equation gives $e^x - 1$. A bootstrap on the component of $\{|c| \le 2r : u_K(c) > \gamma^{-2}\}$ that contains 0 keeps every c-trajectory in $g \le \gamma$; the Gronwall algebra is formalized in Lean (`LemmaH`). Verifier: `SOURCES_HBETA_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (I), (H-HIST)]; corrected: the hypothesis list adds that β_j is the infinite-volume function independent of K, (H-hist) can be weakened to a column-sum bound, the smooth-clamp remark is dropped, and the toy numbers are not reproducible (the bound holds and is nearly sharp). Open inputs: (I) (ledger B12; it holds by construction with the adapted gauge-fixing weight C10 and the frozen thresholds C1, but is not written as a lemma); the real uniform (H-hist) (ledger B13, a verified reduction with open inputs (Q)/c_-, (M_r), CMP 102 (189), and the (M-B) bookkeeping); (E), the uniform positivity of the one-loop $\beta_j(0)$; complex c is not covered and is the strip bound N3 (ledger B18).

`SOURCES_T2_KAPPA2.md`: **ledger B22.** The decay loss $\kappa \to \kappa/2$ of the sourced step occurs at every step in three places (the interpolated inputs of the freezing proof in `SOURCES_D2.md` §6.3, the extraction remainders of `SOURCES_D2C.md` §3, the R-induced terms of Lemma F2 in `SOURCES_D3R.md`), and it is recovered by $L/2$-rescaling without compounding. The conditions are (K1) $\delta \le \frac{1}{10}(1 - 2(1+4\beta)/(\theta L))$, which is $\delta \le 0.0385$ at $L = 13$, $\beta = \frac{1}{4}$, $\theta = \frac{1}{2}$, and (K2) $\theta(1-\beta)L/2 \ge 1 + 3\beta$; at the last scale the final expansion runs at $\kappa_{\text{eff}} = \theta\kappa/(1+4\beta) = \kappa/4$, with the M-condition (K3) $M \ge 12L(4.5\kappa/C_1)^{1/4}$, implied by the existing one of `SOURCES_FX_R2.md`. Bałaban proves only $2d_k(Z) \ge L d_{k+1}(Z')$ (CMP 116 (2.36)), so the gain per step is $L/2 = 6.5$, and his printed choice $(1 - 10\delta)\frac{1}{2}L = 1$ in (2.41) spends all of it; the recovery therefore takes δ smaller, at the cost of a κ about 3.6 times larger, which moves only constants fixed after κ. The R operation drops $4\beta\kappa$ of rate (CMP 122 II (1.66)–(1.100)), the rescaling pays these drops and the T-step fraction 10δ, and the (1.93) volume charge is paid by p_0 through the condition before (1.95). Verifier: `SOURCES_T2_KAPPA2_VERIFY.md`, [VERIFIED REDUCTION]; items 2–4 confirmed on page images, item 1 corrected: the three losses stack within one step, so a literal half per absorption would make (K1) fail at $L = 13$, and the repair reads $\theta := 1 - m_{\text{tot}}/\kappa$ as the kept fraction per cycle (each loss is a polynomial absorption with additive cost; $\theta \ge \frac{1}{2}$ for $\kappa \ge 2m_{\text{tot}}$), with which (K1)–(K3) stand and the new lower bound is $\kappa \ge 1536$ at $d = 4$, $\beta = \frac{1}{4}$. Open inputs: the unprinted large-field T-step rate loss of CMP 119 Sect. 3 (ledger A23), on which the homogeneity of the large-field T step in the input rate rests; CMP 116 (2.36), which Bałaban asserts without proof.

`SOURCES_UN1.md`: **(U), ledger C23.** The scale condition (U) does not conflict with the bridge (B-N1): with $\lambda_K = 1$ (physical units), the r in (U) is the Cauchy margin r_0 of the constant-source family, not a test-function amplitude, so (U) is the constant choice $\rho_1 \le c_* r_0/M^a$, made once with $\rho_1 := \min(c_* r_0/M^a,\, r_0/(4C_h M))$, after r_0 and before K. The bridge evaluates the generating functional only on sources rescaled into the domain of (FX2), at the price of a factor $(4/\rho_1)^n$ per n-point bound, with s and D unchanged. The bridge uses n-linearity, $\kappa_n(f_1,\dots,f_n) = \prod_i \lambda_i\, \kappa_n(f_1/\lambda_1,\dots,f_n/\lambda_n)$, with $\lambda_i = 2N_U(f_i)$ and $N_U(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$, and pairwise σ-separated supports make the sup and the gradient of $\sum_i z_i f_i$ a maximum over single terms, so the whole Cauchy

polydisc stays in the admissible domain. Small widths $h \geq \varepsilon$ are paid by the norm growth $1/h$ per bump, which the flatness of F absorbs, while (U) costs only an h-independent factor; the scale-matched reading of `SOURCES_D2.md` l. 217 would create a real conflict and is dropped. Verifier: `SOURCES_UN1_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON THE (Hβ) GRADIENT BOUND], so (U) is a constant choice (ledger C23), verified compatible with (B-N1); corrections: the (FX2) gradient radius is $\rho_1/2$, hence $\lambda_i = 2N_U(f_i)$, the sup radius of the activity polydisc is $\tau_m(\Gamma) \leq r$, overlapping supports never reach (S), and the (Hβ) bound $|\nabla f_k| \leq \frac{3}{2}\rho_1$ is a verified reduction for real c (Lemma H and the chain rule), not open. Open inputs: (Hβ) for complex c (open); for real c its inputs (I) (ledger B12) and (H-hist) (ledger B13); the $\kappa \to \kappa/2$ recovery (ledger B22, a verified reduction that uses A23).

5.3 Step 3: the (S) clauses and (FX) (ledgers B21, B19)

`SOURCES_B21_S.md`: uniformity of (S), ledger B21. The source bound (S), $|\kappa_n(f_1, \dots, f_n)| \leq (C_S\sigma^{-a})^n (n!)^{1+\eta} \prod_i N_U(f_i)$ for real sources with pairwise separation $\sigma \geq 16\varepsilon$ and $\min_i \operatorname{diam} \operatorname{supp} f_i \leq \sigma$, holds uniformly in the bump width $h \in [\varepsilon, \ell/512]$ and in the position of the bumps, with the C^1 norm $N_U(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$. Every factor of $C^n(n!)^{1+\eta}$ is independent of h. The file traces one rescaled bump through Lemma C of `SOURCES_D4D5.md` and finds one h-dependent term, the cube count $Q_m \leq (2 + D/(M\lambda_m))^4$ at scales finer than the bump. With $D \leq \sigma$ this count is at most $(2+2D/\sigma)^4 u^4$, $u = \sigma/(2M\lambda_m)$, and the tree decay $e^{-3\kappa u/8}$ of the polymers that meet all supports cancels it, while the gradient cost is paid by N_U and the flatness of the test function. Verifier: `SOURCES_B21_S_VERIFY.md`, [VERIFIED REDUCTION], corrected in scope (the inputs must also list the residue of B19 and D2 §4.3) and in one constant (in the bridge $D/\sigma = 3/8$, not $1/16$, so the factor is $(2.75)^4 \approx 57$). Open inputs: S1 at all scales including $m = 0$; (FX) in Bałaban's main form (1.98) (ledger B19 with the choice C13), whose remaining input is N3 (B18); the coarsest-scale (FX2) radius, without which the order is $2 + \eta$ and $D > 2$; D2 §4.3, itself only reduced; the (E14-c) clause of B21, which no file maps to Lemma C.

`SOURCES_FX.md`: hypothesis (FX), ledger B19. (FX), the convergence of the final Mayer expansion of the sum over large-field geometries plus the unit-scale integral, with polymer radius $c \cdot r$ in the source (FX2), uniform in K and ε, is reduced to Bałaban's large-field bounds CMP 122 II (1.79)–(1.89) and (1.97), the convergence criterion of CMP 116 (2.38)–(2.41) and Cammarota (ledger X4), and named local checks. The construction is a final R-step with $Z'' = \emptyset$ followed by the unit-scale cluster expansion of CMP 116 §2, as Dimock does for φ^4_3 (Dimock III, Theorem 2). The file reads CMP 122 I (0.2)–(0.6) and CMP 122 II (1.71)–(1.104) and finds that Bałaban already exponentiates the sum over renormalizable large-field components inside each R step, so that only the components that pay for later steps stay as a sum, and at the last scale there are no later steps. The fine-history sum inside a component is bounded by (1.79), because Dimock's power-decay argument ((209)–(211)) does not transfer to $d = 4$, and the source enters the large-field operations only through the upper bound (1.73). Verifier: `SOURCES_FX_VERIFY.md`, [VERIFIED REDUCTION], corrected: the per-step expansion sums only components of diameter $\leq 100MR_k$ counted by position ((1.96)), so the last step, which sums components of unbounded size, needs the tree decay O1; three page quotes are fixed ((1.80) has R_n^{d+1} and a local K, (1.85) has $A_1^2 p_0^2$, the condition after (1.94) ends in $-\frac{3}{2}p_0(g_k)$); the fallback order is at most $1 + \eta + 2/3$. Open inputs: the verifier left O1(b), O2, and O3 open; O1(b) and O2 are now [PROVED+VERIFIED] (`SOURCES_FX_O1.md`, `SOURCES_FX_O23.md`), R2 is [PROVED+VERIFIED] as a reduction, and O4 is avoided by the main form (C13), so the remaining input of B19 is N3 (ledger B18), through O3.

`SOURCES_FX_O1.md`**: gap O1(b) of (FX), Lemma [P].** Lemma [P]: for a connected union Y of MR_j-cubes, $d'_{j+1}(S(Y)) \geq c_* d'_j(Y)$ with $c_* = [3 \cdot 4^d \sqrt{d}(\rho+2)^d]^{-1}$, $\rho = LR_{j+1}/R_j$, and no additive constant. With the $n = K+1$ term of CMP 122 II (1.80) this gives $\kappa_K(Y) \geq \kappa_{\text{fin}} d_K(Y)$ for every non-renormalizable final component Y, which is Bałaban's assertion of the term $-\kappa_1 d_k(X)$ on p. 387, with $\kappa_1 = \kappa_{\text{fin}}$. The proof combines three elementary facts (a tree of length ℓ meets at most $4^d(1+2\ell)$ closed unit cubes, $d'_j(Y) \leq \sqrt{d}(n_f - 1)$, and a coarse cube meets at most $(\rho+2)^d$ fine cubes) with the count of at least 21^d coarse cubes in $S(Y)$. The halo $S(Y) \setminus Y$ carries no large fields (p. 384), so nothing is absorbed by p_0, and only the first term of (1.80) is used, all its terms being non-negative. Verifier: `SOURCES_FX_O1_VERIFY.md`, [PROVED+VERIFIED], with one remark corrected (the counterexample to the form "factor $1/L$ plus a constant" needs $d \geq 2$, a constant independent of L, and L large) and R3 discharged by Bałaban's own comparison in (1.86)→(1.87). Open inputs: R2 ($\kappa_{\text{fin}} \geq \kappa_{\text{final}}$), the only point left by the verifier, is closed in `SOURCES_FX_R2.md` (choice C14); R1 (grid alignment) changes constants only.

`SOURCES_FX_O23.md`**: gaps O2 and O3 of (FX).** O2: the R-denominator of CMP 122 I (1.100) is one small-field integral per renormalizable component with the bare Wilson action $-g_k^{-2} A(\zeta_i, \cdot)$ and no E-, R-, or B-terms, so it is holomorphic and non-vanishing on a source polydisc of radius $c \cdot r$, with c independent of $|\Lambda_i|$, k, and ε. O3: inside the large-field operations T' the sourced weight obeys $\operatorname{Re}(w + \delta w) \geq (1 - \theta_W)(g''_k)^{-2}$, so the upper bounds (1.73), (1.79), (1.89), (1.92), (1.97), (1.100) hold with $\gamma_0 \to \gamma_0(1-\theta_W)(1-\tau_C) \geq \gamma_0/4$ and $p_0 \geq 16/(\gamma_0 A_1^2)$. For O2 the Wilson part of the source perturbation is $|\sigma_W| \leq \tau \Phi(g_k)$ with $\Phi(g) = C(M) g^2 p_0(g)^2 R(g)^4 \to 0$, and (1.102) also allows an f-independent denominator. For O3 the plaquette terms $P_p \geq 0$ of the real G-valued configuration inside T' (CMP 122 II p. 379) give domination in modulus by a positive operation with a smaller coupling prefactor, while the thresholds stay fixed. Verifier: `SOURCES_FX_O23_VERIFY.md`, [PROVED+VERIFIED] for O2 and [PROVED+VERIFIED, CONDITIONAL ON N3] for O3, since the strip bound $|\delta w| \leq \frac{3}{2}\tau$ is N3; it adds the cleaner condition $p_0 \geq 8/(\gamma_0 A_1^2)$ with $\gamma'_0 \geq \frac{3}{4}\gamma_0$, and corrects O4 (the division by $T'_k(Y_i)1$ in (1.103) has no positivity proof, but (1.103)–(1.104) is an optional final remark, so O4 is avoided by keeping the main form (1.98)). Open inputs: N3 (ledger B18); the main form is the choice C13; O4 binds only if (1.104) were used, with fallback order $2 + \eta$.

`SOURCES_FX_R2.md`**: gap R2 of (FX),** $\kappa_{\text{fin}} \geq \kappa_{\text{final}}$**.** R2 holds as a reduction in volume form: $\kappa_K(Y) \geq \kappa_{\text{vol}} \# M(Y)$ with $\kappa_{\text{vol}} = \sqrt{d}\,\kappa_{\text{fin}} = C_1 M^d R_{K+1}/(3 \cdot 4^d (L+2r)^d)$, $r = R_K/R_{K+1}$, and $\kappa_{\text{vol}} \geq \kappa_{\text{final}} := 2\kappa + \sqrt{d}(1+2\beta)\kappa + c_d$ once $M \geq 12L(3\kappa_{\text{final}}/C_1)^{1/d}$, uniformly in K, R, and g. Here $C_1 := \max(1, C_c)$ is the enlarged constant of (1.80). The file quotes Bałaban's order of constants (L first, $\kappa \geq \kappa_0$ before $M \geq M(\kappa)$, p_0 and γ last), so $\kappa_{\text{final}} = \kappa_{\text{final}}(d, \kappa, \beta)$ is fixed before M (route (a)), after C_1 is enlarged (route (b)). It replaces the tree rate of `SOURCES_FX_O1.md` by the volume rate, because the final expansion charges large-field domains per M-cube and the tree rate alone loses R_K^{d-1}. Verifier: `SOURCES_FX_R2_VERIFY.md`, [PROVED+VERIFIED] as a reduction, corrected in its reasoning: the 2κ is the (1.93) decay-recovery charge $(1 + \beta + \beta/(3 \cdot 2^d))\kappa$ per M-cube, not a Mayer-link cost, and the term $\sqrt{d}(1+2\beta)\kappa$ is surplus. Open inputs: none; the two minor inputs, $R_{K+1} \geq 1$ and C_c independent of g, are verified readings (former ledger A20, `SOURCES_BOOKKEEPING.md`), and κ_{final} is the choice C14, not sharp.

5.4 Steps 4–5: N3, (H-hol), and the large-field inputs (ledgers B18, B16, B17)

`SOURCES_HBETA_N3.md`**: N3, the strip bound, ledger B18.** With the source additive in $u = g^{-2}$ ($w_0 = g_0^{-2} - f$, τ in absolute u-units), N3 is equivalent to Lemma H of `SOURCES_HBETA.md` for complex c on $Q = \{|\operatorname{Re} c| \leq 2r,\ |\operatorname{Im} c| \leq \tau\}$, pointwise in x. Lemma N3 gives $|\delta w_k| \leq \frac{3}{2}\tau$ for all

$k \le K$, uniformly in K and ε, given (I), (E), real (H-hist), (H-hol) on relative polydiscs, and (B-hol), with (H2-hist) added in the history form; the constant $\frac{3}{2}$ can be any $1+\eta$. Step 0 bounds $\partial\beta_j/\partial u$ on the thin strip by Bałaban's real C^1 bound at $\operatorname{Re} u$ plus a Cauchy correction of order $g^4|\operatorname{Im} u|$, and a complex Gronwall step with $\sum_i (\operatorname{Re} u_i)^{-3/2} \le 4\gamma/\beta$ closes a bootstrap in $\operatorname{Im} c$. The file also shows that Cauchy estimates on relative discs alone give a factor $1+Cg^2$ per step and a bound that grows with K, so the real derivative bound is needed. Verifier: `SOURCES_HBETA_N3_VERIFY.md`, [VERIFIED REDUCTION], corrected: condition (a), $3r + \frac{3}{2}\tau < g^{-2} - \gamma^{-2}$, limits r by the gap between $g = g_K$ and γ and does not follow from γ small (settled in `SOURCES_N3A.md`); $d\beta/du = O(g^3)$, or $O(g^4)$ for even β; (H2-hist) is sufficient, not necessary, the minimal input being (H-hist$_C$) on the complex polydisc. Open inputs: (I) (ledger B12), (E), real (H-hist) (B13), (H-hol) in its small-field (B16) and large-field (B17) halves, (H2-hist) or its Route 2(a) (B14), (B-hol), the $h \equiv 0$ booking (C11), and the additive normalization (C12); Step 3 in the D″ form is proved, unverified (B10).

`SOURCES_N3A.md`**: condition (a) of Lemma N3.** With the order of constants $L \to$ Bałaban's constants $\to \gamma \to g \to (r_0, \tau_0) \to K$ and $g = g_K \le \min(\gamma/2, g_1)$, condition (a) holds at the uniform radius $r_0 = \tau_0 = \min(1/8, r_0^{\mathrm{D2}}, r_0^{\mathrm{D3}}, r_0^{\mathrm{FX}}, \gamma^{-2}/10, 1/(6\gamma^2 c_1 c_H))$, since $\frac{9}{2} r_0 \le \frac{9}{16} < 3 \le g^{-2} - \gamma^{-2}$ for $\gamma \le 1$, a margin of 5.33. The radius does not depend on K, and in u-units it is the physical sup-norm radius of the source. The file checks that $g \le \gamma/2$ is admissible for every K (`LEMMA_DPRIME.md` Step 8 in the Lemma E form, `KAPPA_BOUNDED.md` §5.3(b) in the D″ form) and that nothing forces g_K near γ. It refutes the proposed radius $r = \tau = \gamma^{-2}/2$, which satisfies (a) but breaks conditions (b)–(d) of Lemma N3 and the radius bounds of D3R and D3W8. Verifier: `SOURCES_N3A_VERIFY.md`, [PROVED+VERIFIED], with item 1 corrected: "u_k decreasing, g_K largest" holds only in the Lemma E form, while in the D″ form only $u_k \ge u_K - D'$ holds and g_0 can exceed g_K; the D″ sketch has two unnamed gaps. Open inputs: none for condition (a); the order of constants is the choice C5; the inputs of N3 are unchanged, and N3 in the D″ form stays a sketch (ledger B10).

`SOURCES_HHOL.md`**: (H-hol), small field, ledger B16.** For the small-field chain, β_j extends holomorphically to the polydisc $|g_i - x_i| < \rho_0 x_i$ with $|\beta_j| \le M$ uniformly in j, k, and volume, where $\rho_0 = \min(0.1, (LM)^{-2}(8n)^{-1/2})$, divided by $\sqrt{2}$ when the X-modes are counted (ledger C6). This holds given Bałaban's conditions with modified constants: (C1′)–(C4′) with $\varepsilon_1 \to 1.235 c\varepsilon_1$, (C4″) $8n\rho_0^2 (LM)^4 \le 1$ (with $16n\rho_0^2$ once the X-modes are counted), and (2.24′); the radius $\rho \le 1/3$ first asked for is not reached, and Lemma N3 accepts any fixed ρ_0. In reference units $\tilde{B} = xB$ the complex coupling enters only through $w = (x/g)^2$, as the scale of the Gaussian and as the coefficient of the u-prefactored vertices, while the cut $|\tilde{B}| < \varepsilon_1$ does not depend on g. The marginal-domination lemma of the file (its Lemma D) bounds the complex Gaussian by the real one at $\operatorname{Re} w$ at a cost $(|w|/\operatorname{Re} w)^{1/2} \le e^{2\rho^2}$ per real mode, so the volume-uniform bound is made per polymer inside CMP 116 (2.14)–(2.17). Verifier: `SOURCES_HHOL_VERIFY.md`, [VERIFIED REDUCTION], corrected: the per-polymer cost misses the X-modes ($\rho_0 \to \rho_0/\sqrt{2}$), $|\beta_j| \le M$ is a reduction because the majorant covers the Gaussian only, and (C1)–(C5) are not Bałaban's printed labels; single-valuedness is confirmed with no extra assumption. Open inputs: the modified conditions (C1′)–(C4′), (C4″), (2.24′) (the X-mode question is resolved in the row M-C of the dependency table); (F1), reduced to Theorem SM at complex couplings, from which (H2-hist) is derived with $a_m = \lambda_C^{m/2}$; a γ-independent C_1 in real (H-hist) (B13); the large-field half (ledger B17, `SOURCES_HHOL_LF.md`).

`SOURCES_HHOL_LF.md`**: (H-hol), large field, ledger B17.** The large-field half of (H-hol) holds on $P_{\rho_0}(x^*)$ around Bałaban's real trajectory x^*, with the R-ratio density, the thresholds, R_k, and N frozen at x^* (ledger C1): the large-field integrals are entire in $u = g^{-2}$ in finite volume and

are dominated in modulus with $\gamma_0 \to \gamma_0'' \geq 0.82\gamma_0$ at $\rho_0 = 0.1$, and F2′ holds on the polydisc with $32\varepsilon' \to 40\varepsilon'$. This is a reduction to (M-A), (M-B), (M-C), and (M-D). The file shows that the radius $(100MR_k)^{-4}$ of W4 is a radius in the input activities, which the real (1.100) denominator does not contain, and it removes the coupling from the denominator by freezing the ratio at x^*, at a mismatch cost of $O(1)$ per unit cube inside Bałaban's budget $p_0(g) \gtrsim R(g)^{d+2}$. F2′ compares the bound sizes $(x_j^*)^{\kappa_0}$, which are ordered on the real trajectory, so no ordering of complex couplings is needed, and a slot-dependent radius $\rho_j \propto x_j^2$ would have failed Lemma N3. Verifier: `SOURCES_HHOL_LF_VERIFY.md`, [VERIFIED REDUCTION], corrected: R_k is unbounded, of order $(\log K)^r$, but largest at $k = 0$ and non-increasing in k (the file reverses the index); $\theta = 0.174$, not 0.26; (M-D) is not a convention, since the frozen map equals Bałaban's map only at x^* and its derivative there is (H-hist) plus (H-thr). Open inputs: (M-A), reduced to (H−) and CMP 102 (189) (ledger A4, with A18 and A19), with (M-A′) closed; (M-C), a verified reduction with open (I-ε), the (I-V0) residual, and (M1); (M-D), which needs real (H-hist) (B13) and (H-thr) (B15) with (H-hist2) (B14); (M-B) is closed, [PROVED+VERIFIED] with $p_0 \geq 17/(\gamma_0 A_1^2)$ and $\theta_W + \tau_C \leq 0.2$.

`SOURCES_MA.md`**: (M-A), the equipartition bound.** For the cut-off ratio density of CMP 122 I (1.100) at real weight $0 < u \leq x^{-2}$, the file proves $\langle A - A_{\min}\rangle_u \leq (n_i/u)(1 + \eta_x)$, where $n_i = (N^2-1)L_i$ counts the real coordinates on the free links and $\eta_x = 0.68M_0^2x^2p_0(x)^2 + 2c_HM_0^2x\,p_0(x)^3$; the same bound with the full A is false in general, because the background minimum $A_{\min}$ is $O(\varepsilon^2)$ per block. As a corollary, the freeze-mismatch average is at most $e^{0.1906\,n_i(1+\eta_x)}$, that is $e^{0.77(N^2-1)(1+\eta_x)}$ per scale-k unit cube ($e^{2.3}$ for SU(2)), additive in the volume. A virial identity on the convex cutoff ball, centred at the true minimizer X^* of the exponent, bounds $u\langle(X-X^*)\cdot\nabla G\rangle_u$, $G = A - A_{\min}$, by n_i plus a Haar-Jacobian term (Lemma J: $0.68N(N-1)r^2$ per link for $r \leq \frac{1}{2}$). A segment bound from the Hessian lower bound (H−) (Lemma S) turns this into the bound on $\langle G\rangle_u$, and integrating $\frac{d}{du}\log D^G = -\langle G\rangle_u$ gives the corollary. Verifier: `SOURCES_MA_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (H−)]; corrections: the proved constant is $C = 1$, not $\frac{1}{2}$ (sharp when X^* lies on the boundary), the SU(2) toy value 1.87 needs a cutoff norm that the file does not state, and the bookkeeping item (M-A′) is a real gap, closed in `SOURCES_MA2.md`. Open inputs: (H−), the Hessian lower bound along segments with c_H uniform in k, volume, and background; `SOURCES_MA2.md` reduces it to CMP 102 (189) (ledger A4) together with A18, A19, and its conditions (C1), (C2); ledger B17.

`SOURCES_MA2.md`**: (M-A′) and (H−).** The bookkeeping item (M-A′) closes: every normalized ratio of CMP 122 I (1.100) sees only $G = F - A_{\min}$, and the exact identity $N_x/D_x = e^{-uF}\,[e^{uA_{\min}}/D^G(x^{-2})]\,e^{(u-x^{-2})G}$ keeps $e^{uA_{\min}}$ at the coupling u, so the mismatch costs at most $e^{1.07(N^2-1)(1+\eta_x)}$ per unit cube. The Hessian bound (H−) of `SOURCES_MA.md` is derived in the form $D^2F[h,h] \geq -c_H\varepsilon_k\sum_b \|h_b\|_2^2$, with c_H uniform in k, volume, and regular backgrounds. Since $A_{\min}$ does not depend on V', $D^A(s) = e^{-sA_{\min}}D^G(s)$, and the error in `SOURCES_HHOL_LF.md` §3.4(c) was a numerator at u over a normalizer at x. For (H−), the chain rule per plaquette is combined with one Cauchy step on $\delta\mathcal{H}/\delta B$ minus its linear part on the complex ball of CMP 102 (172), where the $1/\varepsilon$ of the radius cancels the ε of CMP 102 (187), and the kernel decay of (189) closes a Schur test. Verifier: `SOURCES_MA2_VERIFY.md`, (M-A′) [PROVED+VERIFIED] and (H−) a [VERIFIED REDUCTION] to CMP 102 (189); corrections: the first Hessian term is ≥ 0, $c_N = O(1)(B_0C_4 + C_2)B_5$, c_H depends on M^5 and M_0, and $A_{\min}$ may be only piecewise analytic in the background, so $e^{uA_{\min}}$ must stay paired with an analytic factor or inside modulus bounds. Open inputs: (C1) $M_0\varepsilon_k \leq C_1\varepsilon_1$; (C2) the localized map U_{k,X_i} obeys the CMP 102 bounds; (C3) CMP 102 (189), announced without proof and already used in Bałaban's own chain (ledger A4, now a verified reduction to A1); ledger A18

(including the k-uniformity of C_2, C_4, B_0) and A19; and the reading that Bałaban's chain removes $e^{uA_{\min}}$ by algebra.

`SOURCES_MB.md`: **(M-B), the large-field bookkeeping.** Bałaban's large-field bookkeeping CMP 122 II (1.73)–(1.99) survives the complex-coupling perturbation with thresholds frozen at the real trajectory x^*: the per-cube factors $\exp(-p_0(x_j^*))$ and the per-component factors $\exp(-2p_0(x_j^*))$ keep the same ε_j, δ_j, δ_j', R_j, and the only new condition is $p_0 \geq 17/(\gamma_0 A_1^2)$ with Bałaban's output factor kept (option B). The cost constant becomes $C_c' = C_c + C_m + C_G + C_E$, a fixed $O(1)$ per site inside the $O(1)\log g_j^{-2}$ budget, and the p. 389 condition $p_0 \geq 8\kappa(100R_k)^d + 4$ is unchanged. Each line of (1.73)–(1.99) that carries γ_0, p_0, ε_j, δ_j, A_1, or a per-cube factor is listed (L1–L24) and redone with $\gamma_0 \to g^*\gamma_0$, which is legitimate because each factor bounds a positive Wilson action or quadratic form whose modulus at complex u is dominated with prefactor $\operatorname{Re} u \geq 0.8264\,(x^*)^{-2}$. The combined factor of p. 381 then needs $p_0 \geq c/(\gamma_0 A_1^2)$ with $c = 2/(g^* - \frac{1}{2})$, and the budget constant $C_1' = \max(1, C_c')$ (route (b) of `SOURCES_FX_R2.md`) multiplies (1.82), (1.88) by a fixed factor that p_0 large and γ small absorb. Verifier: `SOURCES_MB_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (M-A) AND (M-C)]; corrections: the perturbation weights are additive, so $g^* = 0.82645 - \theta_W - \tau_C$ (not a product), and the constant 17 needs the margin $\theta_W + \tau_C \leq 0.2$ (strict limit 0.2088; at $\frac{1}{4}$ one needs $c = 26.2$); p. 383 prints $\frac{1}{2}\gamma_0$ and N^{β_0}, and any fixed $g^* > 0$ is accepted there. Open inputs: (M-A) (the freeze-mismatch cost C_m) and (M-C) (the complex-u activities α^c and E-terms); the link from F2′ to α stays an inference; the margin is ledger C2, and the block feeds ledger B17.

`SOURCES_MC.md`: **(M-C), the complex small-field step.** Lemma MC: if Bałaban's constants satisfy the conditions B1–B23 of CMP 116 §2, then with the shrinks $\varepsilon_1' = \varepsilon_1/(4\lambda_V)$, $\gamma' = \gamma/(8\lambda_V)$, $\alpha' = \alpha/4$, $\lambda_V = \max(c, 1.235)$, the radius $\rho_0 = \min\{0.1, (LM)^{-2}(0.45a/(16n))^{1/2}\}$, and the added condition (M1), the small-field step is holomorphic in complex g with $|g/x - 1| < \rho_0$, for every real $x \leq \gamma'$ and inputs in the ball cE_0, and Lemma 3 (2.38) holds with Bałaban's $C_3\varepsilon_1$. Hence β_j is holomorphic on these discs and bounded uniformly in j, k, and volume. The 23 inequalities of CMP 116 §2 are read on the page images and typed, and each one at complex $w = (x/g)^2$ and primed constants is shown to follow from Bałaban's own. The vertex is affine in w with real-domain coefficients, so its factor λ_V is absorbed by the τ-circle (2.18) with $E_0 \to \lambda_V E_0$ and a smaller ε_1; the cut stays real, and the Gaussian cost $16n\rho^2$ per site and the α_5 items fit into the budget $(LM)^4\alpha_i \leq \frac{1}{4}$ under (M1) and (M2). Verifier: `SOURCES_MC_VERIFY.md`, [VERIFIED REDUCTION] to the printed CMP 116 §2 conditions (B1–B23 and four more), Lemma 2, (1.36), and p. 12 (2.5)–(2.6); corrections: the shrinks are $\varepsilon_1/(4\lambda_V)$ and $\gamma/(8\lambda_V)$, not ε_1/λ_V and γ/λ_V, the total output is $\leq E_0$, not $\frac{1}{2}E_0$, and the open item (I-X) is resolved by p. 12 (the X-measure is the unit Gaussian). Open inputs: (I-ε), that the smaller ε_1, α_0, α_1 keep CMP 116 §1 and the Part I induction valid; the (I-V0) residual (the $\log\sigma$ and Jacobian part); and (M1), an "M large" condition; ledger B17.

5.5 Step 6: thresholds and second history derivatives (ledgers B15, B14)

`SOURCES_HTHR.md`: **(H-thr), first attempt.** With the threshold mismatch $\Delta_j(c) = B_j(u(c); \bar{u}) - B_j(u(c); u(c))$ (integer parameters frozen, convention L3), the file bounds $\sup_c |\Delta_j'(c)| \leq 3(1 + 3r)A\,\Phi(g_j^{-2})$ and sums this along the trajectory to $C\beta^{-1}\gamma^{-2(1+a)}(\log\gamma^{-2})^P$ times a decay factor, uniformly in K; in the verifier's corrected form the decay is $\exp(-(\log\gamma^{-2})^q)$ with $q = 2p_1 - (d+5)r_0 > p_0$, in place of $\exp(-\kappa(\log\gamma^{-2})^{2p_0})$. Lemma H′ then splits the budget: $x \leq \ln(6/5)$ for (H-hist) and $x_{\text{thr}} \leq \frac{1}{4}$ give $\sum_j |B_j'| \leq \frac{1}{2}$ and $\frac{1}{2} \leq |v_k| \leq \frac{3}{2}$. Coarea writes Δ_j as the contribution of the shell between the two thresholds, which reduces Δ_j' to threshold derivatives and mixed threshold–

weight derivatives of B_j, bounded in Lemma S under the model assumptions M1–M4. The sum along the trajectory uses $u_i \geq \gamma^{-2} + \beta m$ and an integral comparison that keeps the factor γ^{-2}/β, and Lemma H′ is an additive Gronwall argument. Verifier: `SOURCES_HTHR_VERIFY.md`, [VERIFIED REDUCTION] of (H-thr) to M2′, M4, M5, with the shell representation, the trajectory sum, and Lemma H′ confirmed; corrections to Lemma S: the fluctuation cutoff is $\delta'_j = g_j A_1 p_1(g_j)$ with $p_1 < p_0$, Bałaban bounds one bond only through the block quadratic form CMP 122 II (1.78) with loss $R_j^{-(d+5)}$, so the per-link density M2 is unsupported, and the mixed derivative needs history holomorphy (M5); M1 is resolved (χ stays inside the integral, so $\Delta_j \neq 0$). Open inputs: M2′ (density form of the block bound), M4 (extraction cost), M5 (history holomorphy), M3, and the frozen-integer convention L3 (ledger B15, C1); `SOURCES_HTHR2.md` replaces M2′ by Lemma T and M5 by (H-hist2).

`SOURCES_HTHR2.md`: **(H-thr), Lemma T.** The shell indicator is at most the large-field function $1 - \chi$ at the lower threshold, so Bałaban's printed tail factor of CMP 122 II (1.77)–(1.78) (with the misprint $p_1 \to p_1^2$) bounds Δ_j itself with exponent at least $\kappa_2(\log s)^q$, $\kappa_2 = \frac{1}{2}L^{-(d+5)}$, $q = 2p_1 - (d+5)r_0 > p_0$ (Lemma T). With a uniform bound D on Δ''_j this gives $\sup_c |\Delta'_j| \leq 2(2AD)^{1/2}\Phi_T(u_{j-1})^{1/2} + (2A/r)\Phi_T(u_{j-1})$, a sum over j that is uniform in K under (G1′)–(G3′), and Lemma H′ of `SOURCES_HTHR.md` unchanged. The zero-order bound on Δ_j uses Lemma T, the history weights M3, and the extraction cost M4, a Cauchy estimate on the analyticity radius $\alpha_{1,j} = g_j C_1(\log g_j^{-2})^{q_1}$ of CMP 119 (2.28), so no history derivative enters. The derivative then comes from the interpolation inequality $\|\Delta'_j\| \leq 2(\|\Delta_j\|\,\|\Delta''_j\|)^{1/2} + \|\Delta_j\|/r$ (Lemma 7.1(a) of `SOURCES_D2.md`), which replaces the history holomorphy M5 by a uniform second-derivative bound (H-hist2). Verifier: `SOURCES_HTHR2_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (H-HIST2)] for the summed bound, with Lemma T and the misprint confirmed and M4 confirmed as reduced; correction: (H-hist2) for β_j contains the mixed threshold–history derivative through the running thresholds, so it is a new hypothesis of the (H-hist) kind, not free by a Cauchy estimate in the history. Open inputs: (H-hist2) (ledger B14, a [VERIFIED REDUCTION] to the real uniform (H-hist), ledger B13); M3; M4 (units, and the small-field analogue of CMP 122 II (1.73)); the transfer of Bałaban's printed large-field factor to small-field shell terms; the frozen-integer convention L3; ledger B15.

`SOURCES_HHIST2.md`: **(H-hist2), second history derivatives.** For the frozen-threshold map $F = B_j(\cdot;\bar{u})$ on $|w - \bar{u}| \leq 3r$ in u-slots, $\sum_{i,l} |\partial_{u_i}\partial_{u_l} F| \leq \varepsilon''_j = C_H(1-\lambda^{1/2})^{-2}u_{j-1}^{-2}$, so $\sum_j \varepsilon''_j \leq C_H(1-\lambda^{1/2})^{-2}\gamma^2/\beta$ uniformly in j, K, and volume, with product decay $\lambda^{m/2}$; the first-derivative half is $\varepsilon'_j \leq \frac{1}{2}C_1(1-\lambda)^{-1}u_{j-1}^{-3/2}$, from the real uniform (H-hist). For β_j with running thresholds the (H-hist2) step needs only a uniform bound, but Lemma H′ still needs a summable first derivative of β_j from the real (H-hist). Route 2(a) differentiates the real chain twice: past slots stay real, the input variable is handled by Cauchy estimates (Theorem SM), and only the own slot is complexified, on one relative u-disc (`SOURCES_SM.md` §4.2). The Jacobians $\partial_{u_i}\partial_{u_l} = \frac{1}{4}g_i^3 g_l^3 \partial_{g_i}\partial_{g_l}$ turn the $1/x$ losses into g^4, which is summable because $u_{j-1} \geq \gamma^{-2} + \beta m$. Verifier: `SOURCES_HHIST2_VERIFY.md`, [VERIFIED REDUCTION] to the real uniform (H-hist), no new input; the Jacobian cancellation and the decay $\lambda^{m/2}$ are confirmed, the product form λ^m is [REFUTED] (toy recursion), and $\varepsilon'' \propto u^{-2}$ follows from the u-disc alone; correction: "β_j needs only a uniform bound" holds for the (H-hist2) step only, since Lemma H′ needs a summable β'_j. Open inputs: the real uniform (H-hist) (ledger B13: (Q)/c_-, (M_r), CMP 102 (189), (M-B) bookkeeping); Theorem SM and the joint u-disc and input-ball holomorphy [R]; the large-field part of the frozen map; for β_j, the mixed threshold–history derivative through Lemma D and M3 (ledger B14).

`SOURCES_SM.md`: **Theorem SM, the step-map Lipschitz bound.** The small-field step map factors as $G = \hat{G} \circ \mathcal{L}$ with $\mathcal{L}$ linear, and Theorem SM gives $|DG(I_0)[W]| \le \theta_0 \sum_{l \le k} L^{-\alpha_*(k-l)} \|W^{(l)}\| \, e^{-\kappa' d_{k+1}(X)}$ with $\theta_0 = 2K_H K_V \varepsilon_1$ and $\alpha_* = \min(1, \alpha)$. With it Lemma HH holds at rate $\lambda = L^{-\alpha_*} + C_V \theta_0$ when $C_V \theta_0 < 1 - L^{-\alpha_*}$, and (Eps) of `SOURCES_D3W8.md` becomes a corollary with $\varepsilon' = \theta_0/(1 - L^{-\alpha_*})$. The per-input bound CMP 116 (1.24) gives the linear stage a fluctuation factor $\alpha_2^{-1}\varepsilon_1$ and a label weight $L^{-(k-l)}$ (Fact A). The map $\hat{G}$ is holomorphic on a ball of radius $R_V = \alpha_4 \alpha_6 e^{-5\kappa}/(4(L+2)^4 O(1))$, free of ε_1 and E_0, by CMP 116 (C1)–(C5) (Fact B), and a Cauchy estimate on this ball gives the contraction. Verifier: `SOURCES_SM_VERIFY.md`, [VERIFIED REDUCTION] (items 1–4 confirmed on page images, items 5–6 corrected); the own-slot constant C_β has the same $1/g$ gap as S_0, so (G-gain) must be applied in both the hop and the own-slot bound, and the per-label decay of the lower-order terms stays reduced. Open inputs: Fact A for the lower-order bracket terms (ledger A10); Fact B, that is (W4) on CMP 116 (C1)–(C5) with α_4, α_6 fixed before ε_1; (I)/L3 (ledger B12); (G-gain) in both slots, carried by `SOURCES_GGAIN.md` and `SOURCES_GGAIN_LF.md` (ledger B13).

5.6 Step 7: history bounds and the gain (ledger B13)

`SOURCES_HHIST.md`: **Lemma HH, (H-hist) in column-sum form.** Given CMP 109 (0.29), uniform own-slot bounds $\|\partial_{g_{l-1}} V^{(l)}\|_l \le S_0$ and $|\partial \beta_l / \partial g_{l-1}| \le C_\beta$, and a step-map Lipschitz bound (SM) with $\theta < 1 - L^{-\alpha}$, Lemma HH gives $|\partial \beta_j / \partial g_i| \le (C_\Pi / C_V) S_0 \lambda^{j-1-i}$ for $i \le j - 2$, with $\lambda = L^{-\alpha} + \theta$. The column sum is then at most $C_\beta + (C_\Pi/C_V) S_0 \lambda/(1-\lambda)$, uniformly in i, j, K, and the volume. The past enters step $k+1$ only through the curly bracket of CMP 109 (2.12) and only through the irrelevant remainders $V^{(l)}$ (the marginal parts cancel, p. 298), and a step-l remainder costs $L^{-\alpha(k-l)}$ per π_k cube by (0.29)–(0.30). An induction on $s_l = \|\partial_{g_i} V^{(l)}\|_l$ with one hop per step then gives the rate λ in place of the one-hop rate $L^{-\alpha m}$. Verifier: `SOURCES_HHIST_N1_VERIFY.md` (Part 1), [VERIFIED REDUCTION]: the quotes, the dependence chain, the cube count, and the slot definition are confirmed; the rate is sharp, but the numerical claim "ratio 1.0" is false ($s_{i+1+n} = (\theta/\lambda)\lambda^n S_0$, so the bound is loose by θ/λ), and Bałaban gives no per-hop Lipschitz constant θ. Open inputs: the own-slot bounds S_0 and C_β (C_β is Bałaban's inductive assumption, ledger A5) and (SM), now carried by `SOURCES_GGAIN.md` and `SOURCES_SM.md`; validity of the step bounds off the trajectory, close to (I) (ledger B12); the large-field contributions, now in `SOURCES_GGAIN_LF.md`; the whole chain is ledger B13.

`SOURCES_GGAIN.md`: **(G-gain), small field.** For every step k and $g_k \in]0, \gamma]$ the own-slot derivatives obey $\|\partial_{g_k} V^{(k+1)}\| \le C_V S_0$ and $|\partial \beta_{k+1} / \partial g_k| \le C_\beta := C_\Pi S_0$, with S_0 free of $1/g_k$ and of ε_1 (Theorem B2-unif). With Theorem SM the column sum of Lemma HH is uniform, and in u-slots it is $O(g_i^3)$, which meets the threshold $C g_i^{2+\delta}$ of `SOURCES_HBETA.md` with $\delta = 1$. By CMP 116 (1.34) the vertex $V_k'(Y, \cdot, B)$ is an analytic function of $x = g_k B$ on a polydisc of g-free radius ε_1, so a pointwise Cauchy estimate in x costs $|B|_Y/\varepsilon_1$, the ε_1 cancels against the vertex size, and the one polynomial insertion is absorbed by the Gaussian at strength α_4. The $P^{(k)}$ part is cubic in B' by (1.39)–(1.40), so $|\partial_g| \le C_P |B|^3$ there, and in the hop the factor $g_k |B|$ of (1.22)–(1.24) gives $\theta_0 \to \theta_g g_k$. Verifier: `SOURCES_GGAIN_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON LEMMA G3 AND THE MARGIN (M_r)], for the small-field step on $]0, \gamma]$ including vacuum subtraction (items 1, 2, 4, 5, 6 confirmed, 2 and 6 with corrections, item 3 corrected); (M_r) is also needed for the $P^{(k)}$ part, Lemma G4 fails as stated and is off-chain, the sharp-cut surface constant is $O(\gamma_2^{-(n_b+1)/2} \varepsilon_1^{-1})$, and Bałaban's p. 264 assumption is proved only in its first-derivative part, on $]0, \gamma]$, for the small-field step. Open inputs: Lemma G3, hence (Q)/c_-; the margin (M_r), which shrinks the χ_k threshold

(ledger C9); the large-field half (`SOURCES_GGAIN_LF.md`); the endpoint $g = 0$ (ledger C19); all inside ledger B13.

`SOURCES_GGAIN_LF.md`: **(G-gain), large field.** With the thresholds frozen at Bałaban's trajectory x^*, the own-slot derivatives of the large-field outputs are uniform in $g_k \in]0, \gamma]$ with $|g_k - x_k^*| \le \rho_0 x_k^*/2$, in k, K, and the volume, for example $|\partial_{g_k} R'^{(k)}(X)| \le 20\gamma^{2A_0-1} e^{-\kappa d_k(X)}$ and $|\partial_{g_k} T_k(Y_i)1| \le 20\gamma^{A_0-1} e^{-\frac{1}{2}p_0(x_k^*)}$ (Theorem B2-full). With `SOURCES_GGAIN.md` and Theorem SM this makes the real uniform (H-hist) a reduction in both halves. On the disc $|g - x| < \rho_0 x$, $\rho_0 = 0.1$, of `SOURCES_HHOL_LF.md` the printed bounds hold with $\gamma_0 \to 0.826\gamma_0$, and one Cauchy step gives $|\partial_g F| \le 20 x^{-1} \sup|F|$. The factor $1/x$ is absorbed by the large-field suppression, since $p_0(x) = A_0(\log x^{-2})^{p_0}$ with p_0 a positive integer and A_0 large (CMP 119 p. 244) gives $x^{-1} e^{-\theta p_0(x)} \le x^{2\theta A_0 - 1}$. Verifier: `SOURCES_GGAIN_LF_VERIFY.md`, [VERIFIED REDUCTION] (items 1, 2, 4, 5 confirmed, item 3 corrected); the mismatch average sits at $u' = 2x^{-2} - u \in [0.892, 1.093]x^{-2}$, which gives the constant $2.48 \le 2.6$, the Gaussian normalization adds a factor $e^{0.01n}$ per unit cube inside (M-B), and "β from E-class terms only" is upgraded to a verified reading of CMP 119 p. 279. Open inputs: (Q)/c_-, the margin (M_r), CMP 102 (189) through (M-A) (ledger A4), and the (M-B) bookkeeping (ledger B13); not covered: (1.103) at complex u, C^1 at $g = 0$, higher derivatives.

`SOURCES_G3.md`: **Lemma G3, the χ_k surface term.** Given (Q) with constants c_- and Λ, for every k and $g_k \in]0, \gamma]$ the g_k-derivative acting on the sharp cut $\chi_{k,Y_0}\chi^c_{k,P}$ of CMP 116 (2.3) in a term of (2.14) is at most $c_1' U(g_k) e^{\delta\kappa d_k(Z_0)} T_{\mathrm{maj}}$, with $U(g) = A_n D_n \varepsilon_1^{-1} e^{-\varepsilon_1^2/(8\Lambda g^2)}$ and $n = N^2 - 1$. The cut adds $S_\chi = c_2 A_n D_n \varepsilon_1^{-1} e^{-\varepsilon_1^2/(8\Lambda\gamma^2)}$ to S_0, uniform in g_k, X, k, and the volume, and $O(\gamma^p)$ for every p. Route 2 keeps Bałaban's cut, a round sphere $|B(b)| = \varepsilon_1/g_k$ per bond, and computes the surface term in polar coordinates, where the Gaussian marginal turns $g^{-1}(\varepsilon_1/g)^n$ into $\varepsilon_1^{-1} t^{n+1} e^{-ct^2}$, $t = \varepsilon_1/g$, with no $1/g$ left. The bond and polymer sums use (2.30), and the smooth cut $\hat\chi$ (Route 3, $\varepsilon_1 \to \varepsilon_1/2$, $\gamma \to \gamma/2$) is the fallback. Verifier: `SOURCES_G3_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (Q)] (items 2, 4, 5 confirmed, items 1 and 3 corrected); the surface term is real in the fixed-B computation that `SOURCES_GGAIN.md` uses (finite-difference check to 10 digits at $n = 8$), (Q) must be read on the joint (X, B) majorant with $a = 0$, the bond count is $c_d|Y_0| + |P|$, the printed γ_k fails under both parses, and the B versus B' cut question the verifier added is closed in `SOURCES_Q.md`. Open inputs: (Q)/c_- (row B-S.Q; ledger B13); numerical values of c_- and Λ; higher derivatives.

`SOURCES_Q.md`: **input (Q), the covariance lower bound $c_- > 0$.** The $B(b)$-block of the covariance of Bałaban's joint Gaussian majorant CMP 116 (2.23) is bounded below by $c_- > 0$ and above by $\Lambda \le 2(1 + \gamma_0^{-1})$ for $\alpha_5(1 + \gamma_0^{-1}) \le \frac{1}{2}$, uniformly in k, the volume, and regular U; after the localization step of `SOURCES_LOC.md`, $c_- = (\gamma_1 + \varepsilon_M)^{-1}$, and in the free case $c_- \ge (a\kappa_d L^{d+1})^{-1}$. The cut of CMP 109 (2.9) is $|B(b)| < \varepsilon_1/g_k$ on every bond except b_0, a round sphere with no Jacobian. The precision of (2.23) is $Q = Q_0 - \alpha_5 \operatorname{diag}(Z, Z_0) \le Q_0$, and inversion reverses order, so the $B(b)$-block of Q^{-1} dominates that of Q_0^{-1}, which is $C_0 + C_0\Gamma\Gamma^* C_0 \ge C_0$ with $C_0 = C^{(k)}(Z_0, 0)$. The remaining norm input $\|C^*\Delta_k C\| \le \gamma_1$, which CMP 116 (2.7) itself uses, is traced to CMP 99 (3.132), and the free-case constants come from CMP 95 (1.100) and CMP 96 (2.125). Verifier: `SOURCES_Q_VERIFY.md`, [VERIFIED REDUCTION] to CMP 99 (3.132), (E_U), and LOC (items 1, 3, 4 confirmed, 1 and 3 with corrections, item 2 corrected in favour of the reduction); Q_0 is coupled, so the block identity is only an inequality, the operator is the $\sigma = 0$ localized one (handled in `SOURCES_LOC.md`), CMP 99 p. 422 prints the kernel bound (3.132), and κ_d depends on a and L unless $a = 1$. Open inputs: the A1 residue (torus-side hypothesis, two conventions, reading $a = 1$);

the two [I] bridges $Q(U) = Q_{\text{free}}(1 + O(\alpha_0))$ and $\|C(U)\|^2 \le L^{d+1}(1 + O(\alpha_0))$ (ledger A18); (E_U), reduced to k-uniform kernel moments that Bałaban does not print; the inputs of LOC.

`SOURCES_LOC.md`: **Lemma LOC, the localized covariance $C^{(k)}(Z_0, 0)$.** Given $\gamma_0 \le C^*\Delta_k C \le \gamma_1$ and $C_0^{-1} = Z_0 C^* \Delta_k C Z_0 - R$ with $\|R\| \le \varepsilon_M \le \gamma_0/2$, the localized covariance $C_0 = C^{(k)}(Z_0, 0)$ of CMP 116 (2.23) has $C_0(b, b) \ge c_- I$ on every bond $b \subset Z_0$ and $\|C_0\| \le 2/\gamma_0$. Here $c_- = (\gamma_1 + \varepsilon_M)^{-1}$ with the symmetry (S) of the truncated walk sum and $c_- = (\gamma_0 - \varepsilon_M)/(\gamma_1 + \varepsilon_M)^2$ without it, and $\varepsilon_M = S_d(O(1)e^{-\delta_0 M/3} + O(\alpha_0 + \alpha_1))$ with $S_d \le d \coth(\delta_0/(4\sqrt{d}))^d$. The printed (2.5) is conditioning on Z_0^c, so $C^{(k)}(Z_0)$ is the inverse of the compression of $C^*\Delta_k C$ and inherits its bounds, while $C^{(k)}(Z_0, 0)$ is the walk sum with the walks meeting σ_0 deleted and is not a compression. Its distance to the compression is Bałaban's bound (2.16) applied to R_2 at $\sigma \equiv 1$, turned into the norm bound ε_M by the Schur test. Verifier: `SOURCES_LOC_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON THE R_2 BOUND (2.16), γ_0, AND γ_1] (item 1 confirmed, items 2–4 corrected); the symmetry (S) is not needed, $\varepsilon_M \to 0$ needs $\alpha_0, \alpha_1 \to 0$ as well as $M \to \infty$, γ_0 sits in the prose after CMP 99 (3.158) and is proved there only at $U = 1$, and the replacement point is the real configuration U. Open inputs: the R_2 bound (2.16), which Bałaban states with "similarly" (ledger A2, now a verified reduction to (W^c), A7, and two [I] items); the upper bound γ_1 through CMP 99 (3.132) and (E_U); γ_0 for general U (ledger A3, now a verified full proof, $\gamma_0 \ge m_L/(48d^2L^d)$, with residue R-$\Delta$2); α_0, α_1 small after M is fixed.

5.7 Step 8: Bałaban-level propagator inputs (ledgers A1–A4)

`SOURCES_A1_KERNEL.md`: **ledger A1, kernel-decay form.** Given (K) and (C) below, the file proves CMP 99 (3.132) in its printed two-scale form, $|(QGQ^*)^{-1}(y, y')| \le 2K_{\text{ext}} L^{d/2+1}(L^j\eta)^{-2}(L^{j'}\eta)^{-d} e^{-\delta_1 d(y,y')}$ for $y \in \Lambda_j$, $y' \in \Lambda_{j'}$, uniformly in j, k, η, the volume and regular U, and the same for $(QG_1Q^*)^{-1}$ plus a diagonal a-term. The rate comes from $\varepsilon = \min\{\delta/2, c_-/(2c_+\Sigma(\delta/2))\}$ and is much smaller than Bałaban's rate. Lemma CT, a Combes–Thomas argument on $\hat{N} = S^{-1/2}(QGQ^*)S^{-1/2}$ on $\ell^2(\mathfrak{B}, \mu)$, turns kernel decay (K) of QGQ^* and a lower bound (C) $\hat{N} \ge c_-$ into decay of the inverse, with no Fourier analysis and no use of A3. (K) is taken from CMP 99 Theorems 3.3 and 3.12 for G, (C) from one feasible extension in the variational form $\langle B, N^{-1}B\rangle = \min\{\langle A, G^{-1}A\rangle : QA = B\}$, and the global perturbation from $U = 1$ and the Demko–Moss–Smith route are shown not to work. Verifier: `SOURCES_A1_A4_VERIFY.md` (Part A), [VERIFIED REDUCTION]; it confirmed Lemma CT, the two-scale form, the non-circular order of the G_1 clause, and both refutations, and it found the file's face-midpoint bumps for (C) not feasible ($Q_j(U)$ is supported on both whole blocks, CMP 98 (14), CMP 99 (3.102)), so its own grade was "reduced to (C), open"; the newer `PLAN.md` entry regrades A1 to a verified reduction once (C) is supplied by `SOURCES_QE_INV.md`, `SOURCES_QE_KE_HQ.md`, and `SOURCES_S2_FORM.md`. Open inputs: the residue of ledger A1, namely the torus-side hypothesis (side $\ge c \cdot RM$ coarsest blocks), two unprinted conventions, and the reading $a = 1$ of CMP 99 Theorem 3.1 [I]; the acceptance of the small rate δ_1 by each downstream consumer (S4) is not re-checked (low risk).

`SOURCES_QE_KE_HQ.md`: **(KE′) and (H_Q), inputs of (C).** (KE′): Bałaban's H of CMP 96 Cor. 2.8 already satisfies $Q_{98}(1)H = I$ on all of $\mathfrak{B}$, interfaces and Λ_0 included, with the decay of CMP 96 (2.151) for H and ∇H, because CMP 96 uses the straight-line average CMP 95 (1.11)/(1.18). (H_Q): $|K_j(U; c, b) - q_j(c, b)\,\text{Ad}\,U(\Gamma_{c_- \leftarrow b})| \le c_H \alpha_0 \ell_j^2 \eta_j^d$ on $B^j(c_-) \cup B^j(c_+)$, $c_H = 2C_1' + c_T(d)L$, uniformly in $j \le k$. For (KE′) the file reads the pages: the invariance group $Q_j'\lambda = 0$ of CMP 96 (2.7) fits only the straight-line average, and CMP 98 p. 27 states that the propagators hold for the average (1.11). For (H_Q) it applies Bałaban's induction (144)–(145) to the difference between $Q_j(U)$ and the

product of the main terms, and it compares the averaged background links with fine straight-line transports and fixed axial contours by a cone-area bound. Verifier: `SOURCES_QE_KE_HQ_VERIFY.md`, [PROVED+VERIFIED], (KE′) by citation on the page images (CMP 95 (1.11), (1.12), (1.18); CMP 96 (2.7), (2.20), (2.35); CMP 98 p. 27 verbatim), so the gap of `SOURCES_QE_INV_VERIFY.md` item 2 was a misreading; corrected: the verifier proved the one [R] step (iii) itself (ribbon of $\le (d-1)L^2$ plaquettes, lasso decomposition, error $\le 9.6(d-1)\alpha_0\ell_i^2$ for $\alpha_0 \le \min(c_2, 1/(8(d-1))))$, the cited (112) is the wrong reference and is not needed, and the corridor fallback $H+d\psi$ is exact within layers but not across interfaces (not needed). Open inputs: the unvalued constants $C_1'(d,L)$ and $c_{\mathrm{cone}}(d)$, and two unprinted conventions ($U(c)$ the straight segment; translation-covariant axial contours), both in the residue of ledger A1.

`SOURCES_QE_INV.md`: **Theorem QE, input (C) of Lemma CT.** For the free multiscale minimizer dressed with axial transports, $\mathcal{E}_U$, the file proves, given (KE) and (H_Q), $\|Q(U)\mathcal{E}_U - I\|_S \le c_Q M\alpha_0 + c_{\mathrm{far}} e^{-\delta_H RM/(16L^2)}$ in the S-weighted norm on $\mathfrak{B}$, uniformly in j, k, the volume, $\{\Omega_j\}$, and regular U. Hence $K_Q \le 2$, and $A = \mathcal{E}_U(Q(U)\mathcal{E}_U)^{-1}B$ is an exact extension, $Q(U)A = B$ and $A = 0$ on Λ_0, with $\langle A, G^{-1}A\rangle \le K_{\mathrm{ext}}\|B\|_S^2$, $K_{\mathrm{ext}} = 4(1+O(M\alpha_0))K_{\mathcal{E}} + a$, which gives (C) with $c_- = 1/K_{\mathrm{ext}}$. The defect of $Q(U)\mathcal{E}_U$ is written exactly as a $q|E|$-weighted sum of loop holonomies minus one, and a cone-area lemma bounds each holonomy linearly in the distance. A Schur test with the decay (KE), a near/far split across layers with far terms e^{-cRM}, and a Neumann series then give $K_Q \le 2$, and a crude absolute-value Schur bound gives the energy. Verifier: `SOURCES_QE_INV_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (KE′), (H_Q), S2, AND A TORUS SIDE $\ge c\cdot RM$]; numerics [NUMERICAL, REPRODUCED] (defect linear in the curvature, slope 0.010 at $d=2$ and 0.019 at $d=4$, rigorous Schur constant 41–76 times the true slope); corrected: ε'' is of order $M\alpha_0$, not α_0, $\alpha_F = 2\cdot O(1)M\alpha_0$ with $O(1) \ge 10$ (so the Neumann threshold reads $O(1)M\alpha_0 \lesssim 0.08$), and a torus-size hypothesis must be added. Open inputs: (KE′), (H_Q), and S2 for G have since been closed (`SOURCES_QE_KE_HQ_VERIFY.md`, `SOURCES_S2_FORM_VERIFY.md`); what stays open is the torus-side hypothesis and the two unprinted conventions of ledger A1.

`SOURCES_S2_FORM.md`: **S2, the form bound for G.** For every bond field A, $|\langle A, \Delta_\pi' A\rangle| \le C_\Pi' M\alpha_0(\|DA\|^2 + \|D^*A\|^2 + \|\ell^{-1}A\|^2)$, hence $G^{-1} \le (1+O(M\alpha_0))(D^*D + DD^*) + O(M\alpha_0)\ell^{-2} + Q^*aQ$ and $\langle \mathcal{E}_U X, G^{-1}\mathcal{E}_U X\rangle \le K(d,L,a)\|X\|_S^2$, uniformly in j, η, the volume, $\{\Omega_j\}$, and regular U. The route through CMP 99 (3.131) stalls, because (3.131) is printed for one-block supports and a tiling then forces sharp cut-offs that cost η^{-1} on the sup side. Route Π instead uses the exact identity (3.119), $\langle A, \Delta_\pi A\rangle = \langle \tilde{A}, \Delta\tilde{A}\rangle$ with $\tilde{A} = A - D\lambda$, $\lambda = G'RD^*A$, the curl of a gradient $\eta^{-2}(\mathrm{Ad}\,U(\partial p) - 1)\lambda$, and the block L^2 bounds (3.46) of Theorem 3.1 with CMP 96 Lemma 2.1 and a Schur test, which give $\|\ell^{-2}\lambda\|, \|\ell^{-1}D\lambda\| \le B_0 c_1 \|D^*A\|$. Verifier: `SOURCES_S2_FORM_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (KE′), $a=1$ IN THEOREM 3.1, AND THE BLOCK GEOMETRY], together with §5.1 and §5.3 of `SOURCES_QE_INV.md`; corrected: Δ' of (3.10) has a second term $\eta^d \,\mathrm{tr}(DA)^2(\mathrm{Re}\,U(\partial p) - 1)$, bounded by $O(M\alpha_0)\|DA\|^2$, so $C_\Pi' = C_\Pi + 2c_R(1+\kappa^2 a_0^2)$ and c_- drops only by a factor $1 - O(M\alpha_0)$; (3.120) carries a minus sign; the normalization of Q^*aQ is settled by (3.16) (one constant a). Open inputs: the reading $a=1$ of Theorem 3.1 and the block-geometry item (both in the residue of ledger A1); the G_1 analogue, left at sketch level here, is closed in `SOURCES_S2_G1_V7.md`.

`SOURCES_S2_G1_V7.md`: **S2 for G_1, and the V7 repair of ledger A2.** Part A proves $|\langle B, \Delta^{(2)}B\rangle| \le c_2 M\alpha_0 \|\ell^{-1}B\|^2$, $c_2 = \dim\mathfrak{g}\cdot C_{137}$, and $|\langle A, \Delta_\pi^{(2)}A\rangle| \le 2c_2 M\alpha_0(\|\ell^{-1}A\|^2 + B_0^2 c_1^2\|D^*A\|^2)$, so $\langle \mathcal{E}_U X, G_1^{-1}\mathcal{E}_U X\rangle \le K_1\|X\|_S^2$ with $K_1 = K(1+O(M\alpha_0))$. Part B proves

$|R_2(b,b')| \le (O(1)e^{-\delta_0 M/3} + O(\alpha_0+\alpha_1))e^{-(1-\theta)\delta_A r}$ for $M \ge \max\{M_*, 96\kappa_1/\delta_0\}$ and $\alpha_0+\alpha_1 \le \alpha_*$. In Part A, (3.135) is the sandwich $(I - DG'RD^*)^*\Delta^{(2)}(I - DG'RD^*)$, so the form is taken at the $\tilde{A}$ of (3.119), and the kernel of $\Delta^{(2)}$ is local by (3.137) and symmetric as a Hessian by (3.136), which allows a Schur test with AM–GM. In Part B, $C(\sigma;U_c) = A_Z(U)^{-1} - W'$ with $W' = W_\sigma(U_c) + [C(1;U_c) - C(1;U)]$, the bracket is bounded by a one-variable Cauchy estimate on the complex walk bound, and a Neumann series around the real $A_Z(U)$ lets $A_Z(U)$ cancel in R_2. Verifier: `SOURCES_S2_G1_V7_VERIFY.md`, Part A [PROVED+VERIFIED, CONDITIONAL] on the conditions of `SOURCES_S2_FORM.md`, Part B [VERIFIED REDUCTION]; corrected: symmetry of $\Delta^{(2)}$ is load-bearing for the column sums, the Cauchy route needs (W^c) on the α'-space ($\alpha' \gg \alpha$), not only on the α-space that W_σ needs (already in the ledger, no new item), the two-constant theorem weakens the complex input to a sup bound but does not remove it, touching σ_0 cubes occur generically, not always, and the two covariances do not coincide on a gauge-fixed subspace; the lower bound for G_1 (A3.5) was not checked. Open inputs: for A2, (W^c) on the α'-space, now proved in the exponential gauge without the Thm 3.15 transfer (`SOURCES_WC_EXPGAUGE.md`, `SOURCES_WC_RESIDUE.md`, both verified) relative to the inputs listed under `SOURCES_A2_R2.md`, the real decay A7 [R], the boundary-contact reading of item (i) [I-forced], and $\mathrm{dist}(Z_0, \Lambda_k^c) > RM$ [I]; for Part A, the inherited reading $a = 1$ and the block geometry.

`SOURCES_A1_WALK.md`**: ledger A1w, walk-expansion form of A1.** Theorem A1-W: for regular U, $(QGQ^*)^{-1} = \sum_\omega T_\omega$ with factors localized in $X \times X$, U-dependence on $\tilde{X}_0^5 \cup \cdots \cup \tilde{X}_n^5$, and $|T_\omega(y,y')| \le (4/c_-)L^{d/2+1}(L^j\eta)^{-1-d/2}(L^{j'}\eta)^{-1-d/2}(K_W/M_1)^{|\omega|}e^{-\frac{1}{2}\varepsilon d(\omega,y,y')}$, absolutely convergent; this is the (3.108) form at rate $\delta_W = \frac{1}{2}\varepsilon \ll \frac{1}{2}\delta_0$, and the same holds for $(QG_1Q^*)^{-1}$ with trees. The local operator on a cube is G's walk sum truncated to $\square^+$, its positivity $\frac{1}{2}c_-$ comes by restriction of the global (C) (not A3), and an accretive Lemma CT gives $\|C_\square\|_\varepsilon \le 4/c_-$, as in Bałaban's own local step (CMP 96 p. 237). The parametrix $P = \sum h_\square C_\square h_\square$ satisfies $\hat{N}P = I + \sum F_\square$ with three remainders (separation, truncation, commutator), each $\le K_F/M_1$, and the Neumann series is the walk series. Verifier: `SOURCES_A1_WALK_VERIFY.md`, [VERIFIED REDUCTION] (construction verified); corrected: (L1) must read "kernel vanishes within about M_1 of ∂X", CMP 96 (2.84) has $e^{-\frac{1}{8}\delta_0 M}$ and (2.82) three error sums, the $\breve{G}_2$ citation is p. 432, the RM condition (2.59) at the walk rate joins the constant order, (W_G) has proof sketches printed only for G', $(Q'G'^2Q'^*)^{-1}$ and G_0 (Theorems 3.7, 3.9, 3.10), and (C) is conditional, not proved. Open inputs: (W_G) [R] (itself reduced in `SOURCES_WG.md` to Bałaban's printed sketches, the Neumann series (3.130), and CMP 96 Lemma 2.1, with $\alpha_0 \le c'M^{-11}$ (WG-1)), (C) with the residue of ledger A1, CMP 96 Lemma 2.1, the [I] items (Lipschitz cut-offs O2, block-locality of Q, κ_1 independent of M), and O3 (Z_0-compressed variant) and O4 (complex α'-space), both now closed in `SOURCES_WC_EXPGAUGE.md` and `SOURCES_WC_RESIDUE.md` (verified), O4 relative to CMP 99 Thm 3.4 [V] and the complex walk form of G [R].

`SOURCES_WG.md`**: $(\mathbf{W}_G)$, walk expansions of G and G_1.** The file writes out Bałaban's walk expansions (W_G) of the final propagators G and G_1 of CMP 99 Sect. E: for regular U, $M \ge M_*(d,L)$ and $M\alpha_0 \le a_*(d,L)$, $G = \sum_\omega T_\omega$ is a walk expansion with bounded branching and rate $(1-\theta)\frac{1}{2}\delta_0$, and G_1 has a tree expansion with rate $(1-\theta)\frac{1}{2}\varepsilon$, set by the walk expansion of $(QGQ^*)^{-1}$. The literal (3.108) form, with a factor M^{-1} per step also at the Δ'_π junctions, needs in addition $\alpha_0 \le c'M^{-11}$ (WG-1). The identity $G^{-1} = G_0^{-1} - \Delta'_\pi$ and the Neumann series (3.130) $G = \sum_n G_0(\Delta'_\pi G_0)^n$ are expanded with the printed walk expansions of G_0, G' and $(Q'G'^2Q'^*)^{-1}$ (CMP 99 Thms 3.10, 3.7, 3.9), one local J-insertion of size $O(M\alpha_0)$ per Δ'_π, and a concatenation lemma that splits the exponent at each junction and sums the junction points by CMP 96 Lemma 2.1. For G_1 the same

scheme runs on $G_1 = \sum_n G(\Delta_\pi^{(2)} G)^n$ with $\Delta_\pi^{(2)} = (I - DRG'D^*)\Delta^{(2)}(I - DG'RD^*)$, whose side branches use the walk expansion A1-W of $(QGQ^*)^{-1}$, and the order $G_0 \to G \to (QGQ^*)^{-1} \to H \to G_1$ has no loop. Verifier: `SOURCES_WG_VERIFY.md`, [VERIFIED REDUCTION]: (W_G) for G reduces to the printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10, the Neumann series, CMP 96 Lemma 2.1 and [I] items, and the G_1 clause to the same plus A1-W (Bałaban himself writes "assuming (3.132)" on p. 426); corrections: WG-1 needs $\alpha_0 \le c' M^{-11}$, not M^{-8}, because a Δ'_π insertion has up to $N_V = 9$ walk blocks, and "after adjusting a definition of δ_0" is on p. 427. Open inputs: WG-1 (admissible in Bałaban's constant order, not implied by $M\alpha_0 \le a_0$); A1-W for the G_1 clause, with (C) conditional on the torus-side hypothesis and two conventions; the [I] items (Q-locality, two-sided derivative bounds, $C^{(2)}$-locality, block counts); all under ledger A1w (O3 and O4 are closed, see `SOURCES_A1_WALK.md` above).

`SOURCES_A2_R2.md`: ledger A2, R_2 and R_3 of CMP 116 (2.16). The precision difference $R_2 := C_0^{-1} - \mathrm{Re}\, C^{(k)}(Z_0, \sigma)^{-1}$ obeys $|R_2(b, b')| \le (O(1)e^{-\delta_0 M/3} + O(\alpha_0 + \alpha_1))e^{-\rho|b-b'|}$ for $|\sigma(\Delta)| \le e^{\kappa_1}$, with the printed rate $\rho = \frac{1}{2}\delta_0$ when the decay rate δ_A of $C^*\Delta_k C$ exceeds $\frac{1}{2}\delta_0$, and a smaller rate $\rho < \delta_A$ otherwise, which (2.21) and LOC tolerate. R_3 holds in two weaker forms (R3a), (R3b), sufficient for its only printed use (2.21), and LOC's input (E) follows directly at $\sigma = 0$ with $\varepsilon_M = O(e^{-\delta_0 M/2})$. A Neumann series around the compression $C(1)^{-1} = Z_0 A Z_0$, $A = C^*\Delta_k C$ (case (a), (2.5)), treats $W_\sigma = C(1) - C(\sigma)$, the walks that meet σ_0, as the perturbation: each such walk makes a round trip to distance $> \frac{2}{3}M$ from Z_0, which gives $O(e^{-\delta_0 M/3})$ once the weight $e^{16\kappa_1}$ is absorbed for $M \ge 96\kappa_1/\delta_0$. The $O(\alpha_0 + \alpha_1)$ part comes from a one-variable Cauchy estimate on the analyticity space, and the verifier's repair V7 expands around the real $A_Z(U)$, so that decay of A is needed at real U only. Verifier: `SOURCES_A2_R2_VERIFY.md`, [VERIFIED REDUCTION] (with `SOURCES_S2_G1_V7.md` and its verifier); corrections: the walk bounds (W) are CMP 99 (3.107)–(3.108), not CMP 109; Lemma N and the Cauchy disc needed decay of A at complex U, repaired by V7; the Cauchy radius must respect the plaquette condition; and "not needed: A1" is wrong, since (W) for $C^{(k)}(Z_0)$ contains the unprinted walk expansion of $(QGQ^*)^{-1}$ (A1w). Open inputs: (W^c), the walk bounds on the complex α'-space, is no longer open: it is proved directly in the exponential gauge, real and complex, with no use of the Thm 3.15 transfer (`SOURCES_WC_EXPGAUGE.md`, `SOURCES_WC_RESIDUE.md`, both verified; ledger A2 regraded to a verified reduction); it rests on the G_1 tree expansion (CMP 99 Thm 3.10, printed sketch), the conditional (C) of A1-W, R2 and R-Δ2 of A3, CMP 109 p. 254 [R], CMP 99 Thm 3.4 [V], and [I] items; the decay of $C^*\Delta_k C$ at real U (A7); the boundary-contact reading of "intersect" [I-forced]; $\mathrm{dist}(Z_0, \Lambda_k^c) > RM$ [I]; all under ledger A2.

`SOURCES_A3_GAMMA0.md`: ledger A3, the Kato route. The lattice Kato (diamagnetic) route to $C^*\Delta_k C \ge \gamma_0$ is refuted, and A3 is reduced instead to Bałaban's two-step mechanism at $U = 1$ made gauge covariant, with $\gamma_0 \ge (48d^2)^{-1}L^{-d-1}$ independent of k, volume and M. This is the verifier's sharpened constant; for $L \ge 10$ it carries the factor Lm_L found later in `SOURCES_A3_GAMMA0_FULL.md`. Step (i) bounds the minimal-extension energy below by the unit-lattice curl energy, $\langle B, \Delta_k B\rangle \ge \|\partial_1 B\|^2$ at $U = 1$ by exact lattice Stokes and Cauchy–Schwarz, and step (ii) is CMP 96 Lemma 2.4, a unit-lattice Poincaré inequality in axial gauge. Both steps are made covariant in a local regular gauge on each region, with the gauge-fixing part removed through (3.115) and relative errors $O(M\alpha_0 + L^2\alpha_0)c(d, L)$ for $M \ge cL$. Verifier: `SOURCES_A3_GAMMA0_VERIFY.md`, [VERIFIED REDUCTION] at sketch level, not a proof; corrections: the example $A = d\delta_x$ fails once $\|RD^*A\|^2$ is included and is replaced by an unfixed mode $\lambda \in \mathrm{Ran}(\Delta^{-2}Q'^*)$ (0 against 4.62), so the Kato route stays refuted; $c_* \le 1$, not 4, so the free constant $(12d^2)^{-1}L^{-d-1}$ stands and the bound sharpens from $(192d^2)^{-1}L^{-d-1}$ to $(48d^2)^{-1}L^{-d-1}$; R1 must be CMP 99 Thms 3.1, 3.3, 3.11, 3.12.

Open inputs: R1 with a uniform bound on the oblique projection $\kappa_\pi = \|I - DG'RD^*\|^2$; R2, the k-step averaging comparison (an A18 piece); the small items R3–R5; the σ-localized clause [I]; all under ledger A3, and taken up in `SOURCES_A3_GAMMA0_FULL.md` and `SOURCES_A3_SIGMA.md`.

`SOURCES_A3_GAMMA0_FULL.md`**: ledger A3, full proof.** Theorem A3: for U obeying (3.35)–(3.36) with $M\alpha_0$ small, $\langle \tilde{B}, C^*\Delta_k C\tilde{B}\rangle \geq \gamma_0\|\tilde{B}\|^2$ with $\gamma_0 = m_L/(48d^2L^d)$, $m_L = \min(2(1-\cos(\pi/L)), 1/L)$, uniform in k, volume and regular U; this equals $(48d^2)^{-1}L^{-d-1}$ for $L \leq 9$ and is 0.7555 times it at $L = 13$. The proof step inside CMP 96 (2.126)–(2.127) is false for $L \geq 10$ (a lost factor L), while the statement of Lemma 2.4 is not refuted and stands with the corrected constant. The failure of (2.127) for every $L \geq 10$ is checked in Lean (`SlabCounterexample.lean`, Appendix A). Lemma 5 shows that the G_1-minimizer already satisfies $RD^*A_* = 0$, because $A \mapsto A - DG'RD^*A$ keeps QA and lowers $\langle A, G_1^{-1}A\rangle$ by exactly $\|RD^*A\|^2$, so the gauge-fixing part drops out and no bound on κ_π is needed; Lemma 5 is checked in Lean from (F1), (F3), (F4) as hypotheses (`GaugeMin.lean`). The two steps of `SOURCES_A3_GAMMA0.md` are then written out in full (step (i) with $c_* = 1$ and a covariant relative error $2(1+\delta^{-1})\varepsilon_2^2$, step (ii) as CMP 96 Lemma 2.4 reproved with m_L in place of $1/L$ and made covariant in one regular gauge per corridor), and the assembly gives $\gamma_0 = \gamma_L/4$ with $\gamma_L = m_L/(12d^2L^d)$. Verifier: `SOURCES_A3_GAMMA0_FULL_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON R-Δ2 AND R3–R5], with CMP 99 Thms 3.11–3.12 published and (H_Q) [PROVED+VERIFIED]; the slab claim against CMP 96 was checked first and confirmed (lowest eigenvalue $L^{-d}\min(2(1-\cos(\pi/L)), 1/L)$); corrections: (2.127) fails from $L = 10$ with free block bonds and from $L = 15$ in axial gauge ($d = 2$), not only from $L = 30$; fact (E) needs a factor $\sqrt{N}$; the reason for R-Δ2 must be the form (3.136) plus a block-average bound on $C^{(2)}$. Open inputs: R-Δ2 [R]; the Schur step of R1 from CMP 99 Thm 3.12, not written out; R3, R4, R5, R-Δ' and R2$'$; the σ-localized clause, reduced in `SOURCES_A3_SIGMA.md`; all under ledger A3.

`SOURCES_A3_SIGMA.md`**: ledger A3, σ-localized clause.** For real $s \in [0,1]^{\sigma_0}$ the s-localized covariance $C_s = C^{(k)}(Z_0, s)$ obeys $\|C_s - C_1\| \leq \eta_M = S_d B_W e^{-\frac{1}{3}\delta_0 M}$ and, with no symmetry assumption, $\langle v, C_s^{-1}v\rangle \geq (\gamma_0 - 2\gamma_1^2\eta_M)\|v\|^2 \geq \frac{1}{2}\gamma_0\|v\|^2$ for $\eta_M \leq \gamma_0/(4\gamma_1^2)$, uniformly in s, Z, Z_0, k, volume and real regular U, with no α-condition. The resolvent bound $\|C_1^{-1} - C_s^{-1}\| \leq 2\gamma_1^2\eta_M$ also gives LOC's input (E) in norm at real points without (2.16). The walk sum of CMP 116 p. 3 expands the covariance, $C_s = \sum_\omega s^\omega T_\omega$, so $C_s - C_1 = -\sum_{\omega \in W_\sigma}(1 - s^\omega)T_\omega$ is dominated termwise by one s-free absolute majorant over the walks that meet σ_0, each of which reaches distance $> \frac{2}{3}M$ from Z_0. A Schur bound gives η_M, and the resolvent identity with $\gamma_0 \leq Z_0 A Z_0 \leq \gamma_1$ turns it into the lower bound on the precision. Verifier: `SOURCES_A3_SIGMA_VERIFY.md`, [VERIFIED REDUCTION] to (W) and (N); corrections: (W) is CMP 99 (3.107)–(3.108) with the Cor. 3.8 counting, not CMP 109; the bound without symmetry is the verifier's strengthening of the author's bound $\frac{2}{3}\gamma_0$ under (S_s), so (S_s) is optional; the uniformity of B_W in Z_0 uses part of the exponential decay (constants only). Open inputs: (W), Bałaban's assertion (p. 13) that the [13] expansion of $C^{(k)}(Z_0)$ with the large domain X_0 obeys (3.108) uniformly in Z_0 (A1w enters here [I]); the interior reading of "intersect" for the zero-length walk, forced by (2.16); (N) $\gamma_0 \leq A \leq \gamma_1$, with γ_0 from ledger A3 and γ_1 through A7; complex σ stays with A2.

`SOURCES_A4_D189.md`**: ledger A4, CMP 102 (189).** CMP 102 (189) bounds the two-point Hessian kernel of the nonlinear action part V at fixed A', and the file derives $|V''(A'; b, b')| \leq c_V\varepsilon_3 e^{-\frac{1}{2}(1-\alpha)^5\delta_0 d(b,b')}$, rate $\geq \frac{1}{4}\delta_0$ for $\alpha \leq 0.129$, volume-free and on the complex domain (77); (190) follows through the Neumann series (188) at rate $\frac{1}{8}\delta_0$ ($\alpha \leq 0.159$). Every operator that CMP 102 inverts has the form I plus a small decaying kernel, so a weighted Schur Neumann series (CMP 96

Lemma 2.1) suffices, without Demko–Moss–Smith and without a Hessian lower bound. Lemma D2 shows that the second derivative of the linearizing map solves $(I+\mathfrak{N})\mathfrak{D}_2 = C''[(I-H\mathfrak{D})\cdot,(I-H\mathfrak{D})\cdot]$ with the operator of (68), with two-sided decay at $\frac{1}{2}(1-\alpha)^4\delta_0$, and composition then gives V''. Verifier: `SOURCES_A1_A4_VERIFY.md` (Part B), [VERIFIED REDUCTION] to A1 (kernel-decay form), published CMP 102 statements, A18, and one bookkeeping item; corrected: the rates are relabelled ($\frac{1}{2}(1-\alpha)^4\delta_0$ for $\mathfrak{D}_2$, $\frac{1}{2}(1-\alpha)^5\delta_0$ for V''), the name $\mathfrak{D}_2$ clashes with Bałaban's (first derivative minus linear term), V'' uses (3.132) directly through (87)–(88) and the decay of $\tilde{G}$ depends on A1 through (3.137) and (3.133), and the CMP 102 rates are fractions of the common $\delta_0 \le \delta_1$. Open inputs: A1 (itself a verified reduction), A18, the second-derivative bookkeeping of the V_0 terms (90)–(97), and the ∇, Hölder and Δ versions of (190) [I].

5.8 Step 9: the alternative (E_U) route

`SOURCES_EU.md`: **(E_U), uniform block extension (alternative route).** (E_U) asks for a fine field A with $Q(U)A = B$ and covariant energy $\le K\|B\|^2$ for every block field B, uniformly in k; in the free case $K_0(4) = 103.25, 146.73, 185.64, 195.19$ at $N = 3, 5, 13, \infty$ ($= \kappa_4 - 1$ at $a = 1$), and for regular U the dressed extension $A = \mathcal{E}_U(Q(U)\mathcal{E}_U)^{-1}B$ gives $K \le K_0(1 + c_{2/3}(M\alpha_0)^{2/3} + c_1 M\alpha_0)$. The claim that transport and back gives $Q(U)A = B$ exactly is false, and the linear form $K \le K_0(1 + cM\alpha_0)$ is not proved. K_0 is computed in momentum space by `numerics/k0_d.py` from the CMP 98 (124)–(125) form of $Q(V_0)$, an equal-weight average over L^d straight lines with an $O(L^2\alpha_0)$ remainder, and is checked in real space at $d = 2$. For regular U the free extension kernel is dressed by parallel transport, the constraint error $\|Q(U)\mathcal{E}_U - I\|$ is bounded linearly in the plaquette bound $\alpha_F \le 20M\alpha_0$ by lattice Stokes, and IMS patching over flat frames of radius $R \propto \alpha_F^{-1/3}$ gives the power 2/3. Verifier: `SOURCES_EU_VERIFY.md`, K_0 [NUMERICAL, REPRODUCED], the bound on K [DERIVED SKETCH], and (E_U) [VERIFIED REDUCTION] to kernel-moment bounds Bałaban does not print; corrections: the $N = \infty$ tail is $\propto 1/M^3$, not $1/M$; (139) is on journal p. 39; K_0 belongs to the straight-line Q_{98}, not to the corridor Q_{95}; $K \to K_0$ is indicated, not proved; the k-step remainder constant is $2C_1'$ by CMP 98 (143). Open inputs: A21 (k-uniform moments and decay of the free kernel E_0); the unprinted constant C_1' (A9, a verified reading); the four [I] items (a)–(d) of `SOURCES_EU_VERIFY.md` §5 (uniform moments Γ_j, ν_j, lattice-sum factors, the boundary cutoff at $\partial\Lambda$, the gauge-fixing term); Q_{95} versus Q_{98}; the linear form. The route is an alternative to A1 and is not binding.

`SOURCES_CDC.md`: **the norm bound γ_1 (ledger A7).** The norm bound $\|C^*\Delta_k C\| \le \gamma_1$, uniform in k and regular U, is reduced: by Woodbury $(QG_1Q^*)^{-1} = N_1^{-1} + a$ exactly, so the term $-a\langle B, B\rangle$ of CMP 99 (3.156) cancels and $\gamma_1 = L^{d+1}[(1+\varepsilon)K + \varepsilon a + c_J]$ with $\varepsilon, c_J = O(M\alpha_0)c(d, L)$ and K the constant of (E_U); in the free case $K \le a(\kappa - 1)$, which is 195.19 at $d = 4$, $a = 1$. The brief's mechanism, a local difference operator for which unitary U suffices, is [REFUTED]: the leading term $(QG_1Q^*)^{-1}$ is an inverse whose naive coefficients scale as L^{2k}, so the regularity (3.35)–(3.36) is needed for the upper bound. Comparing (3.155) with (3.156) identifies $\langle B, (QG_1Q^*)^{-1}B\rangle$ as the minimum of $\langle A, G_1^{-1}A\rangle$ over $QA = B$, and the split $G_1^{-1} = M_1 + aQ^*Q$ of (3.128) gives the exact cancellation of a. The remaining minimal energy is bounded by a trial extension of B (the lemma (E_U)), the Δ_π' and J parts by $O(M\alpha_0)$ perturbation, and the free constant by the CMP 95 Fourier formula (1.99)–(1.101). Verifier: `SOURCES_CDC_VERIFY.md`, [VERIFIED REDUCTION]: the Woodbury cancellation is confirmed (the variational form needs only $G_1 > 0$ and Q onto), but (E_U) is not the last input, since CMP 99 (3.132) prints a k-uniform kernel bound on $(QG_1Q^*)^{-1}$ that the file omits; corrections: $\|C\|^2 \le L^{d+1}$, not $1 + L^{d+1}$; K_0 is an upper bound only; the free case is proved for single-scale $U = 1$ only; J is a second derivative of A; the negative-mode branch is dropped,

and "regularity is needed for γ_0" is unsupported. Open inputs: CMP 99 (3.132) (ledger A1), or in its place (E_U) (A21, alternative route); A18 for the J-term and $\|C(U)\|$; the multiscale correction $e^{-\delta RM}$ to the free constant [I]; the s-localized operator $C^{(k)}(Z_0, 0)$ that Lemma G3 uses, handled by LOC (A2, A3). Through this file ledger A7 is reduced to A1 + A18.

6 Numerical and certified results

This section collects the numerical results that support the hypotheses of Section 3. Sections 6.1–6.3 give the one-loop increments and the per-level quantities of the Route K proof. Sections 6.4–6.6 give the values of (H2), the one-step values, and one rigorous point enclosure. Section 6.7 proves (H2) at $L = 13$ (Theorem 6.1, certified). Section 6.8 gives the original plan for a cell certificate, now an independent cross-check at $L = 13$. SU(2), infinite volume, Balaban's normalized-trace units. Except for Sections 6.6 and 6.7, none of this is certified. Only odd L is inside the proofs; $L = 2, 4$ use corner blocks and are outside them.

6.1 First increment $\beta_1(0)$

Sources: `NUMERICS` §6–§7; `UMK_BETA1.md`.

L	$\beta_1(0)$	how	$b = c \log L$	$\beta_1 - b$	status
2	0.630	$N_g = 12$, $p \to 0$ fit	0.2575	+0.372	[NUMERICAL, REPRODUCED] (`KAPPA_BOUNDED_VERIFY` §4: 0.6296)
3	0.664	$N_g = 8$, $p \to 0$ fit (0.6625–0.6661)	0.4082	+0.256	[NUMERICAL, REPRODUCED]; N_g $6 \to 8$ shift ≈ 0.015
4	0.757	$N_g = 6$, single $p = 0.2$; $O((Lp)^2)$ bias ≈ 0.01	0.5150	+0.242	[NUMERICAL]
5	0.749 ($N_g = 12$, $p = 0.1$); fit 0.743	umklapp code	0.5979	+0.145	[NUMERICAL]
7	0.803	umklapp code, $N_g = 8$, $Lp = 0.5$	0.7229	+0.080	[NUMERICAL]
9	0.833 ($Lp = 0.5$); 0.846 ($Lp = 0.9$), 0.851 ($Lp = 1.0$)	umklapp code, Windows, $N_g = 8$	0.8163	$\approx +0.02$	[NUMERICAL]
11	0.850 ($Lp = 0.5$); 0.868 ($Lp = 1.0$)	umklapp code, Windows, $N_g = 8$; $N_g = 8$ grid adequacy at $L = 11$ unchecked	0.8908	≈ -0.04	[NUMERICAL]

The first-step constant $\beta_1 - c \log L$ is +0.37, +0.26, +0.24, +0.15, +0.08, +0.02, −0.04 for $L = 2, 3, 4, 5, 7, 9, 11$. It decreases with L and changes sign between $L = 9$ and $L = 11$. $\beta_1(0)$ itself stays positive at every L computed. The umklapp code was validated against `ym_momentum` to 10^{-12} per K by its author; no independent verifier for the β_1 values at $L \geq 5$. The queue continues to $L = 15$ on Windows. No value at $L > 11$ exists yet.

6.2 Later increments ($L = 2$ only)

quantity	value	status
$\beta_1 + \beta_2$ (composed, $N_g = 4/6/8$)	1.148 / 0.967 / 0.953	[NUMERICAL]; single $p = 0.2$, Lp bias label
per-level $\beta_2(0)$	≈ 0.32 ($0.953 - 0.630$). Per-level recursion 0.317–0.341 ($N = 8$–12, single $p = 0.2$). Symmetric Nyquist: 0.33334 ($N = 8$, $p_1 = 0.2$), 0.31679 ($N = 12$, $p_1 = 0.2$), 0.32283 ($N = 12$, $p_1 = 0.4$) (`BETA_FP_VERIFY` §7; `BETA_FP_PW_VERIFY` §5–§6). Level-1 values carry N-aliasing of about $\pm 5\%$.	[NUMERICAL]; per-level recursion reproduced in `BETA_FP_VERIFY` §7
$\beta_3(0)$	0.40 ($N = 8$), 0.286 ($N = 12$), 0.261 (verifier, symmetric Nyquist, $N = 8$)	unconverged; not usable
abelian sector S_k^{ab}	$(1 - L^{-4k})/4$, matched to all digits	[PROVED+VERIFIED]; [LEAN] counting part (`AbelianCount.lean`)
(U) test	$\lvert\beta_{j+1}(p) - \beta_{j+1}(0)\rvert \le 0.2p^2$ at every computed point ($L = 2$ levels 0–1, $L = 3$ level 0, $p \le 0.8$); uniformity in j not testable with two levels	[NUMERICAL, RUN BY THE VERIFIER] (`BETA_FP_PW_VERIFY` §6)

All computed increments are positive.

6.3 Quantities of the Route K proof

quantity	value	status
$d_1 = b - \beta_1$, $L = 3$	-0.256 ± 0.01 (fits $-0.254 \cdots - 0.258$)	[NUMERICAL, REPRODUCED]
d_1, κ_1, $L = 2$	-0.372, 1.114	[NUMERICAL, REPRODUCED]
d_2, d_3, $L = 2$	≈ -0.44, ≈ -0.45 (d_3 uses unconverged β_3)	[NUMERICAL]; bounded-looking, not a test
$\rho_1 = \max_K \lVert H_j(K)\rVert / L$, $L = 3$, $j = 1, 2, 3, \infty$	1.652, 1.736, 1.757, 1.760	[NUMERICAL, RUN BY THE VERIFIER, `U3_AW_VERIFY` §6]; refutes (HS); $h^2/L^4 \approx 0.34 < 1$; one-step C^1 factor $\rho_1^2/L > 1$ for $j \ge 2$
ρ_1, $L = 5$, $j = 1, 2, \infty$	1.860, 1.9565, 1.963; $\rho_1^2/L = 0.771$	[NUMERICAL]; $j = 1$ and the bound measured by the verifier (`H2_LARGE_L_VERIFY` §6)
ρ_2 ((H2)), $L = 3, 5, 7, 9$	see Section 6.4: 2.0634, 2.108, $\ge$ 2.1131, $\ge$ 2.1142 at $j = \infty$ against thresholds 3, 5, 7, 9	[NUMERICAL, REPRODUCED] at $L = 3, 5$; [NUMERICAL, RUN BY THE VERIFIER] at $L = 7, 9$
a_*, $L = 5$	0.0107 (binding mode near $K = 0$, gauge/ghost sector)	[NUMERICAL, RUN BY THE VERIFIER] (`H2_LARGE_L_VERIFY` §6)

(H″), $n = 2$, $L = 3$	$2.056 < 2.828$	[NUMERICAL, RUN BY THE VERIFIER], coarse grid, one level pair
$\sigma_j^{(1)}(0)$ J-part, $j = 0, 1, 2$ ($d = 2$ proxy)	0.750000, 0.75015, 0.75051 vs 0.75	[NUMERICAL, RUN BY THE VERIFIER] supports Theorem W
$\sigma_j^{(2)}(K)$ scan	not run: no $d = 4$ constant-field per-momentum code	—

6.4 Values of (H2) [NUMERICAL; (H2) AT $L = 13$ IS CERTIFIED IN SECTION 6.7]

Code: `numerics/h2_scan.py` (umklapp Schur pieces, one colour, Gram form; `H2_LARGE_L` §1). Code structure CONFIRMED (`H2_LARGE_L_VERIFY` grade 1): the one-colour reduction is exact at $U = 1$, the H and Δ_{out} blocks and the Gram identity are correct, the separable Prop. A form is exact. Sign slip in `H2_LARGE_L` §4 ($H = +[\mathfrak{B}^{-1}]_{f,c}$); harmless.

L	j	ρ_2 (scan max)	threshold L	status
3	2	2.00857	3	[NUMERICAL, REPRODUCED] (`B2_TWOSTEP_VERIFY` §4; `H2_LARGE_L_VERIFY` §1)
3	3	2.05601	3	[NUMERICAL, REPRODUCED] (spot value 2.05599 at the argmax)
3	∞	2.0634 (2.06332–2.06377 over l_{max} 2–5)	3	[NUMERICAL, REPRODUCED]; margin 31%; $q_1 \leq 0.473$, $q_2 \leq 0.053$
5	2, 3, 4, ∞	2.098, 2.108, 2.108, 2.108	5	[NUMERICAL, REPRODUCED]: independent dense code gives 2.098283 at $j = 2$; the verifier's own search gives 2.107921 at $j = \infty$, same argmax; margin 58%
7	∞	$\geq$ 2.1131 (face diagonal only)	7	[NUMERICAL, RUN BY THE VERIFIER] (spot 2.11309)
9	∞	$\geq$ 2.1142 (one point, $(2.6, 2.6, \pi, \pi)$)	9	[NUMERICAL, RUN BY THE VERIFIER]; the $j = 2$ point of `H2_LARGE_L` gives ≥ 2.1104

- Shape. The argmax lies on the face $K_3 = K_4 = \pi$, on the diagonal $K_1 = K_2$: $\approx 2.22 \to 2.40$ ($L = 3$, $j = 2 \to \infty$), ≈ 2.54 ($L = 5$). Off-face points are lower: best off-face 1.995 ($L = 3$, $j = 2$), 2.004 ($L = 3$, $j = \infty$), 2.088 ($L = 5$, $j = \infty$). The corner (π, π, π, π) stays below the face peak, but the gap shrinks: 0.0070, 0.0028, ≥ 0.0025, ≥ 0.0023 ($L = 3, 5, 7, 9$). The corner is the obvious place for a competing maximum at large L.

- Level dependence. At $L = 5$, $\rho_2(\infty) - \rho_2(j) = 9.64 \cdot 10^{-3}$, $4.0 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ ($j = 2, 3, 4$); ratios 24.1, 25.0, against $L^2 = 25$. At $L = 3$ the ratios are 7.5, 8.7. Levels $j = 4, \ldots, \infty - 1$ are not sampled at $L = 3$.

- L-trend. $j = \infty$: 2.0633, 2.1079, ≥ 2.1131, ≥ 2.1142 ($L = 3, 5, 7, 9$). This is consistent with a bounded ρ_2 near 2.11–2.14. The rate claim "at least as fast as L^{-2}" is CORRECTED

(`H2_LARGE_L_VERIFY` grade 5): it used a lower bound at $L = 7$ as if it were the sup. "(H2) gets easier with L" is a measured trend for $L \leq 9$, not a result for $L > 11$.

- All values are lower estimates of a sup (grid plus local search).

6.5 One-step values

The arithmetic of Section 3.7.4 shows that $\rho_1^2 < L$ implies (H2). The measured one-step values are as follows.

- $L = 3$: $\rho_1^2 = 3.10 > 3$. The one-step route fails (Section 3.7.2); the two-step map is needed.
- $L = 5$: $\rho_1 = 1.860$ ($j = 1$, Wilson fine kernel, dense, measured by the verifier), 1.9565 ($j = 2$), 1.963 ($j = \infty$). So $\rho_1^2 \leq 3.853 < 5$ and $\rho_1^2/L = 0.771$ [NUMERICAL, CONFIRMED AS A MEASUREMENT, UNCERTIFIED; `H2_LARGE_L_VERIFY` GRADE 4].
- $L = 7$: $\rho_1 \geq 2.032$ (diagonal only), $\rho_1^2/L \approx 0.59$ [NUMERICAL].
- $L = 13$: measured $\sup \rho_1^2 = 4.37$; certified bound $\sup_{j,K} \rho_1^2 \leq 10 < 13$ (Theorem 6.1).

So at $L \geq 5$ the one-step map suffices for Theorem C1, as far as measured. For certification this replaces L^8 blocks per K (two-step map) by L^4 blocks per K (625× fewer at $L = 5$).

6.6 A rigorous point enclosure at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED]

Source: `H2_CERTIFY.md` with the prototype `numerics/h2cert_proto.py`; verifier `H2_CERTIFY_VERIFY.md`. At $L = 3$, $j = \infty$, and the face point $K^* = (2.4035, 2.4035, \pi, \pi)$ near the argmax of Section 6.4, ball arithmetic (python-flint) gives

$$\rho_2(\infty, K^*) \in [1.9993,\ 2.1273] < 3.$$

The width comes from the umklapp tail and scales as N^{-4} in the truncation N. The verifier reran the code at $N = 12$ and obtained [1.5475, 2.5792], which contains the floating-point value 2.063361 and confirms the N^{-4} scaling; at $N = 6$ the code stops with an assertion instead of returning a wrong interval. The verdict is CONFIRMED given the premises P1–P3 of `H2_CERTIFY_VERIFY`; P1 (the loop-family coefficients of the (0.4) row are exact rationals $k/648$, k even) is proved there, and the fidelity of the loop family to Bałaban's paper stays inside P3. Further results of the same two files:

- One-step factorization: $\rho_2(j, K) \leq \rho_1(j, K) \cdot \max_l \rho_1(j-1, k_l)$, exact [PROVED+VERIFIED]. So two-step work is needed only where the right side is $\geq L$, near the face diagonal.
- Oishi–Rump bound $\|\mathfrak{B}^{-1}\| \in [42.7,\ 43.6]$ at K^* (bounds CONFIRMED). The values 177 and 195 of Section 3.9 use the full Δ instead of $\frac{1}{4}\Delta$; this factor reconciles them. No $\|\mathfrak{B}^{-1}\|$ enters ρ_2.
- The umklapp tail enclosure is CONFIRMED for $M = \infty$. For finite M it is CORRECTED: its lower bound must be 0, with class-wise factors $(\pi/2)^{12}$ for two or more large components.
- A uniform Lipschitz grid is refuted as a route: $\Lambda_K \approx 2.8 \cdot 10^4$ (CONFIRMED) would need about 10^{13} cells. The replacement is a per-cell invertibility test with a mean-value enclosure.

- Cost: `H2_CERTIFY` estimates about $1.5 \cdot 10^4$ core-hours for a full certification at $L = 3$. The verifier grades this estimate UNSUPPORTED: at the tested sizes the arb overhead dominates, and the box-sum speedup is overstated.

This is one point at one level. It does not certify the supremum over the Brillouin zone or over j. Section 6.7 does this at $L = 13$.

6.7 (H2) at $L = 13$ [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]

Sources: `H1_ANALYTIC.md` §0–§8 (statement, lemmas, majorant); verifier `H1_ANALYTIC_VERIFY.md` V0–V9 with `numerics/h1analytic/bb_arb.py`; operator check `H1_ANALYTIC_VERIFY2.md` with `numerics/h1analytic/indep_H.py`; symmetry check `H2_CERTIFY_P5_VERIFY.md`. Here $\rho_1(j, K) = \|H_j(K)\|/L$ with $H_j(K)$ the one-step gauge-fixed minimizer map of Section 3.7.

Theorem 6.1 ((H2) at $L = 13$)**.** *For $L = 13$, every level $j \geq 1$ including $j = \infty$, and every coarse momentum $K \in \mathrm{BZ}$,*

$$\rho_1(j, K)^2 \leq 10 < 13.$$

Hence $\rho_2 \leq \sup_{j,K} \rho_1(j, K)^2 \leq 10 < L$ by the factorization $\rho_2 \leq \rho_1(j) \max_l \rho_1(j-1, k_l)$ (Lean `rho2_le`, Section 3.7.4). That is, (H2) holds at $L = 13$.

Proof sketch. Write $s = L^{-2}$, $k_l = (K + 2\pi l)/L$, $M = L^{j-1}$, d^M for the Prop. A symbol at level M, $P_\perp$ for the projection onto $\hat{K}^\perp$, and T, T_2, σ, $Z = (P_\perp T P_\perp)^+$ as in `H1_ANALYTIC` §1.

1. Exact split. $\rho_1(j, K)^2 L^2 = \lambda_{\max}\big(ZT_2Z + cc^\dagger/(\sigma|\hat{K}|^4)\big)$, $c^\dagger = \hat{K}^\dagger(1 - TZ)$: a transverse part and an orthogonal pure-gauge part.

2. Three exact lemmas. Lemma S: the averaging symbol is $E_q(k_l) = E_l^0 - \hat{K} g_l^\dagger$, with E_l^0 diagonal (block average times straight path), $|E_{l,\mu\mu}^0|$ built from $a_{l,\rho} = \sin(K_\rho/2)/(L \sin(k_{l,\rho}/2))$, and $\sum_l \prod_\rho a_{l,\rho}^2 = 1$ (Fejér). Tower identity: $d_\mu^{ML}(K) = L^{-2} \sum_l w_{l,\mu} d_\mu^M(k_l)$, hence $P_\perp T P_\perp = P_\perp \operatorname{diag}(d^{ML}(K)) P_\perp$ and Z is the next-level kernel of Prop. A. Lemma G: a closed form of the gauge row g_l from the path average, so the averaging operator is explicit.

3. Majorant. Lemma T (transverse): $\lambda_{\max}(ZT_2Z)/L^2 \leq \gamma_L F(K)$, $F(K) = 1/(|K|^2 \underline{d}(K))$, $\gamma_L = \operatorname{sinc}(\pi/2L)^{-2}$, from $T_2 \leq d_{\max} T$, $ZTZ = Z$, $d_{\max} \leq L^2 \gamma_L/|K|^2$, and $\|Z\| \leq 1/\underline{d}$. Lemma Gauge: the gauge part is $\leq (1+\tau)^2/(|\hat{K}|^2 S_L)$ with explicit sums V, U and $\tau = (V + |\hat{K}|U)/\underline{d}$. So $\rho_1(j, K)^2 \leq B_L(K) := \gamma_L F(K) + (1+\tau)^2/(|\hat{K}|^2 S_L)$. By Cauchy–Schwarz $F \leq F_{CS}$, a closed form made of one-dimensional factors in each K_ρ.

4. Uniformity in j. B_L does not depend on j. For every $M \in \{1, L, L^2, \ldots, \infty\}$ the fine symbol satisfies $d^M \leq 1/|\tilde{q}_0^M|^2 \leq 1/|\hat{k}|^2$, because the Fejér weights sum to 1. The coarse symbol d^{ML} dominates the truncated $M = \infty$ sum termwise. No monotonicity in j and no O5 rate is used.

5. Branch-and-bound. B_L is invariant under the 384 signed permutations of K by construction, so the sup over the sorted chamber $0 \leq K_1 \leq \cdots \leq K_4 \leq \pi$ is the sup over BZ. The program `numerics/h1analytic/bb_arb.py` encloses $\gamma_L F_{CS} + \bar{G}_L$ on boxes $K = \pi u$ with exact dyadic u, evaluates every one-dimensional value in arb (python-flint), and rounds outward by one ulp after every floating-point operation, with Higham bounds on every sum. Result: $\sup_{\text{chamber}} B_{13} \leq 9.9995277$, 25,169 boxes, 8,075 gauge evaluations, 0 failed leaves, 58 s. Boxes that contain $K = 0$ are enclosed directly; $\rho_1(j, 0) = 1$.

Margins. The measured $\sup \rho_1^2$ is 4.37. The majorant is tight to 2.6% at $|K| \approx 1.5$ ($\min B_{13}/\rho_1^2 = 1.026$ over 896 cases and a local search). The certificate refines boxes only until each is ≤ 10, so its margin to 10 is $5 \cdot 10^{-5}$; its margin to the threshold 13 is 23%.

Verification. `H1_ANALYTIC_VERIFY` rederives the split, Lemmas S, G, the tower identity, Lemma T, and Lemma Gauge on paper and checks them against the production objects to $\leq 5.3 \cdot 10^{-13}$ (V1–V2); it checks $B_{13} \geq \rho_1^2$ at 896 (j, K) cases (V3–V4); it grades the original float program CORRECTED (no directed rounding) and replaces it by `bb_arb.py` (V5–V8). `H1_ANALYTIC_VERIFY2` rebuilds $H_j(K)$ from the definitions (position-space (0.4) paths, sparse KKT, ghost-domain gauge) and reproduces the production ρ_1 to $5.8 \cdot 10^{-9}$ (10^{-14} with the same d symbol); the two parts of the split agree to $3 \cdot 10^{-8}$, and the Lemma S conventions are derived by hand and confirmed to $1.2 \cdot 10^{-13}$. `H2_CERTIFY_P5_VERIFY` proves and checks the invariance of ρ_1 under the 384 signed permutations (P5a, P5b); the analytic certificate does not need it, the box sweep does. Lean: `lean/YmLean/Tower.lean` formalizes the algebraic core of the majorant (tower identity in weighted form, Rayleigh and rank-one bounds, `majorant_core`, `lemmaT_core`); `BlockNorm.rho2_le` is the one-step $\Rightarrow$ two-step factorization.

Residue. The theorem is certified given the following, which no second source has checked.

- The reading of Bałaban's (0.4) (the path family that defines the averaging operator) and the gauge choice (`NUMERICS` §5). The certificate is a statement about H_j so defined; both implementations use the same reading.
- Prop. A at $M \geq 27$ and $M = \infty$: checked only by a second implementation of the same formula (independent recursion at $M = 3, 9$).
- Lemma G and the bound chain: proved on paper and rederived by the verifier, checked numerically at points only. The Fourier/umklapp derivation, Lemmas S and G, and the identification of the split are not in Lean.
- `bb_arb.py`: audited line by line and checked against the float program on 30 boxes; not formally verified. It trusts python-flint arb and IEEE correctly rounded $+, -, \times, \div, \sqrt{}$.
- Independent cross-check: the box sweep of the cell certificate (Section 6.8) is running; its result is not in yet.

6.8 Certification plan and j_0 [DESIGNED; SUPERSEDED AT $L = 13$ BY SECTION 6.7]

At $L = 13$ the analytic route of Section 6.7 replaces this plan: it needs no tail level j_0, no $\|\mathfrak{B}^{-1}\|$ bound, and no Lipschitz constant. The cell certificate below is run at $L = 13$ only as an independent cross-check (box sweep, in progress). `B2_TWOSTEP_VERIFY` grade 6 (CORRECTED in size): certifying (H2) is a computer-assisted proof project, not a check. `H2_LARGE_L` §4 gives the design; `H2_LARGE_L_VERIFY` grades 6–8 check it. Three parts.

(a) Tail $j \geq j_0$. From the resolvent identity, $\rho_2(j) \leq R_\infty + (\rho_1^2/a_*)(\delta_{j-2} + \delta_{j-1})$, where $R_\infty = \rho_2$ at $j = \infty$, $a_* = \inf_K a_j(K)$ with $\|[\mathfrak{B}^{-1}]_{f,f}\| = 1/a_j(K)$ exactly, and $\delta_m = \sup_K \|\Delta_{L^m} - \Delta_\infty\|$ (O5 decay) [derivation CONFIRMED]. Inputs CORRECTED by the verifier: the 2-norm sup of δ_m is at the corner, $\delta = 179$, 92.0, 18.7, 2.39, 0.317 ($M = 1, 3, 9, 27, 81$), 45–90× the assumed value at small m; $a_*(L = 5) = 0.0107$, not 0.05. With these inputs, $j_0 \approx 8$ ($L = 3$) and ≈ 6 ($L = 5$). With the O6 coercivity constant, $j_0 = 25$, which cannot be used in practice (`B2_TWOSTEP_VERIFY` §5 estimated $j \gtrsim 45$ with $\|\mathfrak{B}^{-1}\| \leq b(L) \sim 10^{19}$).

(b) Levels $2 \le j < j_0$ and $j = \infty$. A certified sup over the BZ of $\|H_{(2),j}(K)\|$ (at $L \ge 5$: of $\|H_j(K)\|$, Section 3.7.4). Plan: interval arithmetic with local mean-value enclosures on boxes in the sorted chamber (1/384 of the BZ), 4×4 and 5×5 verified inverses, one 4×4 λ_{max} per K; about $2.5 \cdot 10^3$ boxes at half-width 0.05 at $L = 3$; the analytic form of Δ near $K = 0$, where $\rho_2(0) = 1$. At $j = \infty$ also a certified tail bound for the Prop. A sum (l_{max} truncation; observed effect $4 \cdot 10^{-4}$). There is no explicit global Lipschitz constant today: the A0/O5 strip width is existential, and the constant that can be made explicit is $\sim 10^{40}$. Monotonicity in j is not proved and is not used.

(c) Inputs to certify first: $\|\mathfrak{B}_j(K)^{-1}\| \le B_{num}$ for all j, K (the O6 constant $b(L)$ is unusable), the O5 constant C_0 explicitly (or certified δ_m), certified a_*, ρ_1, R_∞.

The run plan (`H2_LARGE_L_VERIFY` grade 8) uses atomic JSON writes and pre-copied profile caches; the profile build time at $L \ge 11$ is not budgeted. At $L = 13$ the cell certificate runs as the box sweep (`numerics/h1cert_cell2.py`, sorted chamber, valid by P5), an independent cross-check of Theorem 6.1; its result is not in yet.

7 What remains

This section lists every open item of the paper in one place. The items are grouped by the kind of work they need: certification, fixed-level and structural inputs of Theorem 2w, items of the sources block, and verification. The item labels H, 1–13, and B1–B8 are those of the dependency table (Section 4.6). For Theorem 2w ⇒ Theorem 1 the list covers §10 of `THEOREM2_PROOF_v4.md`. The trust ledger (Section 7.7, `TRUST_LEDGER.md`) supersedes this list as the complete record of what the two main theorems take on trust: Bałaban-level statements, external results, our open items, and conventions.

7.1 Certification

Single uniform hypothesis (both routes):

H. (H2) $\rho_2 = \sup_{j \ge 2} \sup_{K \text{ real}} \|(\oplus_l H_{j-1}(k_l)) H_j(K)\|/L^2 < L$. [DONE AT $L = 13$: CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)], Theorem 6.1: $\rho_1(j,K)^2 \le 10 < 13$ for all $j \ge 1$ (including ∞) and all K, hence $\rho_2 \le 10 < 13$ by `rho2_le`. Route K: (H2) ⇒ ($C1\sigma$) ⇒ (B2) [PROVED+VERIFIED, `B2_TWOSTEP_VERIFY` GRADE 3]. Route B: (H2) + (T1) ⇒ (U) [PROVED, UNVERIFIED, `BETA_FP_U` THEOREM 2.4]. Cross-check in progress: the box sweep of the cell certificate at $L = 13$ (Section 6.8). At $L = 3, 5, 9$ (H2) stays [NUMERICAL].

H′. Residue of Theorem 6.1 (Section 6.7): the reading of Bałaban's (0.4) and the gauge choice (single source); Prop. A at $M \ge 27$ and $M = \infty$ (second implementation of the same formula only); Lemma G and the bound chain (point checks only beyond the paper proof); `bb_arb.py` audited, not formally verified.

At $L = 3$ one point is enclosed: $j = \infty$, $K^* = (2.4035, 2.4035, \pi, \pi)$, $\rho_2 \in [1.9993, 2.1273]$ [INTERVAL ENCLOSURE, VERIFIED] (Section 6.6). Certified early steps $\beta_1(0), \ldots, \beta_{j_*}(0) > 0$ at odd $L > 11$ are needed only for literal (0.31) at every K (Section 3.9.2).

7.2 Fixed-level and structural inputs of Theorem 2w

Route K hypothesis set (all must hold):

1. (B2) $\sup_j \sup_K |\sigma_j^{(2)}| < \infty$ (or (B2w)). Reduced to H [VERIFIED REDUCTION]. Older reductions kept: ($C1\sigma$) + (h) [PROVED+VERIFIED]; (H″) [ARGUED]. (HS) is refuted.

2. (PI) probe independence, at each fixed level [ARGUED].

3. (F) lattice = continuum log coefficient at constant field [CITED SOURCE CHECKED (SAVVIDY [22], LEDGER X5); LATTICE TRANSFER STANDARD, NOT PROVED HERE].

4. (M_j) formal = true f^2 coefficient at each fixed level: (S1) $\mathcal{K}_t(f) \geq -C_1|f|$, (S2) density of states $\leq C_2|f|$ near 0, (S3) relative bound of the level difference, all on the lattice: [VERIFIED REDUCTION] (`SOURCES_T2_S123.md` + VERIFY, 25 confirmed, 2 corrected): (S1), (S3), $(S2)|_{t=0}$, (S2c) are proved at fixed j given the Peierls input (I3), and (M_j) reduces to (I3), the open 2D gap (G'_A), and the region-1 step of Prop. 4.4 [ARGUED].

5. V1 order-g^0 completeness of CMP 109 (1.3)–(1.5): $\beta_{j+1}(0)$ is the Maxwell coefficient of the per-step Gaussian determinant [OPEN; SUPPORTED AT ORDER p^2].

6. (G2) per-cell infinite-volume densities [FORMAL].

Route B hypothesis set (alternative to 1–5; per `BETA_FP_U` §4; all [PROVED, UNVERIFIED] as implications, no verifier file for `BETA_FP_U`):

B1. (T1) $\sup_j \sup_{q \text{ real}} |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of Theorem A3 [ARGUED; THE LANDAU-SYMMETRIC GAUGE CONVERSION AND STEPS 2–5 OF A3 FOR (2.1) ARE NOT WRITTEN].

B2′. (H2) = item H.

B3. (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$, the gauge-fixing difference at the plane wave [HYPOTHESIS, ARGUED]. With the exact split $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ (`BETA_FP_U` Prop. 3.1) there is no Y-defect; all non-stationarity sits in $\mathfrak{d}_F$. This replaces the earlier $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ items for this route (Section 3.6.2).

B4. Lüscher–Weisz (5.7) [CITED].

B5. A6 (the label of v4; not ledger A6): the plane-wave increment equals Balaban's β up to the expansion point, uniformly in k [ARGUED]. A uniform-in-k input, of the same kind as (DF).

B6. (H0) = item 7′ (Lemma D″ (i)).

B7. $\theta_* < 1$, so that $c_2 \ln(1/\theta_*^2) \geq 0$ in C_B [TRIVIAL CONDITION ON THE CHOICE OF θ_*].

Route B also needs 6 ((G2)), item 7′, and the items 8 and 10–13 (item 9 is closed). It does not need (PI), (M_j), V1; it needs (DF) and A6 instead.

Common to both routes:

7′. The own-slot bound $|\beta_{k+1}(h, g) - \beta_{k+1}(0)| \leq C_\beta g$ on $]0, \gamma]^k \times [0, \gamma]$, for every past h, monotone or not; the lift [PROVED+VERIFIED] (`SOURCES_GGAIN_NONMONO_VERIFY`), 7′ itself conditional on (H0) or on the (G-gain) inputs. Item 7 asked for (H0), Bałaban's smoothness in g with bounds uniform in j and history (CMP 109 inductive assumption at odd $L > 11$; "uniform in j" is our reading of p. 264, not printed, `R2_VERIFY` row 12; uniformity in the history is not printed; ledger A5). The shooting evaluates β on clamped, non-monotone pasts, and `SOURCES_T2_KUNIF_VERIFY` found that (G-gain) proves C_β only on monotone pasts. `SOURCES_GGAIN_NONMONO.md` lifts the restriction with no change of constant: the only use of monotonicity is a comparison of bound sizes in Lemma F2′, and the running maximum $G_k = \max_{i \leq k} g_i$ closes the induction for every past. So 7′ follows from (H0) or from (G-gain) with this lift. The proof uses the Lipschitz form on $]0, \gamma]$ only.

8. L5 history dependence [OPEN]. (a) The clamp proof of Lemmas D′/D″ is valid only if $\beta_{k+1}(g_0, \dots, g_k)$ is defined and continuous on the whole cube $]0, \gamma]^{k+1}$; Balaban does not supply this. (b) Balaban's β_{k+1} exists only on histories where his induction has run; that domain must carry (c′) + (S), not only $g_i \in]0, \gamma]$. Clamped histories can violate the growth side of (c′). (c) True trajectories with all $u_i \geq \gamma^{-2}$ inherit (c′) from (a″). A nested-interval variant of Lemma D″ on these trajectories would avoid the gap [NOT WRITTEN]; the variant is written only for Lemma D (`LEMMAS_DE_VERIFY` §1.5). Lean covers one-argument β only (Lemmas D′, D″, D, E).

9. K-independence of $\beta_j(g)$ [CLOSED AS A REDUCTION; VERIFIED] (`SOURCES_T2_KUNIF.md` + VERIFY). Item 9 is a volume statement, not a history identity: K enters β_j only through the torus size. CMP 119 (2.24), p. 249, and p. 279 define β_j as the whole-lattice coefficient, so the β_j of the recursion is K-free by definition. Lemmas D′/D″ run at one fixed K and never compare trajectories. The window bound needs only (K1) (history-free at zero own coupling), (K2) C_β (item 7′), and (K3) the window bound (5.1) of Theorem K. The residual volume question (the finite-volume step equals the whole-lattice one, ledger A15, B8) is an input of Theorem 1, not of the shooting.

11. Odd L. All one-loop proofs need odd $L \geq 3$ (centered blocks: U3 Lemma 1.2, O5/Prop. A, evenness, Lemma 4.1 of LEMMA43). Balaban assumes L odd > 11 throughout CMP 109, so (H0) and $\beta_j(g)$ at $g > 0$ exist only there. Theorem 2w is stated for odd $L > 11$, or with (H0) at this L as a hypothesis. Numerics: $\beta_1(0)$ up to $L = 11$ (Section 6.1); (H2) at $L = 3, 5$ (scans), 7 (face diagonal), 9 (one point); ρ_1 at $L = 3, \dots, 13$. Only $L = 13$ is inside Bałaban's range; there (H2) is certified (Theorem 6.1).

12. SU(2) only. The charged/neutral split of Lemma 5.1 and (W) are SU(2) statements; $c = 11N^2/(12\pi^2)$ is general but the chain is not checked for $N > 2$.

7.3 Items of the sources block

These items stand between the trajectory of Section 3 and the bound (H-U) of Section 4.2. Tags and files are those of the dependency table.

- (B-S) The sourced version of Bałaban's Theorem 1, (E14-a, b, c), is [OPEN] as a whole.

- (H-hist) via (G-gain). The small-field step is done: pointwise Cauchy and uniform C_β on $]0, \gamma]$ [PROVED+VERIFIED]. Lemma G3 (the χ_k surface term) holds given (Q); (Q) is a verified reduction to CMP 99 (3.132), (E_U), and LOC. The large-field half is a [VERIFIED REDUCTION] (`SOURCES_GGAIN_LF.md` + VERIFY): one Cauchy step on the large-field disc, with the factor $1/g$ absorbed by the exponent of p_0 (CMP 119 p. 244). The real uniform (H-hist) is therefore reduced [R] in both halves; open inputs: (Q)/c_-, the margin (M_r), CMP 102 (189), and the (M-B) bookkeeping. Lemma G4 is off-chain (`SOURCES_G4.md`). Second derivatives: (H-hist2).

- Bałaban-level core (round 3). The propagator and covariance inputs A1–A4 of the ledger reduce to published theorems:

  - A1, kernel-decay form: [VERIFIED REDUCTION] via Lemma CT to the printed CMP 99 Theorems 3.1, 3.3, and the G part of 3.12, with (KE′) proved by citation (CMP 95 (1.11)/(1.18), CMP 96 (2.35)), (H_Q) proved, and S2 closed for G. Residue: a stated torus-side hypothesis (side $\geq c \cdot RM$ coarsest blocks), two unprinted conventions, and

the reading $a = 1$ of Theorem 3.1 ([I]); S2 for G_1 is closed (`SOURCES_S2_G1_V7.md` + VERIFY).

- A1w, walk-expansion form of A1 (CMP 99 Theorem 3.15, "analyzed in the same way"): [VERIFIED REDUCTION] (construction verified, `SOURCES_A1_WALK.md` + VERIFY): local Combes–Thomas and Bałaban's parametrix (CMP 96 (2.82)) give a walk expansion of $(QGQ^*)^{-1}$ with (3.108)-form bounds at rate $\frac{1}{2}\varepsilon$, given (W_G), Bałaban's walk expansions of the final G and G_1 (proof sketches printed only for G', $(Q'G'^2Q'^*)^{-1}$, G_0), input (C), and CMP 96 Lemma 2.1; open: the Z_0-compressed variant (O3) and the complex α'-space (O4). (W_G) is itself reduced (`SOURCES_WG.md` + VERIFY) to Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10, the Neumann series (3.130), and CMP 96 Lemma 2.1, with the constant order $\alpha_0 \le c'M^{-11}$ (WG-1).
- A2 (R_2, R_3 of CMP 116 (2.16)): [VERIFIED REDUCTION] (`SOURCES_A2_R2.md`, `SOURCES_S2_G1_V7.md` + VERIFY): R_2 by a Neumann series around the compression with the complex-U repair V7, to the complex walk bounds (W^c) on the α'-space in the exponential-gauge version of CMP 99 Sect. E (the transfer from the axial gauge is Bałaban's assertion, Thm 3.15; [R]; it contains A1w), the decay of $C^*\Delta_k C$ at real U (A7), the boundary-contact reading (item (i), [I-forced]), and $\mathrm{dist}(Z_0, \Lambda_k^c) > RM$ [I]; R_3 in weaker forms sufficient for (2.21).
- A3 ($C^*\Delta_k C \ge \gamma_0$): full proof, verified (`SOURCES_A3_GAMMA0_FULL.md` + VERIFY), from the printed CMP 99 Theorems 3.11–3.12, (H_Q) [PROVED+VERIFIED], R-Δ2 [R], and items R3–R5: $\gamma_0 \ge m_L/(48d^2L^d)$ with $m_L = \min(2(1-\cos(\pi/L)), 1/L)$, uniform in k, volume, and regular U, for $M \ge L/2$ (this equals $(48d^2)^{-1}L^{-d-1}$ for $L \le 9$ and is 0.7555 times it at $L = 13$); the σ-localized clause reduces to (W) + (N), without symmetry. The proof step inside CMP 96 (2.126)–(2.127) is false for $L \ge 10$ (a lost factor L); the statement of Lemma 2.4 stands with the corrected constant.
- A4 (CMP 102 (189)): [VERIFIED REDUCTION] to A1 + published CMP 102 statements + A18 + one bookkeeping item.

So the Bałaban-level core now rests on published propagator theorems, (W_G) (through A1w), (W^c) and A7 (through A2), A18, R-Δ2, two [I] items of A2, the two conventions, and the torus-side hypothesis.

- (H-hist2), on which the threshold-shell bound (H-thr) depends: [VERIFIED REDUCTION] to the real uniform (H-hist), no new input; with it the M3 history weights and the frozen-integer convention (L3).

- (FX). O1(b), O2, and R2 are [PROVED+VERIFIED]; O3 is [PROVED+VERIFIED, CONDITIONAL ON N3]; O4 is avoided by Bałaban's main form (1.98). N3, the HBETA strip bound $|\delta w| \le \frac{3}{2}\tau$, is reduced to (H-hol) + real (H-hist) + (H2-hist) (condition (a) settled); (H-hol) is a verified reduction to (M-A) (via (H−) and CMP 102 (189)), (M-B) (verified), and (M-C). (M-C) is a [VERIFIED REDUCTION] (`SOURCES_MC.md` + VERIFY) to the printed conditions of CMP 116 §2 (B1–B23 and four more), Lemma 2, (1.36), and p. 12 (2.5)–(2.6), with shrinks $\varepsilon_1/(4\lambda_V)$, $\gamma/(8\lambda_V)$; open: (I-ε), the residual of (I-V0), and "M large" (M1). The two minor inputs of R2 ($R_{K+1} \ge L$, C_c independent of g) are verified readings (`SOURCES_BOOKKEEPING.md`). (FX2) is conditional on O3, so also on N3.

- (B-S.D2C, B-S.Hβ) The domain hypothesis (D), and the conditions (I) and (H-hist) of the two-loop bound.

- (B-S.D2, B-S.D3) The recovery of the decay loss $\kappa \to \kappa/2$ [VERIFIED REDUCTION] by $L/2$-rescaling, $\kappa \geq 1536$ at $d = 4$ (ledger B22; it uses the unprinted large-field T-step rate loss, ledger A23). The scale condition (U) ($\rho_1 \leq c_* r_0/M^a$ at the unit scale) is a constant choice (ledger C23), verified compatible with (B-N1) (`SOURCES_UN1.md`), and does not appear in Theorem B.

- (B-N1) The bridge from the source form to (H-U) derived given (S) (`ASSEMBLY_N1.md`, confirmed); the uniformity of (S) in support size and in ε is a [VERIFIED REDUCTION] (`SOURCES_B21_S.md` + VERIFY): every factor of $C^n(n!)^{1+\eta}$ is independent of h, given Lemma C, S1 at all scales, (FX) in the main form, and the coarsest-scale (FX2) radius; open: the (E14-c) clause. Given (FX), the step ($\mathrm{S}_{\mathrm{src}}$) $\Rightarrow$ (H-U) with $s = 6$, $D > 1$ is confirmed.

- (NT) = (E14-f), for nontriviality (Theorem 4.3); at every layer for part (ii). The constant $c > 0$ must be computed in Bałaban's normalization.

- E13, full-sequence convergence: not needed for Theorem B; [OPEN].

7.4 Verification debt

10. Balaban readings: the sweep (Section 3.4.3) is grep plus page images, not line-by-line, with 4 of 19 rows checked by a second reader; "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) not located; "β_0 can be chosen arbitrarily small" is on CMP 119 p. 255, and β_0 is fixed before γ, as used (verified reading, `SOURCES_BOOKKEEPING.md`; former ledger A12); the readings of `BALABAN_EXCERPTS_CHECK` item 11 (II (4.37)/(5.1), CMP 119 (2.42), (2.12)/(2.23), the "L odd" page) are verified, with corrections, in `SOURCES_BOOKKEEPING.md` (former ledger A22); the CMP 119 Sect. 3 residue is the unprinted T-step rate loss (ledger A23); Theorem 1 itself rests on Balaban's published proof, which uses the Group A items of Section 7.7.

B8. Verification: `BETA_FP_U` has no verifier file. `BETA_FP_PW_VERIFY` §6 items 1–4 grade the logic of Theorem B [PROVED+VERIFIED, CONDITIONAL] (tag decision, `PLAN.md`); the inputs (U), (DF), (T1) stay [PROVED, UNVERIFIED].

13. Lean formalization of the history form of Lemmas D′/D″ [OPEN]. Lemma D″ in the one-argument form is done (Section 3.4.2).

- Results tagged [PROVED, UNVERIFIED] need a verifier pass; Appendix B lists them. They include the sweep, the step Theorem 2w $\Rightarrow$ Theorem 1, the `BETA_FP_U` reductions, `BETA_FP_M` Prop. 4.3, the Prop. 3.6 assembly, and H2_CERTIFY_L13 (its cell plan is superseded at $L = 13$ by Theorem 6.1; the box sweep runs as a cross-check). H2_CERTIFY_P4 has partial support: identity (D) is proved in Lean (`P4_divergence`), Lemma S is rederived in `H1_ANALYTIC_VERIFY.md` V1, and the rebuilt H_j of `H1_ANALYTIC_VERIFY2.md` confirms it numerically; the Fourier step has no independent verifier.

- The paper is not verified as one document. Its statements carry the tags of the project files.

7.5 Open for stronger statements

Not needed for Theorem 1, open for the stronger statements: (A) in full and Lemma 5.1 for $m = 2$ (convergence, rate); C^2, C^3 of $\sigma^{(1)}$ under (H2) (not written); certified $\beta_1(0), \ldots, \beta_{j_*}(0) > 0$ at odd $L > 11$ (literal (0.31) at every K); explicit j_*; (N).

7.6 Why the mass gap is not on this route

Write β_W for the coefficient of $\sum_p(1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p)$ in the Wilson action. Bałaban's action is $g^{-2}\sum_p(1 - \operatorname{Re}\operatorname{tr} U_p)$ with the normalized trace (CMP 109 (0.2)), so $\beta_W = 1/g_B^2$. Shen, Zhu, and Zhu [24] prove a mass gap for SU(N) lattice Yang–Mills with action $N\beta \operatorname{Re}\sum_p \operatorname{Tr} Q_p$ (their (1.2)) when $|\beta| < 1/(16(d-1))$ (their (1.3)); with $\beta_W = N^2\beta$, $N = 2$, and $d = 4$, this is $\beta_W < 1/12$. Bałaban's construction requires $-2(1+\beta_0)^{-1}p_0(g_k) + 1 + 2\kappa(100R_k)^d \le -\frac{3}{2}p_0(g_k)$ (CMP 122 II p. 389). With $d = 4$ and $R_K \ge 1$, this gives $p_0(g_K) > 4\kappa \cdot 10^8 + 2$ for every $\beta_0 > 0$ (twice this at Bałaban's $\beta_0 = 1/7$). The unit-scale threshold $g_K p_0(g_K)$ must be below 2, the largest value of $\|U - 1\|$ on SU(2); with $\kappa \ge 1$ this gives $\beta_W = 1/g_K^2 > p_0(g_K)^2/4 > 4 \times 10^{16}$ at the unit scale. The two regimes are therefore separated by at least seventeen orders of magnitude in β_W, and no choice of L narrows this, because L does not enter the condition. A proof of a mass gap needs a mechanism outside the renormalization-group construction, and Bałaban's papers do not address it.

7.7 Trust ledger

The file `TRUST_LEDGER.md`, corrected by `TRUST_LEDGER_VERIFY.md` (Steps 1–28), lists every statement that the two main theorems use and that this project does not prove with a verifier's confirmation. The three tables below condense it to one row per item. Group A holds Bałaban-level statements and external results (X), Group B holds our own open items, and Group C holds conventions and choices. The IDs are those of the ledger. Ledger B1–B22 are not the Route B items B1–B7 or item B8 of this section, and ledger A6 is not the item "A6" of Route B (that label comes from v4 §4.2a). Sources are cited by paper and page; "[I]" marks an inference step of ours that no verifier proved.

Group A: Bałaban-level statements and external results

ID	statement	source	status
A1	Kernel bounds on $(QGQ^*)^{-1}$, $(QG_1Q^*)^{-1}$, uniform in k and regular U	CMP 99 p. 422 (3.132)	asserted, "not repeated"; kernel-decay form a verified reduction via Lemma CT to CMP 99 Thms 3.1, 3.3, 3.12 + (KE′) + S2; residue: torus-side hypothesis, two conventions, the reading $a = 1$ (G_1 clause closed)
A1w	Walk-expansion form of A1 (builds the walk expansion of $C^{(k)}(Z_0)$, bounds (3.107)–(3.108))	CMP 99 Thm 3.15, p. 416	asserted, "analyzed in the same way"; verified reduction (construction) to (W_G) (reduced to printed sketches, WG + VERIFY) + (C) + CMP 96 Lemma 2.1; open O3, O4
A2	R_2 and R_3 obey (2.16)	CMP 116 p. 16	asserted, "similarly"; verified reduction (V7 repair) to (W^c) in the exponential gauge [R] + A7 at real U + item (i) [I-forced] + RM distance [I]; R_3 in weaker forms

Group A: Bałaban-level statements and external results

ID	statement	source	status
A3	$C^*\Delta_k C \geq \gamma_0 > 0$, uniform in k and U	CMP 99 p. 428	asserted, "we can prove"; full proof, verified, from CMP 99 Thms 3.11–3.12 + (H_Q) + R-Δ2 [R]; $\gamma_0 \geq m_L/(48d^2L^d)$; σ-localized clause to (W) + (N); CMP 96 (2.126)–(2.127) proof step false for $L \geq 10$
A4	Kernel decay (189) of $\delta^2 V/\delta A'^2$ and its consequence (190)	CMP 102 p. 308, Prop. 9	announced; verified reduction to A1 + published CMP 102 + A18 + one bookkeeping item
A5	(H0): $E^{(j)}$, β_{j+1} smooth in g with bounded derivatives	CMP 109 pp. 263–264	inductive assumption; "uniform in j" is our reading, uniformity in the history not printed; Theorem 2w uses only item 7′ (Lipschitz form on $]0,\gamma]$, every past), which also follows from (G-gain) with the running-maximum lift [verified (`SOURCES_GGAIN_NONMONO_VERIFY`)]
A6	$E^{(j)}$, β_j analytic in the couplings	CMP 109 pp. 264, 266	asserted; not used
A7	$\|C^*\Delta_k C\| \leq \gamma_1$ and uniform decay of $C^*\Delta_k C$	CMP 116 p. 13 (2.7); CMP 99 p. 428	premise; reduced to A1 + A18
A8	Positive Gaussian measure for the real pair $(U, 0)$; complex case "perturbative"	CMP 116 p. 15	asserted
A10	Lower-order bracket terms "bounded similarly as in (1.24)"	CMP 116 pp. 8–9	asserted, "similarly"
A11	"Similar inequalities hold for other constants"	CMP 119 p. 256	asserted; the later inequalities not located
A13	Theorem 1	CMP 122 II p. 355; CMP 116, 119, 122 I–II	proved, conditional on $g_k \in]0,\gamma]$; uses A1–A4, A1w, A7, A8, A10, A14, A15, A17, A18
A14	Induction closure recovers E_0	CMP 109 p. 280	asserted, constant unspecified; inside A13 only
A15	Small-field terms and β_j are whole-lattice quantities, independent of Λ_{k+1}	CMP 119 p. 278, (2.24), p. 249; CMP 109 (4.35)–(4.37)	asserted, small-field part only; R-term part is B8
A16	Propagators may be replaced by operators with the same bounds	CMP 116 p. 3	asserted for (1.3)–(1.4); our use for C_f is an extension [I]
A17	$\operatorname{supp}\chi_k$ lies in the analyticity domain of (1.34)	used in CMP 116 Lemma 1	implicit; inside A13; C9 changes it
A18	Backgrounds lie in the regular class of CMP 99/102; C-maps agree	used in CMP 109, 116, 122 I–II	implicit; pieces unsupported on the read pages
A19	(190) at complex B	CMP 102 Prop. 9 (silent)	implicit
A21	Uniform-in-k moments of the free extension kernel E_0	not printed	implicit; numerical at $N = 3, 5$; alternative route only

Group A: Bałaban-level statements and external results

ID	statement	source	status
A23	Large-field T-step rate loss	CMP 119 Sect. 3 (defers to CMP 109 Sects. 6–7, CMP 116 (2.33)–(2.41))	not printed; use consistent; used by B22
X1	Lüscher–Weisz (5.7) one-loop coefficient	Lüscher–Weisz [16]	cited; Route B only
X2	OS II growth condition and reconstruction from torus data	OS II [20]	cited
X3	Reflection positivity of the Wilson action, link planes	Osterwalder–Seiler [21]	cited, text not fetched
X4	Convergence criterion of the final cluster expansion	Cammarota, CMP 85 (1982)	cited
X5	(F): continuum log coefficient at constant field	Savvidy [22]	cited, source checked

Group B: our own open items

ID	statement	source	status
B1	(H2) $\rho_2 < L$	none	certified at $L = 13$ (Theorem 6.1); residue: reading of (0.4), gauge choice
B2	(PI) probe independence	none	argued
B3	(F) lattice coefficient = continuum coefficient	X5	source checked; lattice transfer not proved
B4	(M_j) formal = true f^2 coefficient, (S1)–(S3)	none	reduced to (I3), (G'_A), and Prop. 4.4 region 1 (argued); (S1), (S3) proved at fixed j given (I3)
B5	V1: $\beta_{j+1}(0)$ is the Maxwell coefficient	CMP 109 (1.3)–(1.5)	open; supported at order p^2
B6	(G2) per-cell infinite-volume densities	→ A15	formal
B7	L5: β_{k+1} on the clamped cube, or nested-interval Lemma D″	CMP 109 p. 298	open
B8	K-independence of $\beta_j(g)$, R-term part: finite-volume step = whole-lattice step	→ A15	open; input of Theorem 1, not of the shooting (item 9 closed)
B9	Route B inputs (T1), (DF), v4 "A6", `BETA_FP_U` Theorem 2.4	none	argued; Route B only
B10	Proved, unverified results in the chains (sweep, `BETA_FP_M` Prop. 4.3, Prop. 3.6, N3 in D″ form)	→ A11	proved, unverified
B11	(B-S): sourced Theorem 1, (E14-a, b, c)	→ A13	open as a whole; parts B12–B20, B22
B12	(I): $B_j(c) = \beta_j(u(c))$	none	by construction (C1, C10); not written as a lemma
B13	Real uniform (H-hist)	→ A1–A3, A1w, A7, A8, A10, A17, A18	verified reduction, both halves (large field: `SOURCES_GGAIN_LF`); open inputs (Q)/c_-, (M_r), CMP 102 (189), (M-B) bookkeeping; G4 off-chain
B14	(H-hist2)	none	verified reduction
B15	(H-thr) threshold mismatch	CMP 122 II p. 383	reduced to M2′, M4, M5
B16	(H-hol), small field	conditions (C1)–(C5)	verified reduction

Group B: our own open items

ID	statement	source	status
B17	(H-hol), large field	→ A4, A18, A19	verified reduction; (M-C) reduced to printed CMP 116 §2 conditions + Lemma 2; open (I-ε), (I-V0) residual, (M1)
B18	N3 strip bound	none	proved+verified given B12–B17
B19	(FX) final Mayer expansion	not in CMP 116–122	reduced; remaining input B18
B20	Sourced step bookkeeping: (W4), (P8), F2′, (D), Λ_j	→ A4, A10, A16	reduced; Λ_j bookkeeping open
B21	(B-N1) bridge and (S)	none	(S) clauses: verified reduction, h-independent; (E14-c) clause open; (S_{src}) ⇒ (H-U) proved+verified; compatible with (U) (C23)
B22	Recovery of $\kappa \to \kappa/2$ in the sourced step (only; (U) moved to C23)	none	verified reduction [R] by $L/2$-rescaling, $\kappa \geq 1536$; uses A23

Group C: conventions and choices

ID	choice	source	status
C1	Freeze thresholds and partitions at Bałaban's trajectory x^*	`SOURCES_D3`, `SOURCES_HTHR`	free; derivative needs B15
C2	Option B margin, $\theta_W + \tau_C \leq 0.2$	`SOURCES_MB_VERIFY`	free; costs a smaller γ
C3	L odd, $L > 11$; target $L = 13$	v4 §1.1	loses generality
C4	$G = SU(2)$, $d = 4$	Theorem 2	loses generality
C5	Order of constants	`SOURCES_N3A`	free; settled
C6	Holomorphy radius ρ_0	`SOURCES_HHOL`	free
C7	Cut on B, not B'	CMP 109 (2.9), (2.12)	reading; closed
C8	Sharp χ_k; smooth fallback	`SOURCES_G3`	free
C9	Shrunk χ_k threshold	`SOURCES_GGAIN_VERIFY`	changes the construction; A13 must hold for it
C10	Gauge-fixing weight $1/u_k(c)$	`SOURCES_D2C_VERIFY`	changes the construction; A13 must hold for it
C11	$h \equiv 0$ booking; freezing set	`SOURCES_D2C`	free
C12	Additive source normalization	`SOURCES_HBETA_N3`	free
C13	Main R-step form (1.90)–(1.98)	`SOURCES_FX_O23_VERIFY`	free; removes O4
C14	κ_{final} fixed before M	`SOURCES_FX_R2`	free; not sharp
C15	$2p_1 - (d+5)r_0 > p_0$	CMP 122 II p. 383	condition (Bałaban's own)
C16	$32\varepsilon' \leq 1$, $S \geq 1$, $\kappa_0 > 2$	`SOURCES_D3W8_VERIFY`	changes the construction; A13 must hold for it
C17	(S): $\gamma^{-2} \geq c_{sw} D'$	v4 §1.2	condition; needs β_0 fixed before γ (verified reading)
C18	ℓ even; thermal state; plaquette observable; test space D_n; origin at a site; compactness form	`ASSEMBLY` §0–§1	loses generality
C19	Coupling domain $]0, \gamma]$	`SOURCES_GGAIN_VERIFY`	free
C20	Route K primary, Route B alternative	v4 §1.3	free

Group C: conventions and choices

ID	choice	source	status
C21	$\theta_* < 1$ in Route B	`BETA_FP_PW_VERIFY`	free; trivial
C22	$g_0(K, g)$: any shooting solution	Lemma D	loses generality
C23	(U): $\rho_1 \le c_* r_0 / M^a$ at $\lambda_K = 1$	`SOURCES_UN1` (+ VERIFY)	constant choice, not a hypothesis on test functions; verified compatible with B-N1; C^n cost per n-point bound

Verified readings. Four former Group A rows are now verified readings and appear in no set (`SOURCES_BOOKKEEPING.md`): A9 (the constant $C_1'(d, L)$ is unprinted and not derivable from the printed lines; its value is not used), A12 (β_0 "arbitrarily small" is on CMP 119 p. 255; β_0 is fixed before γ), A20 ($R_{K+1} \ge L$ follows from CMP 119 (2.5); C_c is free of g), and A22 (Π (4.37)/(5.1), CMP 119 (2.42) corrected, (2.12)/(2.23), the "L odd" page; the Sect. 3 residue is A23).

Binding sets. *Theorem 2w* (Theorem A(i), Route K) rests on A5, A15, B1–B7, and X5. Item 9 is closed, so B8 is no longer an input of the shooting. Item 7′ comes from A5 or from (G-gain) with the running-maximum lift (the lift [PROVED+VERIFIED]); in the second case the (G-gain) inputs of B13 enter Theorem 2w. Route B replaces B2–B5 by B9 and X5 by X1.

Theorem 2w ⇒ *Theorem 1* (Theorem A(ii)) rests on 4 Group A items (A5, A11, A13, A15), 9 Group B items (B1–B8, B10), and X5. Inside the proof of Theorem 1 (A13), Bałaban also uses A1–A4, A1w, A7, A8, A10, A14, and the items A15, A17, A18, which he uses without a separate proof.

Theorem B rests on 17 Group A items (A1–A5, A1w, A7, A8, A10, A11, A13, A15–A19, A23), 4 externals (X2–X5), and 21 Group B items (B1–B8, B10–B22). C9 does not remove A17, because A13 uses A17. Beyond A13, Theorem B adds only A16, A19, A23 to the Bałaban-level trust. Of the 17, A1–A4 are verified reductions to published theorems of CMP 99 and CMP 102; their joint residue is (W_G) under A1w (O3, O4 open), the complex walk bound (W^c) in the exponential gauge, A7 at real U, A18, R-Δ2, two [I] items of A2, two unprinted conventions, and a stated torus-side hypothesis; the G_1 clause is closed.

Counts: Group A has 20 items (A1w added; A9, A12, A20, A22 moved to verified readings; A23 added in the previous pass). 14 are printed by Bałaban, and 13 of these are printed without a proof (11 of the 17 behind Theorem B, 3 of the 4 behind Theorem A); 4 of the 13 (A1–A4) now carry verified reductions. There are 5 externals, 22 Group B items, and 23 Group C items. In Group B, B13, B17, B21 ((S) clauses), and B22 are now verified reductions. Of Group C, C3, C4, C18, C22 lose generality, C23 ((U)) is a constant choice, and C9, C10, C16 change Bałaban's construction.

A Lean formalization

The shooting lemmas of Section 3.4, the arithmetic of Theorem K and Theorem C1, and the bridge lemmas of the certifier are formalized in Lean 4 with Mathlib (25 files in `lean/YmLean/`); seven files formalize lemmas of the sources block (Schur test, block-valued Schur test, Combes–Thomas, interpolation and rescaling, resolvent and weighted Neumann bounds, the gauge condition of the constrained minimizer, the slab counterexample to CMP 96 (2.127)). The inventory is as follows.

[LEAN] (no sorry; axioms propext, Classical.choice, Quot.sound)

- Lemma D′ (a)–(d) (`YmLean.lemma_DPrime`), Step 8 (`lemma_DPrime_step8`), Lemma D, Lemma E (Cg^2 form), Lemma D+E (`lean/YmLean/LemmaDPrime.lean`, `LemmaDE.lean`). One-argument β only.

- Lemma D″ (`YmLean.lemma_DDoublePrime`, `lean/YmLean/LemmaDDoublePrime.lean`, 146 lines): (a″) bracket $(3b/4)(K-k) - 2R \le T(k,K) \le (5b/4)(K-k) + 2R$; (b) shooting for $g \in]0, g_1]$ with the clamp inactive; (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, for every trajectory with couplings in $]0, \gamma]$; (d) for $n - m \ge 8R/b$, lower constant $b/2$. Hypotheses weaker than the note (continuity, any admissible γ, $R \ge 0$ derived). One-argument β only; L5 not formalized.

- Full inventory (`lean/README.md`; file, main theorems, source note):

  1. `LemmaDPrime`: `lemma_DPrime`; `LEMMA_DPRIME.md` §1–§2.
  2. `LemmaDE`: `lemma_D`, `lemma_E`, `lemma_DE`, `lemma_DPrime_step8`; `LEMMAS.md`.
  3. `LemmaH`: `gronwall_telescope`, `lemma_H`; `SOURCES_HBETA.md` Lemma H.
  4. `DetIdentities`: `det_bordered`, `det_restrict_add_gauge`; `numerics/NUMERICS.md` §1.
  5. `Telescoping`: `telescoping_det`, `telescoping_log`; `R2_TELESCOPING.md` Prop. 3.1.
  6. `AbelianCount`: `su2_count`, `abelian_share`; `U1_INFINITE_LATTICE.md` Prop. 9.1 (used here only as a cross-check, Section 6.2).
  7. `LemmaDDoublePrime`: `lemma_DDoublePrime`; `KAPPA_BOUNDED.md` §5.3.
  8. `TheoremK`: `telescoping_bound`, `window_of_partial` (the window bound (5.1) from partial sums); `KAPPA_BOUNDED.md` §5.
  9. `Gronwall`: `discrete_gronwall`, `contraction_bound` ($N_{j+1} \le qN_j + Z$); `KAPPA_BOUNDED.md` Prop. 4.1.
  10. `CubeCount`: `cube_count_sharp`, `cube_count_lower` (Lemma O1-R′ arithmetic); `SOURCES_FX_O1.md`.
  11. `BlockNorm`: `rho2_le`, the exact factorization $\rho_2 \le \rho_1(j) \max_l \rho_1(j-1, k_l)$; `H2_CERTIFY.md`.
  12. `OishiRump`: `oishi_rump`, `oishi_rump_eta`, `frobenius_power_bound`; `H2_CERTIFY.md` §5.
  13. `Envelope`: `envelope_nontrivial`, `envelope_not_scale_invariant` ((B-NT) in log form); `ASSEMBLY.md` §2.9.
  14. `Divergence`: `profile_divergence`, `P4_divergence` (identity (D) of premise P4); `H2_CERTIFY_P4.md`.
  15. `TheoremC1`: `c1_two_step`, `c1_kappa_bound`, `c1_window`; `B2_TWOSTEP.md` §2.
  16. `StripBound`: `strip_bound_linear`, `strip_bound_quadratic` (the $\frac{3}{2}\tau$ bootstrap); `SOURCES_HBETA_N3.md` §4.
  17. `CertBridge`: `cell_invertible`, `cell_sup_bound`, `vertex_bound` (interval-to-norm bridge of the cell certifier); `numerics/h1cert_cell.py`, `h1cert_cell2.py`.
  18. `Tower`: `tower_bounds`, `I1_compress`, `majorant_core`, `lemmaT_core` (algebraic core of the one-step majorant, Theorem 6.1); `H1_ANALYTIC.md`.
  19. `Schur`: `schur_test_weighted`, `schur_test` (weighted Schur test for the spectral norm); `SOURCES_LOC.md` §3.3, `SOURCES_S2_FORM.md` §2, `SOURCES_A3_SIGMA.md` 2.6.
  20. `CombesThomas`: `norm_entry_le`, `coercive_inv`, `combes_thomas`, `combes_thomas_exp`, `combes_thomas_decay`, `combes_thomas_schur` (the proof structure of Lemma CT, the kernel-decay reduction of ledger A1); `SOURCES_A1_KERNEL.md` §7. The identification with the block operators, the kernel-to-row-sum step, (K)/(C), and the rescaling are not formalized (`lean/VERIFY_CombesThomas.md`); the block-valued Schur test is in `BlockSchur`.

21. `Interp`: `multilinear_convex`, `monomial_affine` (`SOURCES_A3_SIGMA.md` 3.2); `window_perturb`, `window_perturb_partial` (`SOURCES_T2_KUNIF.md` 5.2, chained to Theorem K); `rescale_exp`, `rescale_rate`, `K1_iff`, `K1_numeric` (`SOURCES_T2_KAPPA2.md`); `xexp_le`, `xexp_absorb` (`SOURCES_GGAIN_LF.md` §2).
22. `Resolvent`: `resolvent_identity`, `resolvent_bound`, `lower_transfer`, `neumann_perturb`, `resolvent_chain` (`SOURCES_A3_SIGMA.md` §4); `wnorm_mul_le`, `wnorm_neumann`, `pwnorm_mul_le`, `pwnorm_neumann` (weighted Neumann series, row half; `SOURCES_A2_R2.md` §2).
23. `SlabCounterexample`: `slab_fails_10`–`slab_fails_13`, `slab_fails`, `lambda1_lt` (the proof step CMP 96 (2.127) fails for every $L \geq 10$); `SOURCES_A3_GAMMA0_FULL.md` §3.2–§3.3.
24. `BlockSchur`: `block_row_le`, `block_schur_test` (Schur test with block norms $\|K_{xy}\|_2$ in place of $|K|$); `SOURCES_A1_KERNEL.md` §7, `SOURCES_S2_FORM.md` §2 Π3.
25. `GaugeMin`: `variational`, `constrMin_existsUnique`, `lemma5`, `lemma5_full` ($RD^*A_* = 0$ for the constrained minimizer, from (F1), (F3), (F4) as hypotheses); `SOURCES_A3_GAMMA0_FULL.md` §5.

Not formalized: interval arithmetic itself, the enclosure premises P1′–P4, the identity $\rho_1^2 L^2 = \lambda_{\max}(G)$, and the history form of the shooting lemmas.

B Index of results by status

This appendix indexes the results of the paper by status tag; the tags match the verifier files. Here v1–v4 denote the successive versions `THEOREM2_PROOF.md`, `THEOREM2_PROOF_v2.md`, `THEOREM2_PROOF_v3.md`, and `THEOREM2_PROOF_v4.md` of the Theorem 2 notes. The statements that the main theorems take on trust are listed by item in the trust ledger `TRUST_LEDGER.md`, condensed in Section 7.7; ledger IDs are given below where an entry corresponds to a ledger item.

[PROVED+VERIFIED]

- Lemma D′ (a)–(d), K_0, Remark 4 (`LEMMA_DPRIME_VERIFY` grades 1, 2, 5).
- Lemma D (corrected interval), Lemma E (Cg^2 form), nested-interval variant of Lemma D (`LEMMAS_DE_VERIFY`). Lemma E C^1 form: trivial variant, same proof.
- Lemma D″ (tail-sum form, no J) (`KAPPA_BOUNDED_VERIFY` §6).
- Telescoping $S_k = \sum \beta_j(0)$ for the $p \to 0$ coefficient (`BETA_FP_VERIFY` item 1, corrected scope).
- Prop. 2.1 anchor-free identity; exact at the constant field (`BETA_FP_LEMMA43_VERIFY` item 1; `BETA_FP_H_VERIFY` item 1).
- $c = 11N^2/(12\pi^2)$, normalization chain; (F) source (Savvidy eq. (1.4); ledger X5) (`BETA_FP_LEMMA43_VERIFY` item 2; `BETA_FP_H_VERIFY` §4).
- KAPPA Prop. 1.1 as an identity given its fixed-level inputs; Lemma 2.1 (integration by parts); Prop. 2.2; level-independence of the degree -4 part; Theorem K and Theorem 2w as conditional reductions (`KAPPA_BOUNDED_VERIFY` (i)–(v)).
- $\sup_j |\sigma_j^{(2)}(0)| < \infty$; Prop. 4.1 algebra (`KAPPA_BOUNDED_VERIFY` (vi), reason for A4 failure corrected: obstruction = $\partial\sigma^{(1)}$ in Q_j and in (W-2), §7 there).

- `B2_TWOSTEP` two-step recursion; Theorem C1 ((H2) $\Rightarrow$ ($C1\sigma$), factor ρ_2^2/L^2) and Theorem B2 ((H2) $\Rightarrow$ (B2), factor ρ_2^2/L^4), conditional on (H2); threshold $\rho_2 < L$ (no strip loss); (H2) weaker than (H$''$) at $n = 2$ (`B2_TWOSTEP_VERIFY` grades 1, 3, 4). Form of Q_j by read-off (grade 8; `U3_LEMMA51` (3.3) not re-derived).

- `h2_scan` code structure, one-colour reduction, H and Δ_{out} blocks, Gram identity, separable Prop. A form (`H2_LARGE_L_VERIFY` grade 1); tail-bound derivation (4.1)–(4.3) (grade 6, inputs corrected).

- Logic of Theorem B and Lemma 1.3 of `BETA_FP_PW`, as conditional implications (`BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED: doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B; Lemma 4.1, conditional). Its inputs (U), (DF), (T1) are listed under [PROVED, UNVERIFIED] and [ARGUED].

- Prop. E evenness, Theorem W0/W, Theorem A3, Lemma A4, A0, Prop. A2 (`U3_AW_VERIFY` grades 1–4).

- U3 Theorem 4.1 + O5, Prop. A, Prop. B, O6, Theorem 6.1 (given V1; Taylor-data norm), Lemma 5.1 neutral value bound, single-block $m = 2$ failure in $d = 4$, one-step recursion, R2 midpoint identity (`U3_VERIFY`, `U3_REPAIRS_VERIFY`, `U3_LEMMA51_VERIFY`, `U3_LEMMA51_R23_VERIFY`).

- Exact one-step factorization $\rho_2 \le \rho_1(j) \max_l \rho_1(j-1, k_l)$ (`H2_CERTIFY_VERIFY`; Lean `rho2_le`).

- Sources block: Lemma G3 given (Q) (`SOURCES_G3_VERIFY`); LOC (`SOURCES_LOC_VERIFY`); (M-A) given (H$-$) and (M-A$'$) (`SOURCES_MA_VERIFY`, `SOURCES_MA2_VERIFY`); (M-B) with $p_0 \ge 17/(\gamma_0 A_1^2)$ (`SOURCES_MB_VERIFY`); condition (a) of Lemma N3 (`SOURCES_N3A_VERIFY`); the sourced large-field draft B-S.D3, with corrections (`SOURCES_D3_VERIFY`, `SOURCES_D3R_VERIFY`, `SOURCES_D3W8_VERIFY`); (B-N1) given (S) (`SOURCES_HHIST_N1_VERIFY`).

- Cesàro: given convergence, $\kappa_k = o(k) \Leftrightarrow \beta_\infty = c \log L$ (equivalence, not a weaker form) (`THEOREM2_PROOF_VERIFY` Point 4).

- Item 7$'$: the running-maximum lift of C_β to non-monotone pasts (`SOURCES_GGAIN_NONMONO_VERIFY`).

- (KE') by citation and (H_Q) (`SOURCES_QE_KE_HQ_VERIFY`); S2 for G and G_1 (`SOURCES_S2_FORM_VERIFY`, `SOURCES_S2_G1_V7_VERIFY`); fixed-j (S1), (S3), (S2)$|_{t=0}$, (S2c) given (I3) (`SOURCES_T2_S123_VERIFY`).

[PROVED, UNVERIFIED]

- `BETA_FP_U`, no verifier file: Theorem 2.4 ((U) $\Leftarrow$ (T1) + (H2)), Prop. 3.1 ($\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$), Prop. 3.2 ($\gamma_{rec}$ bounded), Corollary 4.1, Theorem 2w by the plane-wave route given (T1), (H2), (DF), LW (5.7), A6, (H0); the argued step "(H2) closes ($C1\sigma$)" of §3.5 there is superseded by `B2_TWOSTEP` Theorem C1 [PROVED+VERIFIED].

- `LEMMA_DPRIME_SWEEP`: 19 cross-scale uses covered by (b) + (c$'$) + (S) (4 of 19 table rows, 11, 12, 16, 18, verified on page images by `LEMMA_DPRIME_VERIFY` comparisons A–E).

- Step "Theorem 2w ⇒ Balaban's Theorem 1" (depends on the sweep and on L5 in the sharpened form of Section 3.4.5).
- Prop. 3.6 assembly with $C(1+j)^2 L^{-2j}$; neutral-sector K-derivative paragraph; $\sigma^{(1)}$ convergence sketch jL^{-2j}.
- Necessity of individual positivity for literal (0.31) at every K (elementary).
- `BETA_FP_M` Prop. 4.3 ($\log|\det|$ bound in the N–O window, given (S1)–(S3)); no verifier file.
- Premise P4 of the one-step certificate (`H2_CERTIFY_P4`; identity (D) in Lean `P4_divergence`); the $L = 13$ certificate sizing after fix G1 (`H2_CERTIFY_L13_G1`).

[NUMERICAL]

- $\rho_1(\infty) = 1.760, \ldots, 2.0895$ at $L = 3, \ldots, 13$; at $L = 13$ the measured $\sup \rho_1^2 = 4.37$ (`H2_CERTIFY_L13`); the certified bound is below.

[CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]

- (H2) at $L = 13$ (Theorem 6.1; ledger B1): $\rho_1(j, K)^2 \le 10 < 13$ for all $j \ge 1$ (including ∞) and all K, hence $\rho_2 \le 10 < 13$ by `rho2_le` (`H1_ANALYTIC`; `H1_ANALYTIC_VERIFY` V0–V9, `H1_ANALYTIC_VERIFY2`, `H2_CERTIFY_P5_VERIFY`). Residue as in Section 6.7.

[VERIFIED REDUCTION]

- (Q) to CMP 99 (3.132), (E_U), and LOC (`SOURCES_Q_VERIFY`, `SOURCES_CDC_VERIFY`); (E_U) to kernel-moment bounds Bałaban does not print (`SOURCES_EU_VERIFY`; the bound $K \le K_0(1 + \ldots)$ is a [DERIVED SKETCH]).
- N3 to (H-hol) + real (H-hist) + (H2-hist) (`SOURCES_HBETA_N3_VERIFY`).
- (H-hol), small field, to Bałaban's conditions with modified constants (`SOURCES_HHOL_VERIFY`); large field to (M-A), (M-B), (M-C), (H-hist2) (`SOURCES_HHOL_LF_VERIFY`); (M-D) to (H-hist) + (H-thr).
- (H−) to CMP 102 (189), an existing trust item (`SOURCES_MA2_VERIFY`).
- (H-hist2) to the real uniform (H-hist), no new input (`SOURCES_HHIST2_VERIFY`).
- Round 3 (ledger A1–A4, B13, B17, B21, B22): A1 kernel-decay form via Lemma CT to CMP 99 Thms 3.1, 3.3, 3.12 + (KE′) + (H_Q) + S2 (`SOURCES_QE_KE_HQ_VERIFY`, `SOURCES_S2_FORM_VERIFY`); A1w to (W_G) + (C) + CMP 96 Lemma 2.1 (`SOURCES_A1_WALK_VERIFY`); A2 with the V7 repair to (W^c) + A7 + two [I] (`SOURCES_A2_R2_VERIFY`, `SOURCES_S2_G1_V7_VERIFY`); A3, full proof with $\gamma_0 \ge m_L/(48d^2L^d)$, and its σ-localized clause (`SOURCES_A3_GAMMA0_FULL_VERIFY`, `SOURCES_A3_SIGMA_VERIFY`); (M_j) to (I3), (G'_A), Prop. 4.4 region 1 (`SOURCES_T2_S123_VERIFY`); A4 to A1 + CMP 102 (`SOURCES_A1_A4_VERIFY`); large-field (H-hist) (`SOURCES_GGAIN_LF_VERIFY`); (M-C) to the printed CMP 116 §2 conditions (`SOURCES_MC_VERIFY`); the (S) clauses of B21 (`SOURCES_B21_S_VERIFY`); the $\kappa \to \kappa/2$ recovery B22 (`SOURCES_T2_KAPPA2_VERIFY`).
- Item 9 (K-independence in the shooting) closed: β_j is the whole-lattice coefficient by definition, Lemmas D′/D″ run at fixed K (`SOURCES_T2_KUNIF_VERIFY`).

[INTERVAL ENCLOSURE, VERIFIED]

- Point enclosure $\rho_2(\infty, K^*) \in [1.9993, 2.1273] < 3$ at $L = 3$; P1 (loop-family coefficients exact rationals); Oishi–Rump bound on $\|\mathfrak{B}^{-1}\|$ (`H2_CERTIFY_VERIFY`).

[ARGUED]

- (PI); (M_j) (Prop. 4.4 of `BETA_FP_M`); (T1) (plane-wave form of A3, `BETA_FP_U` §2.3); (DF) (`BETA_FP_U` §3.4); A6 (v4 label, not ledger A6) = plane-wave vs constrained-minimizer expansion point (uniform in k); $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ bounds; (H″) ⇒ (A) in full ⇒ (B2) (`U3_AW` §4.7 sketch); C^2, C^3 of $\sigma^{(1)}$ under (H2) (not written); j_* size; the rate $O(\Lambda^{-2} \log \Lambda)$.

[OPEN]

- (H2) at $L \neq 13$ (numerical at $L = 3, 5, 9$; certified at $L = 13$, see above); (B2) is reduced to it [VERIFIED REDUCTION]. The residue of A1w: (W_G) for the final G, G_1 (reduced to printed sketches with WG-1, `SOURCES_WG_VERIFY`), the Z_0-compressed variant (O3), and the complex α'-space (O4). The residual inputs of (M-C): (I-ε), (I-V0), (M1). (T1), (DF), A6 (v4 label) for Route B (ledger B9). V1. The residue of (M_j): (I3), (G'_A), Prop. 4.4 region 1. L5 (domain of β; nested-interval D″ [NOT WRITTEN]). The finite-volume identification of the step, R-term part (ledger B8; an input of Theorem 1). The (E14-c) clause of B21. (A) in full. See Section 7.

[CORRECTED BY VERIFIERS SINCE V3]

- `B2_TWOSTEP` G1: the certification of (H2) is a computer-assisted proof project, not a check (`B2_TWOSTEP_VERIFY` grade 6).
- `H2_LARGE_L`: L-trend rate claim (grade 5); j_0 inputs, δ_m 45–90× larger, $a_*(L = 5) = 0.0107$ (grade 6), giving $j_0 \approx 8$ ($L = 3$), ≈ 6 ($L = 5$).
- `BETA_FP_PW`: comparison and numerics (`BETA_FP_PW_VERIFY` grades 5, 6).

[REFUTED]

- (HS) $\sup \|H_j\| < L\sqrt{L-1}$ at $L = 3$.
- Not a refuted statement, but a failed method: the one-step sup-norm C^1 recursion at $L = 3$ (factor 1.005–1.033 for $j \geq 2$; Section 3.7.2).
- `U3_LEMMA51` Theorem 4.1 for $m = 2$ with a single block-M constraint in $d = 4$.
- v1 claims withdrawn: "every early step must be certified for Theorem 1" (false: Lemma D′/D″); "(H′) is weaker than $\beta_\infty = c \log L$" (it is equivalent); "the whole $\log \Lambda$ sits in the fine piece" as a fact (it is equivalent to the open statement); "one-loop parts hold for SU(N)" (Lemma 5.1's charged/neutral split is the SU(2) Cartan split); (3.2) with Cj^2L^{-2j} (0 at $j = 0$); the (M) resolvent bound $\|\mathcal{K}_t^{-1}\| \leq C/|f|$ (Nielsen–Olesen zero); (H) with IR weights as stated (UNSUPPORTED).

[Balaban]

- (H0) smoothness of β_{j+1} in g (CMP 109 (1.22), pp. 263–264), at odd $L > 11$ only; uniformity in j is our reading and uniformity in the history is not printed (ledger A5); Theorem 1 and the inductive bounds of CMP 109, 116, 119, 122 and the propagator papers CMP 95, 96, 98, 99, 102. Uniform level bounds partly cited from CMP 95/96/99 (O6 replaced the main citation). Each Bałaban input that is printed without proof, or used without being printed, is a Group A row of the trust ledger.

[verified quote]

- Theorem 2 and Theorem 1 wording, "separate paper", p. 298 history sentence, (0.1) with $1/g_k^2$ and χ_k, (1.3)–(1.5) (print misprint $E^{(j+k)}$ read as $E^{(j+1)}$), (2.6)–(2.9) of CMP 119, items A–E of CMP 122 (`BALABAN_EXCERPTS_CHECK`; `LEMMA_DPRIME_VERIFY`; `LEMMA_DPRIME_SWEEP` page images). `BALABAN_EXCERPTS_CHECK` item 10 (CMP 116 has no C^m-in-g statement) is unchanged since v3; the item 11 readings are verified, with corrections, in `SOURCES_BOOKKEEPING.md` (former ledger A22).

[numerical] see Section 6.

C Label map

The numbered environments of this paper carry the names used in the project files. The first table maps them. The second table maps the section numbers of `THEOREM2_PROOF_v4.md` ("v4") and `ASSEMBLY.md` to the sections of this paper; section references inside the project files use the old numbers.

this paper	name in the project files	source
Theorem A	Route K summary: Theorem 2w ⇒ Theorem 1	`THEOREM2_PROOF_v4` §6, §10
Theorem B	Theorem 1.3 (summary)	`ASSEMBLY` §1.3
Theorem 3.1	Theorem 2w	`THEOREM2_PROOF_v4` §1.3
Lemma 3.2	Lemma D′	`LEMMA_DPRIME`; v4 §2.1
Remark 3.3	Remark 4	`LEMMA_DPRIME`; v4 §2.1
Lemma 3.4	Lemma D″	`KAPPA_BOUNDED` §5.3; v4 §2.2
Lemma 3.5	Lemma D	`LEMMAS`; v4 §2.4
Lemma 3.6	Lemma E	`LEMMAS`; v4 §2.4
Proposition 3.7	Prop. 2.1 (anchor-free identity)	`BETA_FP_LEMMA43`; v4 §3.2
Theorem 3.8	Theorem K	`KAPPA_BOUNDED`; v4 §4.1
Theorem 3.9 (Theorem B_{PW})	Theorem B = Theorem 3.2	`BETA_FP_PW`; v4 §4.2
Theorem 3.10	Theorem C1	`B2_TWOSTEP`; v4 §5.3
Theorem 3.11	Theorem B2	`B2_TWOSTEP`; v4 §5.3
Definition 4.1	(H-U)	`ASSEMBLY` §1.2
Theorem 4.2	Theorem 1.3	`ASSEMBLY` §1.3
Theorem 4.3	Theorem 1.4	`ASSEMBLY` §1.4
Proposition 4.4	Proposition of (B-C)	`ASSEMBLY` §2.4
Lemma 4.5	finite-form lemma of (B-RP)	`ASSEMBLY` §2.7
(GC1)–(GC3)	(G1)–(G3) of the dependency table	`ASSEMBLY` §3
(S_{src})	(S), final sourced bound	`SOURCES_D4D5`

this paper	name in the project files	source
items H, 1–13, B1–B8	items of the open list	`THEOREM2_PROOF_v4` §10

old section	this paper
v4 §1.1	Section 1.2
v4 §1.2	Section 3.1
v4 §1.3	Section 3.2
v4 §1.4	Section 3.3
v4 §2 (notation)	Section 2.3
v4 §2.1	Section 3.4.1
v4 §2.2	Section 3.4.2
v4 §2.3	Section 3.4.3
v4 §2.4	Section 3.4.4
v4 §2.4 (L5)	Section 3.4.5
v4 §3.1	Section 3.5.1
v4 §3.2	Section 3.5.2
v4 §3.3	Section 3.5.3
v4 §3.4	Section 3.5.4
v4 §3.5	Section 3.5.5
v4 §4	Section 3.6
v4 §4.1	Section 3.6.1
v4 §4.2	Section 3.6.2
v4 §4.2a	Section 3.6.3
v4 §4.3	Section 3.6.4
v4 §5	Section 3.7
v4 §5.1	Section 3.7.1
v4 §5.2	Section 3.7.2
v4 §5.3	Section 3.7.3
v4 §5.4	Section 6.4
v4 §5.5 (arithmetic)	Section 3.7.4
v4 §5.5 (values)	Section 6.5
v4 §5.6	Section 6.8
v4 §5.7	Section 3.7.5
v4 §6	Section 3.8
v4 §7	Section 3.9
v4 §8.1	Section 6.1
v4 §8.2	Section 6.2
v4 §8.3	Section 6.3
v4 §9 (Lean)	Appendix A
v4 §9 (other tags)	Appendix B
v4 §10 (open list)	Section 7
v4 §10 (hypothesis table)	Section 4.6
v4 §11	Section 1.5
v4 preamble (tag legend)	Section 1.4
v4 preamble (range of L)	Section 2.2
v4 change logs	Appendix D
`ASSEMBLY` §0 (notation)	Section 2.1
`ASSEMBLY` §0 (observables)	Section 2.5
`ASSEMBLY` §1.1	Section 4.1
`ASSEMBLY` §1.2	Section 4.2
`ASSEMBLY` §1.3	Section 4.3

old section	this paper
ASSEMBLY §1.4	Section 4.4
ASSEMBLY §2	Section 4.5
ASSEMBLY §2.1	Section 4.5.1
ASSEMBLY §2.2	Section 4.5.2
ASSEMBLY §2.3	Section 4.5.3
ASSEMBLY §2.4	Section 4.5.4
ASSEMBLY §2.5	Section 4.5.5
ASSEMBLY §2.6	Section 4.5.6
ASSEMBLY §2.7	Section 4.5.7
ASSEMBLY §2.8	Section 4.5.8
ASSEMBLY §2.9	Section 4.5.9
ASSEMBLY §3	Section 4.6
ASSEMBLY §4	Section 4.7
ASSEMBLY corrections	Section 4.8

D Revision history

- *Theorem 2 notes, v3.* Built on THEOREM2_PROOF_v2.md with the corrections of THEOREM2_PROOF_v2_VERIFY (cited there as v2_VERIFY): the range of L (odd, $L > 11$, or (H0) at this L as a hypothesis); L5 sharpened (the clamped shooting leaves Bałaban's domain; the nested-interval variant of Lemma D$''$ is [NOT WRITTEN]); (H$''$) $\Rightarrow$ (B2) tagged [ARGUED]; "independent" routes replaced by "formally distinct", with the points $\theta_* < 1$, k-uniform β identification, and the first-order vertex smoothness of (U); the independence record completed (ONLINE_SWEEP §2 as the first channel; the O(3) paper cited); open list extended (item 9, B5, item 10, hypothesis table); sweep count "4 of 19 table rows"; γ_* depends on R through B'. At that time BETA_FP_PW_VERIFY.md had no grade, and Route B stayed [PROVED, UNVERIFIED]. The grades of LEMMA_DPRIME_VERIFY and KAPPA_BOUNDED_VERIFY §9(iv) were cited explicitly.

- *Theorem 2 notes, v4.* Added the material verified after v3: B2_TWOSTEP.md with B2_TWOSTEP_VERIFY.md, H2_LARGE_L.md with H2_LARGE_L_VERIFY.md, lean/README.md (Lemma D$''$ in Lean), BETA_FP_U.md (Route B inputs), and numerics/NUMERICS.md §7. The section on (B2) was rewritten around (H2) (Theorems C1, B2; threshold $\rho_2 < L$; one-step sufficiency at $L \geq 5$; certification plan; corrected L-trend and j_0 inputs). The ledger and open list were updated, with item H as the single uniform hypothesis of both routes. KAPPA_BOUNDED_VERIFY states the (B2) obstruction in its §7 and §9(vi). BALABAN_EXCERPTS_CHECK had no change since v3.

- *Tag change in the LaTeX version of v4.* The logic of Theorem $\mathrm{B_{PW}}$ is tagged [PROVED+VERIFIED, CONDITIONAL], citing BETA_FP_PW_VERIFY §6 items 1–4, per the tag decision in PLAN.md (2026-10-28). The Markdown v4 kept [PROVED, UNVERIFIED]. The inputs (U), (DF), (T1) from BETA_FP_U stay [PROVED, UNVERIFIED].

- *Assembly.* ASSEMBLY.md with the corrections of ASSEMBLY_VERIFY.md (Section 4.8). The verifier also found the dependency table stale in five rows (grade 7; no overclaim): B-T2.2, B-T2.3, B-T2.4, B-S.D2C, B-S.D3 lagged PLAN.md. The table was rebuilt with a tag for every block and the sub-blocks added later (Route K, Route B, (H2), (H-thr), (G-gain), (FX)). Minor items: SOURCES.md §1.3 gives $c_j = 1 + O(g_j^4)$, while this paper uses $O(g^3)$ per SOURCES_VERIFY item 8 (the corrected value); the variance estimate drops a harmless g_0^4 (logarithmic) factor.

- *This version.* The Theorem 2 chapter and the assembly chapter are merged into one article: one notation section, one status legend, numbered environments (Appendix C), one numerical section, one list of open items, and the hypothesis table of v4 §10 merged into the dependency table. Three names were changed to avoid collisions (Section 2.6). The dependency table and the open list were brought to the current state of `PLAN.md`: (FX) is reduced to N3 (O2 and R2 [PROVED+VERIFIED], O3 [PROVED+VERIFIED, CONDITIONAL ON N3], O4 avoided by the main form; previously O2, O3 [PROVED, UNVERIFIED] and R2, O4 [OPEN]), and the point enclosure of (H2) at $L = 3$ is verified (Section 6.6). No theorem, lemma, or proof changed.

- *Round 3.* A1–A4 regraded to verified reductions (A3 a full proof with a corrected constant), A1w added and reduced, item 9 closed (B8 moved to the input set of Theorem 1), item 7 restated as $7'$, G4 off-chain, B13, B17, B21, B22 regraded, Lean files 21 → 22 (`TRUST_LEDGER.md` round 3; `PLAN.md`).

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